Paying to Be Paid Early: Debt Bondage in India's Brick Kilns

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October 23, 2026

Latest draft available here

Abstract

Bonded workers take credit from their employer, and voluntarily forgo the freedom to leave until this is repaid. This paper asks why, using India's brick kilns as a setting, to which 15 million workers migrate annually on advance payment contracts. The charge on the advance is not stated: it is built into a lower piece rate, and I test whether workers make mistakes when calculating it. Partnering with employers, I advertise real vacancies with randomised terms to 1,494 Odisha households. Workers favour high advances by 10 percentage points, implicitly borrowing at 19.5% per month. Demand concentrates amongst credit constrained workers and spikes after negative income shocks, as a rational borrower's would. However, workers fail the arithmetic the trade-off requires, and three quarters wrongly believe high advance jobs pay more. Making the charge explicit in an information intervention fails to correct these deeply entrenched beliefs or dampen advance demand, suggesting regulation may be necessary to protect workers from costly mistakes.

*This project is registered in the AEA RCT Registry (0016828, 0019111) and received ethical approval from the University of Warwick.

1 Introduction

In many labour markets, workers make choices which leave them worse off. They deny the health hazards of their work to avoid cognitive dissonance (Akerlof and Dickens, 1982), stay in low wage jobs because they underestimate what other employers pay (Jäger et al., 2024), and sign away their right to work for a competitor in clauses they see only after accepting a job (Starr et al., 2021). Governments introduce legislation to override these choices, but do so at the risk of removing options which were genuinely efficient.[1]

Of all the choices workers make, none is more contested than voluntary servitude. Bonded workers trade freedom for credit, pledging labour to their employer until the loan is repaid. Human rights campaigns and legislative efforts date back centuries.[2] Nonetheless, this form of modern slavery remains a blight on migrants in developing countries, including India's 15 million brick kiln workers (Eil et al., 2020). Whether they need protecting from these choices has never been tested empirically.

A ban protects no one if better jobs are not available. However, my observational evidence on brick kiln contracts suggests otherwise: some employers offer lower advances paired with higher piece rates. Assuming the same worker can access both, choosing the high advance amounts to borrowing at a steep price.

Nor do bans protect bonded workers who can access better jobs, but rationally prefer bondage. When the poor face credit constraints, pledging labour may be their only means of smoothing consumption. At brick kilns, the advance is large enough to absorb major shocks: the median household took 90,000 Rs last season, more than ten times its savings for 80% of households. Under this 'market failure' explanation for debt bondage, a ban removes a lifeline.

The case for a ban rests instead on the other leading explanation: cognitive biases distort the choice to enter servitude. Present biased workers may pay more for upfront cash than their ex ante selves would prefer. Workers may also misjudge the cost itself, especially when it is implicit in contract terms rather than stated. Low financial literacy raises this risk. Under this 'behavioural' explanation, a ban protects workers from their own mistakes.

The costs of bonded labour contracts have only been studied theoretically, and without attention to the biases which determine the optimal policy response. I fill these gaps by answering three research questions. First, how much future income do workers forgo in exchange for advance payments? Second, how does demand for advances vary with access to alternative sources of credit? Third, do behavioural and cognitive biases distort workers' choices?

I answer these questions through incentivised recruitment experiments. These are embedded within household surveys: respondents are informed about an upcoming vacancy at a partner brick kiln in Hyderabad. If they apply, contact details are passed on to employers. Surveys take place during

[1] For the examples provided, legislative efforts include mandatory safety standards, pay transparency laws requiring employers to disclose salary ranges, and bans on non-compete clauses.

[2] See Genicot (2002) and von Lilienfeld-Toal and Mookherjee (2010) for a review of these bans.

the brick kiln off-season in 71 villages of Western Odisha, a hotspot for out-migration to kilns.

Round 1 takes place in September 2025 and samples 1,494 households. Those randomly assigned to a 'Choice' arm choose between two contracts. One pays a high advance. The other pays a low advance but a higher piece rate, and therefore more overall. For an average productivity worker, choosing high over low advance amounts to borrowing at a 19.5% monthly interest rate. Workers who choose it are paying for bondage, not settling for it. How many do so at this price answers the first research question.

The second research question asks whether demand reflects rational borrowing. I test for heterogeneity in contract preferences by credit access, exploiting stark variation across households. In a follow-up one month later, respondents can revise baseline application decisions. This provides a better identified test. Using a triple difference, I ask whether unexpected income shocks shift applications towards high advance.

This follow-up also tests whether debt bonds trigger present bias, the first of two risks under the third research question. The follow-up takes place in the days before the recruitment meetings at which advances are paid. For those whose incomes have been stable since baseline, present bias predicts a shift towards high advance.

The second risk is that workers underestimate what debt bonds cost. Two additional treatment arms test this. Those randomly assigned to 'high advance' receive a single high advance offer. 'Low advance' workers receive a lower advance, higher piece rate offer. Both arms estimate future earnings under their assigned contract, and I test whether the average difference accurately reflects contract terms. Round 2 builds on this, testing whether misperceptions can be corrected. I revisit 1,078 Round 1 households in August 2026. Respondents can apply for a vacancy for the new brick season, choosing high or low advance. Before deciding, half are randomly told what each would pay. If workers prefer large debt bonds because they underestimate their costs, information should increase low advance applications.

I find that workers pay dearly to be paid upfront. Households are 10 percentage points more likely to apply for high advance. The low advance alternative which they forgo has higher net present value at any monthly interest rate below 19.5%. This partly reflects credit market failures: high advance applications concentrate amongst credit constrained workers, and rise by 37 percentage points after negative income shocks. However, pledging labour for credit prevents workers from seeing what this costs, through two channels. First, workers struggle to calculate pay from contract terms, and instead anchor on past earnings. As a result, the gain from low advance contracts barely registers. Three quarters believe the high advance offer pays more overall, when the opposite is true. Second, needing liquidity makes workers unwilling to hear what advances cost. Telling workers what each offer pays raises demand for the high advance option by 4 percentage points. This is driven by credit constrained workers, who are motivated to dismiss news that advances are costly. By contrast, present bias finds no support: demand for high advances does not strengthen when pay-out is imminent.

Debt bonds therefore relieve and exploit workers at the same time. Workers accept lower pay in exchange for liquidity, but cannot judge how much lower. Employers can exploit this on two margins, misleading workers to raise profits. At recruitment, they can offer front-loaded contracts, knowing that deductions in future pay go largely unnoticed. Once the contract is agreed, the same numeracy failures conceal underpayment. Whereas a fixed wage worker can spot theft immediately, a bonded worker can only detect it by deducting debt repayments from aggregate earnings. Indeed, over half of migrants in my sample took home far less last season than their own contract terms imply. Most nonetheless believe their pay was fair. Low earnings then become the anchor for pricing new contracts, so workers see only a fraction of the gain from better paying options. Employers therefore gain little from competing on pay. My information experiment suggests that simply telling workers how much more alternative contracts pay can even backfire.

I now set out the evidence for each of these findings in more detail, starting with the costs workers pay for advances. In the Choice arm, the low advance contract carries a higher piece rate, and so pays more overall. At average expected output, choosing the high advance means giving up around 2 Rs in future pay per 1 Rs of additional advance. Nonetheless, 41.7% of households apply with a preference for the high advance contract, against 32% for low advance ($p = 0.019$). Costly high advance contracts are also more popular in a hypothetical multiple price list (MPL) covering the full sample. Over half prefer the high advance contract to every low advance alternative shown. Declining the most generous of these is only rational at a monthly interest rate above 24.3%.

Only households with no cheaper way to borrow would rationally accept interest rates this high. Access to credit varies widely in my sample. 40% of households have outstanding loans, at a median monthly rate of just 1.5%, a fraction of what the high advance costs. At the other extreme, 52% have no access to credit, making advances their only source of liquidity. In every elicitation, demand for the high advance contract is concentrated amongst this group. In the single offer arms, for example, those who can borrow are 11.9 percentage points less likely to apply under the high advance contract. No such gap exists amongst those who cannot borrow ($p < 0.01$ for the difference). I complement this descriptive evidence with a triple difference, exploiting unexpected income shocks between survey rounds. A negative shock, such as crop failure or illness, raises the chance that a household needs more credit than it can obtain. Households hit by such a shortfall raise high advance applications by 36.8 percentage points ($p < 0.01$). Those receiving an unexpected inflow raise low advance applications by 25.8 percentage points ($p < 0.01$). In a placebo test, shocks twelve months prior produce no such shifts, and the estimates survive controls for baseline characteristics which predict shock exposure.

Demand therefore responds to liquidity, as a rational decision maker's would. A rational decision maker must also know what the loan costs, which brings me to the third research question. The cost of an advance is never stated. Recovering it means multiplying a piece rate over hundreds of thousands of bricks, deducting the debt, and repeating this for each contract. A calculation module shows that this is beyond most workers. Rather than calculating pay directly from contract terms,

workers anchor estimates in what they took home last season. This anchor is a low one. Most migrants work under high advance contracts which leave little to collect at season's end, and over half are paid less than even those terms promise. Against earnings this low, the low advance contract advertised in Round 2 is an outlier: almost no migrant in my sample took home as much last season as this would pay. Control respondents estimate pay under each contract at their own expected output. They understate the gain from low advance by more than half.

If these errors were purely informational, performing the calculations on workers' behalf should correct pay expectations and raise low advance applications. Instead, minutes after being told what each contract would pay at their own expected output, treated respondents give pay estimates no different to those in control, still anchored in low past pay. Job applications follow the same pattern. Low advance applications fall by 4 percentage points, and high advance applications rise by the same amount, despite the intervention showing this contract to be *less* lucrative. The dismissal of the information also appears partly motivated, as it is concentrated amongst those for whom liquidity is most valuable. High advance applications rise by 8 percentage points among households who cannot borrow, and by 11 percentage points among those hit by a recent income shortfall. They do not change among households with credit access or stable incomes.

By contrast, the other distortion which I test, that advances trigger present bias, finds no support. The evidence comes from the Choice arm revisit, held in the days before advances are disbursed. This is when the temptation of immediate cash should be strongest. Among households whose circumstances remained stable between the two surveys, preferences do not strengthen for the high advance contract. If anything, they move away from it. Moreover, survey based measures of time inconsistency do not predict demand for upfront pay at baseline.

The status quo in the seasonal labour market which supplies India's kilns therefore involves workers willingly bonding themselves to employers in exchange for early pay, even at a steep price. This price is only partly seen, meaning employers can raise it without workers noticing. An enforced ban would end this, but at the cost of the genuine relief which advances provide to households with no other way to borrow. An alternative is to make the cost visible, as disclosure rules do for consumer credit. However, my information experiment shows how easily this can go wrong. Handing over a correct final number proves counterproductive where numeracy is low and the underlying mental model is entrenched. Those in most urgent need of cash move further towards the high advance contract. An effective intervention may instead involve a broader education programme, designed for those with limited formal schooling, which teaches how earnings are calculated rather than stating the answer. A worker who can do this calculation can make a better informed choice at recruitment. They can also detect underpayment once hired, which my results suggest is widespread but unnoticed.

The literature on bonded labour has to date been almost entirely theoretical. One strand rationalises why bonded labourers might accept lower wages, pointing to what these contracts provide workers where credit markets are incomplete. In particular, pledging future labour as collateral

turns even the poorest households into viable borrowers (Srinivasan, 1989; von Lilienfeld-Toal and Mookherjee, 2010). A wage spread across both seasons also smooths consumption against seasonal income fluctuations (Bardhan, 1979, 1983; Mukherjee and Ray, 1995).[3] A second strand extends these models to general equilibrium. Permitting bondage can then reduce welfare, by crowding out institutional lenders (Genicot, 2002) or by raising lenders' rents (von Lilienfeld-Toal and Mookherjee, 2010). Arguments of this kind would motivate a ban even if workers were perfectly rational, which both strands take for granted. Three gaps follow. First, the size of the pay discount which bonded labourers accept has never been measured. To my knowledge, this paper provides the first such evidence. I advertise real contracts and recover the interest rate which workers will accept for upfront pay. Second, the liquidity value of bonded labour arrangements has never been documented empirically. Third, the rationality assumption has never been investigated. I test it directly, against the two distortions most likely to apply to a large upfront payment made for labour supplied later.

A second, broader literature shows that this same market failure distorts labour supply outside bonded labour. Previous studies have shown that credit constrained workers forgo profitable investments, including seasonal migration (Bryan et al., 2014; Kleemans, 2023) and better job matches (He and le Maire, 2023). Most relevant to my setting, they also give up lucrative employment opportunities for alternatives which pay sooner. For example, workers take up lean season casual work at the expense of their own farms (Fink et al., 2020), choose immediate-paying self-employment jobs over wage jobs (Scarelli and Margolis, 2025), and favour daily or same-day pay contracts at the cost of lower earnings (Kapoor, 2026; Scarelli, 2025). The pay-in-advance contracts which I study go further still, converting a season of future labour into a loan large enough to smooth major shocks. I show that demand for advances varies with liquidity as this literature would predict, concentrated among households who cannot borrow and rising in response to unexpected income shortfalls. This literature focuses on the liquidity benefits of earlier pay. I also test the behavioural and cognitive risks which may accompany it, to which I now turn.

A third literature concerns self-control in intertemporal financial decisions. Present bias sustains costly borrowing (Meier and Sprenger, 2010). Workers who know this will pay to protect themselves against it. Kenyan dairy farmers accept a lower milk price for monthly rather than daily payment (Casaburi and Macchiavello, 2019), Malawian tea workers defer wages without interest (Brune et al., 2021), and time inconsistent savers take up commitment products (Ashraf et al., 2006; Bauer et al., 2012). Advances are the mirror image of these devices, bringing income forward into a single large payment rather than pushing it into the future. They share this structure with Earned Wage Access services, which allow workers to draw wages ahead of payday and are expanding across developed and developing countries alike (International Labour Organization, 2025). These settings are primed for self-control failures, yet I find that workers exhibit none. Preferences do

[3]This literature has also modelled other benefits of bonded labour contracts, such as securing a fixed wage, thereby insuring workers against wage fluctuations in casual labour markets (Bardhan, 1979, 1983; Mukherjee and Ray, 1995; Basu, 2002). These are less relevant here, as my focus is the trade-off between bonded labour contracts differing in the size of the advance.

not shift towards the high advance contract when its disbursal is imminent, in contrast to the reversals which Giné et al. (2018) document as pay-out dates approach. Nor do survey measures of present bias predict demand for advances. This is consistent with evidence that time inconsistency is considerably weaker over money than over effort (Augenblick et al., 2015).

A fourth literature documents the costly financial mistakes which follow from cognitive limitations. In the United States, employees choose dominated health plans (Bhargava et al., 2017) and households understate the compounding of interest (Stango and Zinman, 2009, 2011). Comparable evidence has more recently emerged from poorer countries, with Zambian farmers underestimating their future spending because familiar expenses fail to come to mind (Augenblick et al., 2023). None of this work, however, concerns the labour contracts through which poor households earn most of their income. These distortions are a particular risk under bonded labour contracts, as the earnings they imply are never stated. As piece rates rise and debts fall, my respondents adjust their estimates by only a fraction of the correct amount. They anchor instead on past earnings, themselves depressed by underpayment which they lack the numeracy to detect. Anchoring of this kind has been shown to hold workers in low wage jobs elsewhere (Jäger et al., 2024).

These errors point to an obvious remedy, which my final experiment tests: an information treatment which performs the arithmetic on the worker's behalf. Similar interventions have curbed payday borrowing (Bertrand and Morse, 2011) and raised the working hours of mothers who had underestimated the cost of part-time work (Costa-Ramón et al., 2026). In my setting the treatment does not shift choices, and instead entrenches demand for large advances among those who most need the liquidity these contracts provide. This is in line with evidence that misspecified mental models survive abundant feedback (Hanna et al., 2014; Esponda et al., 2024). People also discount information which contradicts beliefs they are motivated to hold (Kunda, 1990; Thaler, 2024), to the point where it hardens their initial position (Fryer et al., 2019). The market failure and the cognitive failure therefore compound one another: those who most want advances are the least willing to hear what they cost. The literature on preferences over payment timing usually treats liquidity needs and behavioural biases apart. My results show why they belong together.

The rest of this paper proceeds as follows. Section 2 provides contextual information about India's brick kiln industry. Section 3 presents a simple model of contract choice under advance payment. Section 4 describes my experimental design and data collection, and Section 5 presents my empirical specifications. Section 6 presents results on demand for advances, and Section 7 tests whether this demand reflects credit constraints or cognitive limitations. Section 8 investigates the origins of contract misvaluations, and tests empirically whether an intervention can correct these. Section 9 discusses and concludes.

2 Context and Setting

This section serves two purposes. Section 2.1 places brick kilns within the wider family of bonded labour arrangements, highlighting two features which these contracts share: an upfront payment which serves both parties, and a cost implicit in contract terms which may be hard to quantify. Section 2.2 then uses my own survey data collected in brick kiln out-migration hotspots of Western Odisha to establish three facts about the setting on which the rest of the paper builds: kiln work fills the agricultural off-season, advances are large and appear to come at the cost of lower piece rates, and workers return season after season despite harsh conditions.

2.1 Bonded Labour and Brick Kilns

Brick kiln contracts are one instance of an older and wider practice, in which workers receive upfront payments which they are then obliged to repay out of earnings. The practice persists despite bans, as debt bonds can solve problems for both parties. For employers, advances can lock in a workforce for peak periods, a strategy used by India's farmers when recruiting labourers ahead of the main agricultural season (Bardhan, 1979; Mukherjee and Ray, 1995). For asset poor households, they extend access to credit: under Nepal's Kamaiya and Kamlari systems, landless families pledge members as agricultural or domestic workers in exchange for loans (World Organization Against Torture, 2006; KidsRights Foundation, 2014). Debt bonds can also improve job matching by overcoming fixed costs. Europeans covered passage costs to the Americas through indentured servitude agreements, and a 'credit-ticket' system later brought Chinese labourers to California (Galenson, 1984; Cloud and Galenson, 1987). Today, Thai fishermen are advanced recruitment fees by brokers (Human Rights Watch, 2018).

These benefits come at a cost: debts are deducted from future earnings, potentially with interest, which will be lower still if bonded contracts offer reduced wages or piece rates. Only once this debt is repaid can a worker leave a job. In this sense, debt bondage differs fundamentally from other labour supply decisions: estimating income requires projecting earnings over an extended time horizon, and comparing these against total debt. Aggregating earnings can be cognitively difficult. For example, brick kiln workers must multiply a piece rate over hundreds of thousands of bricks. Interest charges introduce further complications, and individuals have been shown to systematically underestimate these for regular loans (Stango and Zinman, 2009). Together, this introduces the risk that workers misvalue debt bonds. Simultaneously, the knowledge that workers struggle to track net income creates the opportunity for employers to covertly siphon pay. For example, reports describe landlords fabricating Kamaiya debt accounts (World Organization Against Torture, 2006) and Thai brokers and boatswains inflating debts and making fraudulent deductions from wages (Human Rights Watch, 2018).

Brick kilns in South Asia combine these features, and operate on a scale which has centred them within the contemporary bonded labour debate. India is the world's second largest clay fired

brick producer, with an estimated 144,000 kilns employing 15 million workers (Eil et al., 2020), predominantly from lower caste backgrounds (Maithel, 2013). Western Odisha, the setting of my data collection, sends hundreds of thousands of migrants to kilns each year (Odisha Labour Directorate, 2024), during the dry months when kilns operate. Far from being celebrated as a vital source of employment for poor households, kilns draw widespread criticism for their reliance on debt bondage for recruitment. They attract human rights abuse allegations from international organisations and journalists (Anti-Slavery International and Volunteers for Social Justice, 2017; Mitra and Valette, 2017; Press Trust of India, 2017; Deccan Herald, 2023), are the subject of field research arguing that advances trap workers in recurring debt (Guerin et al., 2015; Kumari, 2018; Majumder, 2015; Sengupta and Vijay, 2017), and in 2024 prompted Odisha to launch a taskforce to curb migration to kilns (Times of India, 2024).

The three facts which follow describe the setting in which I quantify demand for these contracts. Of these, Section 2.2.2 matters most for the design: the cost of a kiln advance appears to take a single form, a lower piece rate on the job, which permits a clean test of the premise behind criticisms, that workers bond themselves at a significant monetary cost which they may only partially see.

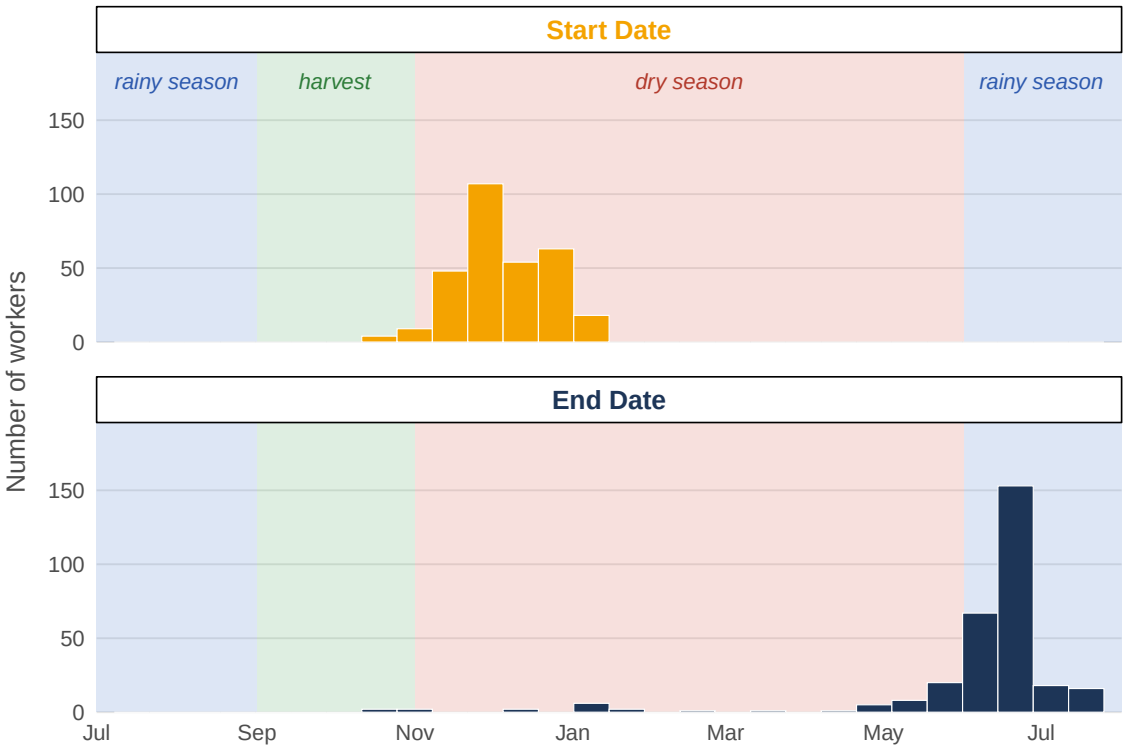
2.2 Three Facts about Kiln Migration from Western Odisha

2.2.1 A Natural Complement to Monsoon-Dependent Agriculture

Figure 1 plots the expected work period for sample brick kiln workers. Most migrate soon after the "Kharif" (July-September) harvest at the onset of the dry "Rabi" season. They then return in June, with the start of the next Kharif. This seasonality is a function of the production process specific to the traditional kilns which dominate South Asia: bricks are moulded from wet clay and then left to dry in the sun before firing, and any rain destroys bricks during this drying process (Mitra and Valette, 2017) (see Figure A1 for photographs depicting these steps). This seasonality also creates a natural complement with the time of year when rainfed agriculture is least productive, particularly in Western Odisha where droughts are commonplace (Odisha State Disaster Management Authority, 2025). In this region, 38% of gross cropped area is irrigated against a national figure of 49%, collapsing the crop calendar into a single Kharif paddy season for many farmers (Mitra et al., 2024).

Employers tap into this large temporarily unemployed agricultural workforce by hiring middlemen in affected rural areas ahead of each production season. Around the onset of the monsoon harvest, middlemen approach workers with offers of employment for the upcoming Rabi season under terms set by their partner employer. I next describe how advances figure in these offers.

Figure 1: Seasonality of Brick Kiln Migration



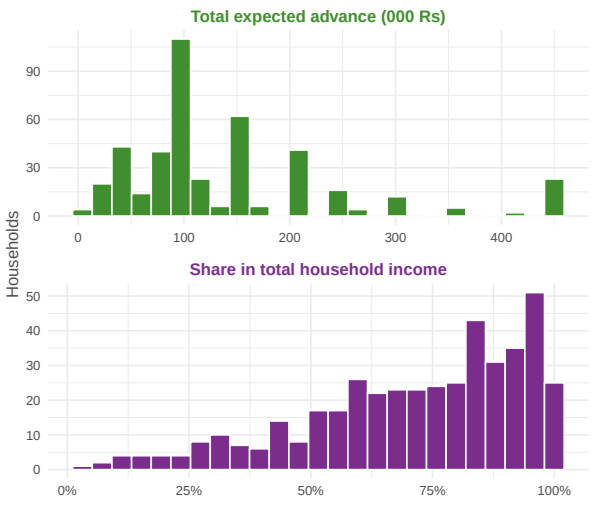
Note: This figure plots the distribution of the dates at which brick kiln workers report that they expect to start and finish work during the upcoming brick season ($N = 315$). These have been overlaid with illustrative dates for the rainy, harvest and dry seasons in the study region of Western Odisha.

2.2.2 Advance Payments at a Steep Cost

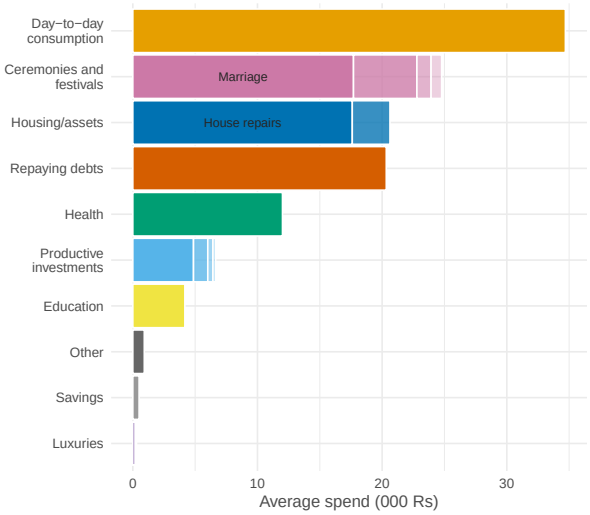
For the vast majority of workers, a work agreement is sealed by an advance, paid by employers but distributed by middlemen pre-migration. Indeed, 88.1% of sampled brick kiln workers for the 2025-26 season report having received one. Workers are recruited as a family: multiple members migrate together, then work and receive pay as a unit. A household's advance scales with the number of migrating members. Figure 2(a) plots the distribution of the expected total transfer for households planning to take an advance in the upcoming season, as well as its share in earnings from other sources over the coming year. This underlines the size of the intertemporal earnings redistribution: the median household expects an advance of 100,000 Rs or 77% of its annual income.

Figure 2: The Size and Use of Brick Kiln Advances

(a) Size of Advance and Share in Annual Income



(b) How Advances are Spent

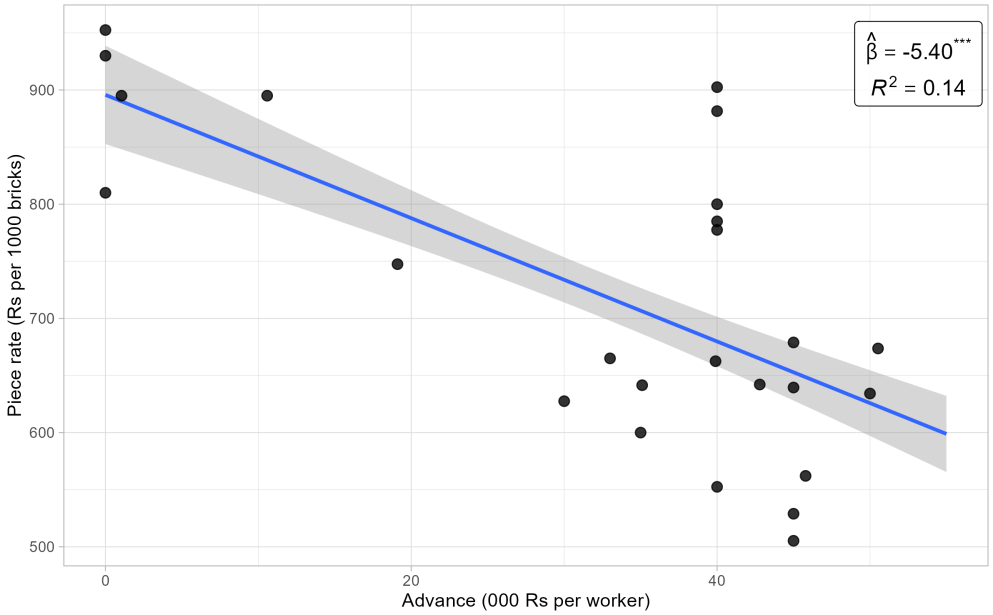


Note: Panel (a): households intending to migrate to a kiln with an advance in the coming season, November 2026 to June 2027 ($N = 434$). The total advance is the advance per worker times the number of migrating household members, winsorised at the 95th percentile. Household income is total expected earnings over the coming year, including the advance. Panel (b): households who took an advance for the 2025-26 season ($N = 525$). Respondents divided ten stones across up to five spending categories. Bars show the average rupee amount of the household's advance, winsorised at the 95th percentile, allocated to each area. Segments averaging at least 10,000 Rs are labelled.

Figure 2(b) breaks down the spending of advances for the 2025-26 production season, and shows that this was dominated by basic needs. In particular, households dedicate 35,000 Rs of the advance on average towards day-to-day consumption (31% as an average share of the advance), 20,000 Rs towards the repayment of existing debts (14%) and 18,000 Rs towards house repairs (13%). Ceremonies are also important, absorbing 25,000 Rs on average (18%), of which marriage accounts for 18,000 Rs. This partly reflects the timing of advance payments: disbursement takes place around the Kharif harvest, when major local festivals are celebrated. Weddings also peak in winter months. Far less common are business expenses, which comprise investments and day-to-day outgoings in agricultural and non-agricultural businesses, averaging together under 7,000 Rs.

For most workers, advances are interest free on paper: the amount which employers promise to

Figure 3: Advance vs. Piece Rate for the Previous Brick Kiln Season



Note: Binscatter of the per worker advances and piece rates that brick kiln migrants report having earned in the previous season ($N = 492$ respondents who migrated to a kiln last year on piece rate contracts). The linear fit is plotted with 95% confidence intervals, and reflects an estimated coefficient of -5.4 ($p < 0.001$) and R squared of 0.14.

deduct out of eventual earnings is equal to the amount paid upfront.[4] The charge instead appears to emerge in the lower piece rates which those receiving larger advances are paid on the job. Figure 3 is a binned scatterplot depicting the correlation between the advances and piece rates which brick kiln migrants report that they earned during the previous season. I overlay the fitted line from the corresponding Ordinary Least Squares (OLS) regression. I find that a 1,000 Rs increase in the per-worker advance earned last season is associated with a decrease in the pay per 1,000 bricks by 5.40 rupees ($p < 0.001$).[5] This relationship is directionally understood by most workers: 83.5% said they would expect higher (lower) on-the-job pay if migrating without (with) an advance, with the median change in the piece rate or wage standing at 30%.

If taken causally, and for a worker producing the average level of expected bricks (123,000 across the season), the detected correlation would imply that each additional 1,000 Rs of advance costs 1,664 Rs in future pay: the 1,000 Rs debt repayment plus 664 Rs in forgone piece rate earnings. Spread over the period between the receipt of the advance pre-migration and its deduction from earnings at the end of the work period, this amounts to a monthly interest rate of 7.0%.[6] However, given that I do not observe the contracts which each worker chose between, this correlation cannot

[4] In my 2025 survey, 70.9% of brick kiln workers reported that they expected that their advances would be deducted from pay without interest. This matches the description of focus group participants and employers.

[5] This correlation is robust to the inclusion of village fixed effects: the coefficient falls from -5.40 to -3.50 but remains highly significant ($p < 0.001$), indicating the trade-off holds both within and across villages.

[6] I convert the loss in piece rate earnings per 1,000 Rs increase in advance to a monthly interest rate assuming monthly compounding over the 7.5 months between the advance being paid, shortly after recruitment, and being repaid out of end-of-season earnings.

speak to how much pay they forgo in order to access larger advances. As Section 4 sets out, my data collection exercise overcomes this by advertising real vacancies at different advance-piece rate pairs which speak to the trade-off between upfront pay and overall earnings.

2.2.3 Repeat Migration Despite Harsh Conditions

Globally, poor people often work under deeply undesirable conditions, including long hours and health and safety risks (International Labour Organization, 2023), and brick kilns are no exception.[7] The median migrant in my sample anticipates a 70-hour work week. Production takes place in the open air, meaning sustained exposure to dangerous heat (Lundgren-Kownacki et al., 2018). Children often accompany parents to kilns, and in 28% of migrating families, parents report that their children were directly involved in production, potentially an underestimate of the true prevalence of child labour.[8] Even where children are exempt from work, migration severely disrupts education, due to language barriers at the destination and difficulties accessing services. Instances of more serious abuses are also reported, including the siphoning of wages by employers, as will be discussed in further detail in Section 8, and physical violence (Deccan Herald, 2023, 2016).

What sets brick kilns apart from other hazardous workplaces is that bonded labourers cannot simply leave if they decide they cannot sustain the conditions, before repaying their debt. Employers also place tight restrictions on workers' freedoms in an attempt to prevent these early departures (Kumari, 2018; Guerin et al., 2015). Indeed, only half of my sample respondents expected to be permitted to leave the work site in an emergency.

In spite of these poor conditions, workers migrate to kilns with remarkable regularity. The average migrant in my sample has worked for 8.8 previous seasons at a kiln, or around 40% of their adult years. This suggests that the poor working environment which characterises kilns is not simply sustained by deception, where recruiters misrepresent the job to uninformed workers. Instead, migrants appear to repeatedly enter bonded labour arrangements with knowledge of and experience in the strenuous and sometimes hazardous conditions that they are bonding themselves to.

3 Theoretical Framework

In this section, I model a worker's choice between bonded labour contracts for a fixed job, which differ in the balance between upfront and on-the-job pay. After describing the basic setup, I first show that a rational and unconstrained worker optimally takes whichever contract offers the highest income, in present value terms. Second, I show how credit constraints may rationalise preferences for high advance contracts, even if they do not meet this threshold. Third, I describe

[7] Each year, an estimated 2.93 million workers die from work-related injuries and diseases, with long working hours alone linked to close to 745,000 deaths, and informal workers such as waste pickers exposed to hazardous substances and pathogens with little or no protection.

[8] The rate of migration to kilns for children of brick kiln workers stands at 40% for girls and 33% for boys.

two distortions which could generate these same preferences for advances, cognitive limitations in earnings calculations and present bias. This will motivate my research design, set out in Section 4.

3.1 Setup

I consider a two period setup. In $t = 1$, the worker chooses a contract j governing the structure of payments for a job which will take place in $t = 2$. In $t = 1$, the worker will receive an advance $d_j \geq 0$. This advance binds them to the employer in $t = 2$, at the end of which they receive their earnings net of the repayment of this debt. I model earnings as a piece rate p_j multiplied by output o, and assume that advances are repaid without interest, as is the case for the majority of workers in my sample.[9] In the basic setup, I assume o is known when the contract is chosen and does not vary across contracts. Appendix A relaxes the first assumption by allowing output to be stochastic, and Section 7.3 shows empirically that expected output does not vary with the contract offered. The income stream generated by contract j is therefore

$$(y_1, y_2) = (d_j,\ p_j o - d_j)$$

I denote period 2 pay net of the advance by $f_j \equiv p_j o - d_j$.

The worker has initial wealth a and faces an interest rate r in the outside credit market. a encompasses all other resources over which consumption is smoothed across periods, and is assumed to have been realised at the point when workers choose between contracts. Let s_j denote savings carried from period 1 into period 2, with $s_j < 0$ indicating borrowing. Consumption in each period is then

$$c_1 = a + d_j - s_j, \qquad c_2 = f_j + (1 + r)s_j$$

Eliminating s_j, the present value of consumption $c_1 + \frac{c_2}{1+r}$ must equal the present value of lifetime resources available under contract j:

$$W_j = a + d_j + \frac{p_j o - d_j}{1 + r} \tag{1}$$

In what follows, I focus on the choice between a high advance option (d_{ha}, p_{ha}) and a low advance, high piece rate alternative (d_{la}, p_{la}), with $d_{ha} > d_{la}$ and $p_{ha} < p_{la}$. This is in keeping with the negative advance-piece rate correlation which I observe empirically. As a benchmark, I consider the case where the low advance is exactly compensated by the higher piece rate, such that the contracts are equivalent in present value terms.

Finally, equation 2 is the log utility function with discount factor δ.

[9] As noted in Section 2.2.2, 70.9% of brick kiln workers in my 2025 survey expected their advances to be deducted from pay without interest. In the same survey, 67% of respondents expected to be paid on a piece rate basis, as opposed to a fixed regular wage.

$$u(c_1, c_2) = \log(c_1) + \delta \log(c_2) \tag{2}$$

3.2 Utility Maximisation Under Perfect Markets

With unrestricted borrowing and saving, the worker chooses s_j to maximise equation 2. Optimal savings are $s_j^* = a + d_j - \frac{W_j}{1+\delta}$, which implies the consumption rule

$$c_1^j = \frac{W_j}{1+\delta}, \qquad c_2^j = \frac{\delta(1+r)W_j}{1+\delta}$$

so that indirect utility under contract j takes the form $(1+\delta)\log(W_j) + K$, with K a function of δ and r. The worker therefore chooses the high advance contract if and only if $W_{ha} \geq W_{la}$, a comparison of present discounted values. Crucially, the advance d_j enters the decision only through its contribution to W_j, and plays no independent role in facilitating consumption smoothing, which is achieved through credit markets. In the next section, I show the implications of removing this assumption.

3.3 Credit Market Imperfections

Suppose the worker can borrow at most $-b$ from outside lenders, so that savings must satisfy $s_j \geq b$ with $b \leq 0$. A worker with $b = 0$ cannot borrow at all, as is the case for 52% of my sample. The constraint binds under contract j whenever desired savings s_j^* lie below b. Two contracts of equal present value imply the same optimal consumption path, so desired savings differ only by the gap in advances: $s_{ha}^* - s_{la}^* = d_{ha} - d_{la} > 0$. The constraint therefore binds first under the low advance contract. For brevity, I focus on the key case $s_{la}^* < b \leq s_{ha}^*$, in which the borrowing constraint binds under the low advance contract, but not the high advance alternative, leaving the remaining cases to Appendix A.

Under the high advance contract, the worker attains the unconstrained indirect utility $(1+\delta)\log(W_{ha})+K$. Under low advance, savings are held at b, and utility is below its unconstrained level by

$$\Lambda_{la} \equiv (1+\delta)\log(W_{la}) + K - [\log(a + d_{la} - b) + \delta \log(f_{la} + (1+r)b)] > 0 \tag{3}$$

Λ_{la} is the utility cost of consuming less in period 1, and more in period 2, than the worker would choose if able to borrow freely. The worker chooses the high advance contract when $(1+\delta)\log(W_{ha}) + K \geq (1+\delta)\log(W_{la}) + K - \Lambda_{la}$, which rearranges to

$$j^* = \begin{cases} \text{ha} & \text{if } W_{\text{ha}} \geq W_{\text{la}} \exp\left(-\frac{\Lambda_{la}}{1+\delta}\right) \\ \text{la} & \text{otherwise} \end{cases} \tag{4}$$

With a binding constraint, the right hand side is the level of lifetime wealth which, with unrestricted borrowing, would leave the worker as well off as the low advance contract does under the constraint. The worker will give up a share of up to $1 - \exp\left(-\frac{\Lambda_{la}}{1+\delta}\right)$ of lifetime resources W_{la} in exchange for the high advance. The borrowing constraint therefore introduces a wedge in favour of the high advance contract, which facilitates consumption smoothing beyond the level credit markets now permit. Appendix A shows that the high advance contract becomes relatively more attractive as the borrowing limit tightens and as period 1 resources a fall, the two sources of variation which I exploit in Section 7.1.

3.4 Misperceived Earnings and Present Bias

The previous section rationalised preferences for high advance contracts in the presence of market failures, even if these are attached to lower income streams. Two departures from the model could generate this same preference: workers may misperceive what each contract pays, or they may be present biased.

3.4.1 Misperceived Earnings

The derivations above assume that, for a given output o, the worker perceives future pay f_j correctly. In practice, the underlying calculation is cognitively demanding: the worker must multiply the piece rate by total season output, which amounts to hundreds of thousands of bricks, and then subtract the advance. A worker who cannot perform this calculation must rely on a perceived value $\tilde{f}_j \neq f_j$, and ranks contracts by perceived lifetime resources $\hat{W}_j = a + d_j + \frac{\tilde{f}_j}{1+r}$. This can distort decisions by altering the perceived ranking of contracts, in a direction which depends on the form of the errors. I elicit $\tilde{f}_j$ directly and compare it with f_j at the respondent's own expected output (Sections 7.3 and 8).

3.4.2 Present Bias

Present bias instead concerns how the worker weighs period 2 against period 1 once pay has been calculated. When making an intertemporal consumption decision in $t = 1$, a present biased individual applies an additional $\beta \leq 1$ discount to future utility:

$$u(c_1, c_2) = \log(c_1) + \beta\delta \log(c_2)$$

This drives up consumption in the first period, relative to when the decision is made at an earlier date and the two periods are weighted by the standard factor $\delta > \beta\delta$. However, whether this impacts contract choices depends on the incidence of credit constraints. Under perfect credit markets, the choice reduces to a comparison of present values, which does not depend on the discount factor. If instead the constraint binds under low advance, the share of lifetime resources which the worker

will give up for the high advance is decreasing in the discount factor (Appendix A). This share is therefore larger when the choice is made as the advance is received than when it is made at an earlier date, and the worker sacrifices higher total income in exchange for immediate cash to a greater extent than their ex ante self would prefer. Section 7.2 tests this by comparing contract choices of the same workers at two points in time, at only the second of which is the advance's receipt imminent.

3.5 Further Extensions

Appendix A relaxes the assumption that output is known when the contract is chosen. The advance is paid before output is realised, whereas piece rate earnings are not, so uncertain output imposes consumption risk which rises with the piece rate. A risk averse worker without access to insurance therefore prefers the high advance contract to a low advance alternative of equal expected present value.

I do not model effort. If output responded to contract terms, the income effect of a larger advance and the substitution effect of a lower piece rate would both predict lower output under the high advance contract. The low advance contract would then pay more on the job than the constant output calculation implies, so the interest rates necessary to justify a preference for high advance would rise further. Empirically, I do not find that workers' expected output levels vary according to the contract offered (Section 7.3).

4 Survey Design and Data Collection

This section describes two rounds of data collection, each built around the advertisement of real vacancies at partner brick kilns. Figure 4 presents the full timeline. Round 1 begins in September 2025, ahead of the 2025-26 brick kiln production season. Respondents are randomly assigned to different arms which dictate the contract terms attached to the advertised vacancy. In addition to a baseline survey covering all 1,494 respondents, Round 1 comprises a revisit for one treatment arm, discussed in further detail below. Round 2, the information experiment, instead begins in July 2026, and advertises vacancies for the 2026-27 season. Before application decisions are recorded, half of respondents are randomised into an information treatment which provides accurate earnings projections for the advertised vacancies.

4.1 Research Collaboration with Kiln Employers

My research questions require observing how a worker chooses between contracts which differ in the split between upfront advance and on-the-job pay. Observational data does not permit this analysis: workers might struggle to recall all interactions with recruiters, and pay terms in job

offers received might be correlated with other features of those offers, such as job location, which also drive preferences. I therefore construct the choice set myself, advertising vacancies differing only in these payment terms.

For application decisions to carry real consequences, these vacancies had to be genuine. This meant finding employers willing to offer payment terms agreed together with the research team, and willing to engage in recruitment events with prospective workers identified by the research team. For this outreach, I therefore engaged the support of my partner NGO, with whom I had already delivered focus group discussions with migrant workers. This NGO works alongside employers to guarantee safe migration, and has a significant presence in kilns in Hyderabad, a brick production hub. We therefore approached employers in Hyderabad with whom the NGO had the longest standing relationship, who would be most open to this form of collaboration.

Under the agreement reached with these employers, the research team advertises jobs on their behalf under pre-agreed contractual arrangements, and the contact details of interested applicants are shared with employers. The employers then commit to holding recruitment events with interested workers, and honouring the advertised payment terms for any workers hired through this pipeline. On the research side, this permits the collection of *incentivised* data on interest in job opportunities under different payment structures, as respondents' answers are linked to real offers: a worker who does not apply will not be invited to attend recruitment events.

4.2 Round 1: Vacancies at Randomised Pay Terms

4.2.1 Sample

Standard brick kiln recruitment practices involve approaching households in origin villages during the wet season, when brick kilns temporarily close, with offers of employment for the upcoming dry season. My survey therefore does the same. Ensuring that this would yield a sufficient pool of interested applicants meant choosing sample villages with high pre-existing rates of migration to kilns in Hyderabad, the site of my partner employers. Odisha's Western Interior is the natural choice, for two reasons. First, with a high concentration of workers migrating to this city, Odia workers had sufficient familiarity with Hyderabad, and my partner Hyderabad kilns sufficient familiarity with Odia workers. Second, my NGO's work in origin villages is concentrated in this region, as part of which they collect village-level data on brick kiln migration. I was able to use this data to identify high migration candidate villages, of which the final sample included 71 across the districts of Bolangir and Nuapada.

The logical unit of analysis for this research is the household, as this is the basis on which brick kilns typically recruit. In each village, I randomly sampled 18 to 23 households for participation, giving a total sample of 1,494 households.[10] This limited per-village sample helped reduce the scope for

[10] I targeted 20 households in each village, but exceeded this by 1 to 3 households in 90% of villages.

spillovers across treatment arms within villages, whilst guaranteeing a minimum level of statistical power. The survey was carried out with the household head, or a secondary decision-maker if this person was unavailable.

Bolangir and Nuapada are part of the historically underdeveloped KBK (Kalahandi-Balangir-Koraput) region, with poverty rates more than double the national average of 22% (World Bank, 2016). 99.6% of my sample belong to marginalised caste backgrounds, namely Scheduled Tribe (41.8%), Other Backward Class (29.4%) or Scheduled Caste (28.4%). Their reliance on brick kiln work is borne out in the data: a staggering 89.7% of randomly sampled households had migrated to a kiln in the past, and the average respondent had worked at a kiln for 7.9 previous seasons.

4.2.2 Experimental Job Offers

Round 1 serves three research questions: how much future pay workers give up for a larger advance, how this demand responds to liquidity needs, and whether it is distorted by present bias or by errors in calculating earnings. Households in 21 villages are assigned to a Choice arm, in which they rank two contracts against one another. Households in the remaining 50 villages are assigned to one of two single offer arms, in which they are shown one contract only. Table 1 sets out the payment terms, and Section 4.3.1 describes the randomisation.

Table 1: Experimental Arms

	Arm	**Advance per worker**	**Piece rate per 1,000 bricks**	**Households**	**Villages**
1	Choice			434	21
	Option A	45,000	1,100		
	Option B	10,000	1,400		
2	High advance	45,000	1,100	534	50
3	Low advance	10,000	1,250	526	50

The cleanest demonstration that workers pay for advances is choosing a high advance contract from a menu which also includes an option paying less upfront but more overall. The Choice arm embeds this menu. Respondents can apply for a vacancy with a choice between two contracts. In Row 1 of Table 1, option A offers an advance of 45,000 Rs per worker, just above the median among respondents who migrated to a kiln in the year prior at 40,000 Rs, and a piece rate of 1,100 Rs per 1,000 bricks. Option B offers an advance of only 10,000 Rs per worker, for which respondents are compensated with a higher piece rate of 1,400 Rs. At the average expected productivity, a worker would have to face a monthly interest rate of over 19.5% for option A to match B in present value.[11] Section 3 demonstrates that a preference for A can thus be read as a symptom of

[11] Average expected productivity of 123,000 bricks per worker per season, meaning option B pays 35,000 Rs less upfront and 71,900 Rs more at the end of the season, around 2 Rs of future pay for each rupee of advance forgone. Even at the 25th percentile of expected productivity, an interest rate of at least 15.2% would be required for the

market or behavioural distortions. As Section 4.2.3 discusses, contract rankings are only observed for applicants, a concern mitigated by the generosity of advertised piece rates relative to outside options.[12] Nonetheless, I complement the incentivised preference analysis with a hypothetical MPL in which all respondents choose between high and low advance contracts, including non-applicants and those in other arms, to which I now turn.

The single offer arms provide a second incentivised measure of advance demand, this time weighing one contract against workers' outside option: those assigned to high advance receive an offer matching Choice option A, and those in low advance are offered an advance of 10,000 Rs and piece rate of 1,250 Rs. As pre-registered, this piece rate premium was calibrated so that these contracts would be roughly equal in present value.[13] However, respondents' output expectations were higher than initially assumed, so the low advance contract is worth more for most respondents and should generate at least as many applicants in the unconstrained model.[14] Given the generosity of the experimental piece rates, either offer may dominate the outside option, even in a population with tight liquidity constraints on average. As such, these arms yield a less direct measure of paying a cost for an advance, and predominantly serve the expectations analysis discussed below.

Around one month after baseline, enumerators revisit the Choice arm and give respondents the opportunity to revise their initial application decision and reported contract preference. As Section 4.3 will discuss, analysis of shifts in applications between waves will serve the second and third research questions, which study the roles of credit constraints and present bias in demand for advances. The former will be complemented by an analysis of heterogeneity in contract preferences by borrowing capacity at baseline, described in Section 5.

The remaining distortion is misperceived earnings, for which the single offer arms provide a clean test. Each respondent estimates take-home pay, net of advance repayments, under the contract they were shown. Average estimates should differ across arms to the extent that their own output projections imply: at average output, take-home pay is 53,000 Rs per worker higher under low advance. By comparing estimates across respondents, rather than asking each to produce estimates for multiple contracts, I avoid the risk of earnings differences being made unnaturally salient. Respondents are also asked about expectations of other job features, including output, wage theft and amenities. If these do not differ across arms, gaps in estimated pay can more safely be attributed to calculation errors, and gaps in demand to preferences over payment timing.

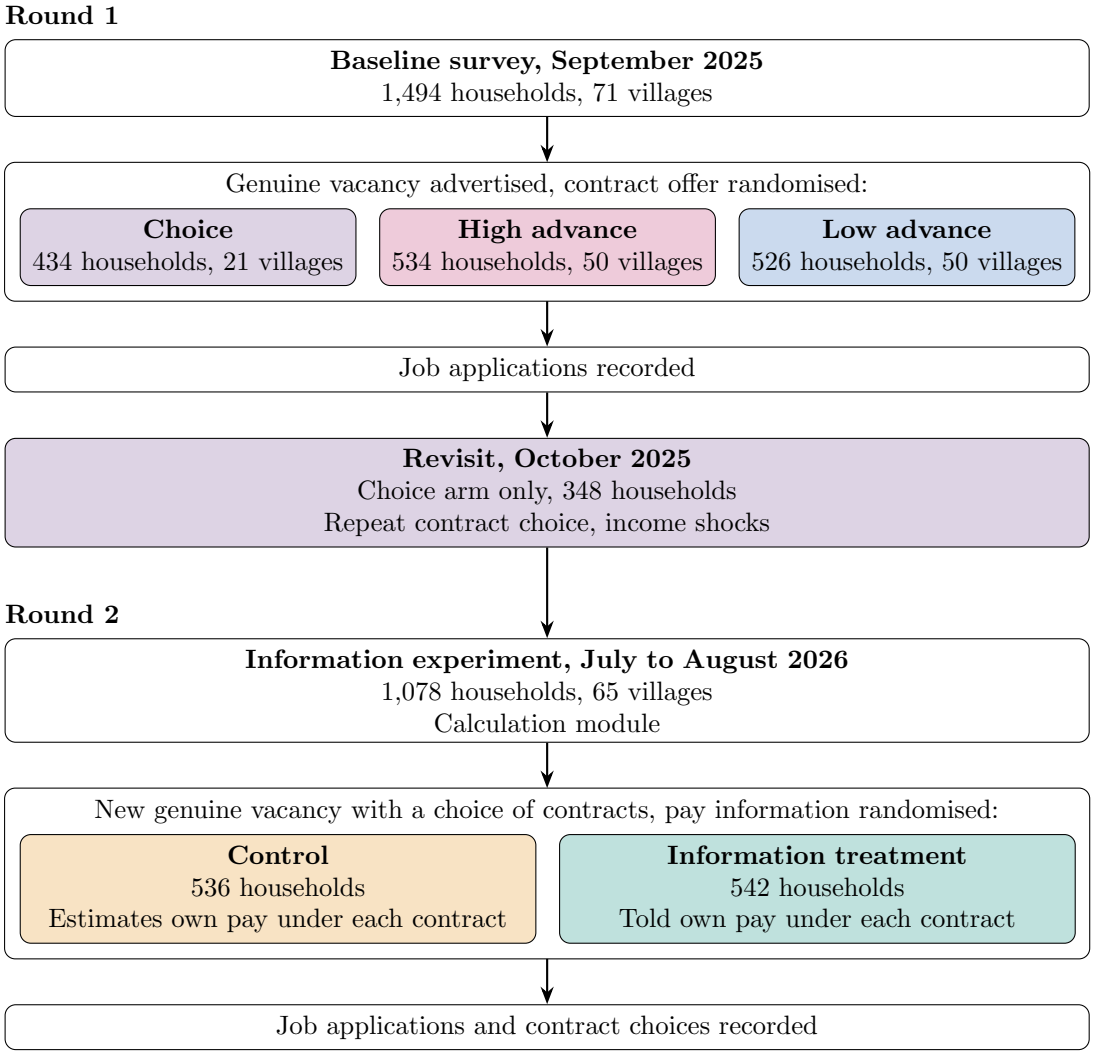
high advance contract to dominate in present value terms, far above the median rate of 1.5% reported by indebted households in my sample.

[12]The lowest experimental piece rate of 1,100 Rs per 1,000 bricks stands at the 85th percentile of those which migrants expect to earn in other brick kilns.

[13]The piece rate premium for Option B in the Choice arm was set to be larger, so that this would have higher present value.

[14]At the average expected output, the low advance contract is worth more at any monthly interest rate below 11%, and below 8.4% at the 25th percentile.

Figure 4: Timeline of the Study Design



Note: This figure plots the components of the study in chronological order. Shaded boxes mark randomised treatment arms. The revisit shares the shading of the baseline Choice arm, to which it was restricted.

4.2.3 Interest Registration Procedure

The vacancy is advertised, and interest in it registered, during the household survey itself (see Figure A3 for the full sequence of vacancy questions). Participants are shown a leaflet containing basic information about the vacancy, and the enumerator reads a script containing further details. Besides payment terms, leaflets are identical across arms. In the Choice arm, the two contracts appear side by side on the same leaflet.

After answering questions about their expectations of the job, participants are asked whether they would like to be added to the list of applicants. This is explained as the basis for invitation to the recruitment events, with places allocated at random in the event of over-subscription. Before deciding, they are told that registering involves answering some additional questions. This yields the first outcome, an indicator for asking to be added to the list.

For those who do apply, the additional registration questions then follow. In the Choice arm, they are first asked which contracts they would accept, and can name the high or low advance contract or both. Those naming both are asked which they prefer, and told that this preference will be prioritised if places are available under both. In all arms, respondents then confirm three standard requirements for brick kiln work: the household can form a group of three or more physically fit workers, no group member has committed to another employer, and the group agrees to migrate to Hyderabad. Respondents must also consent to contact details being shared with the employer. Households passing these checks enter the list passed to employers, yielding the second outcome: successful interest registration. In the Choice arm, both outcomes are contract specific: a respondent reports interest or successfully registers with a preference for a given contract if it fulfils the respective requirements and names that contract alone, or names both and ranks it first.

This registration procedure is characterised by a *hard* incentive not to under-report interest: only those who successfully register could be invited to recruitment events. The same holds for each contract in the Choice arm: an employer hiring under only one contract would not invite applicants who had named only the other, so a respondent loses out by omitting a contract they would accept. By contrast, over-reporting is relatively cheap, costing only a few minutes of additional questions, and in the Choice arm, naming both contracts instead of only one costs nothing further. Reported interest in the vacancy itself may therefore overstate true demand, whether through low stakes or a wish to please the enumerator, an acknowledged limitation of this study.[15] However, three considerations limit this concern. First, my analysis rests on comparisons across contracts, for which over-reporting is only a concern if it affects some contracts disproportionately. Second, in the Choice arm, each option was described in neutral language from a fixed script, with no suggestion of either being better, limiting the scope for experimental demand. Third, only 4.4% of applicants in the Choice arm said they would accept both contracts, although doing so was costless.

[15] Downstream outcomes, including attendance at recruitment meetings and job offers accepted, are beyond the scope of this study. This trial of alternative contract structures was intended to be limited in size, and so the sample size is insufficient for robust analysis.

This suggests that respondents treated the choice of contract as consequential, and means that the ranking I observe comes almost entirely from respondents who named only one contract.

4.2.4 Hypothetical Multiple Price List

After interest registration (Figure A3), respondents complete a hypothetical MPL. Here, they trade off a fixed high advance contract against low advance alternatives at piece rates ranging from 950 to 1,500 Rs, displayed in Table 2. Whilst unincentivised, these questions yield contract rankings for the entire sample, including non-applicants and single-offer respondents. To address anchoring on the first comparison, the order in which low advance contracts are presented, ascending or descending in piece rate, is randomised at the respondent level.

The positioning of the MPL after interest registration protects incentivised outcomes from anchoring in these hypotheticals, at the risk of introducing it in the reverse direction. One specific concern is depressing interest in low advance MPL options with less generous piece rates than the advertised offers. I cannot test a common effect across arms, an acknowledged limitation. However, I do not find that those assigned to arms shown a real high piece rate, low advance offer are less likely to prefer hypothetical low piece rate, low advance offers specifically (Table A1), though I do detect modest level differences across arms.

Table 2: Contract Terms for Hypothetical Multiple Price List Exercise

High advance: 45,000 Rs		Low advance: 10,000 Rs	
	Piece rate (Rs per 1,000 bricks)		Piece rate (Rs per 1,000 bricks)
A	1,100	B1	950
		B2	1,050
		B3	1,100
		B4	1,150
		B5	1,200
		B6	1,500

4.3 Revisit Survey for the Choice Treatment Arm

I revisit the Choice arm one month after baseline. This serves the second research question and part of the third, which study the roles of credit constraints and present bias.

As shown theoretically in Section 3, present biased respondents who are credit constrained will have a stronger propensity to pay large costs to access high advances when their receipt is imminent, relative to when they commit to a contract ahead of advance receipt. I construct this comparison by holding the revisit survey in the days directly prior to the recruitment meetings when advances are paid out, giving respondents the opportunity to revise both their initial decision of whether to

apply and their registered contract preference. Respondents were not informed at baseline that this revisit would take place, and so should have expected their initial choices to be binding. Conditional on financial circumstances being unchanged, preferences should be stable across waves. A reversal of preferences in favour of the high advance is therefore the signal of present bias. The exact specification is outlined in Section 5.

Questions on income shocks define the relevant stable-income subsample for the test of present bias, and also form the basis of the credit constraints analysis. After the interest registration module, respondents are asked whether they have experienced any unexpected financial difficulties since baseline. They are given a list of examples and asked to report which of these apply, and can mention additional events outside this list. Said shocks increase the likelihood of experiencing a need for credit which outstrips borrowing capacity, and may increase reliance on advances for consumption smoothing, even at the significant cost embedded in the Choice arm's payment terms. This would predict a shift towards high advances for affected households. A subsequent question then addresses unexpected positive financial changes, which might reduce a respondent's willingness to incur these costs, shifting preferences away from high advances.

In total, 348 of the 434 Choice households were successfully revisited. Selective attrition would reduce the external validity of this analysis without violating my identification assumptions, which rely only on individual-level comparisons, as Section 5 outlines in full. Table A9 finds that non-attriting households are poorer along several dimensions, and I reject the test for overall balance with a p value below 0.01. Behaviour consistent with present bias and credit constraints might therefore be more prominent in my analysis than in the overall sample, an acknowledged limitation of the study.

4.3.1 Randomisation

Randomisation occurred in two stages. First, 21 out of the 71 villages were randomly allocated to the Choice arm, stratifying by district. Second, in the remaining 50 single offer villages, households were allocated to either the High or Low Advance arm. This second randomisation occurred at the household level, stratifying by village. Enumerators were given a pre-assigned ID for each household, which, once entered into the survey software, would automatically reveal the assigned contract at the relevant stage of the survey. This ensured equal allocation to each treatment per village, whilst keeping the randomisation blind to enumerators and limiting the scope for non-compliance. Randomising within village, however, introduced a risk of contamination across neighbouring households, which I mitigated by treating only a very small number of households per village (18 to 23). Table 1 breaks down the realised sample size by arm.

The identification strategy for comparing outcomes across the single-offer arms requires balance in potential outcomes. As shown in Table A2, I detect imbalances on household size and certain characteristics of the head, and reject the overall balance test with a p value of 0.013. I therefore

control for imbalanced variables throughout and test for robustness to a post-double-selection lasso specification, which selects controls from a larger candidate set (Cilliers et al., 2024).

4.4 Round 2: The Information Experiment

As Sections 6 and 7 will show, Round 1 finds that respondents pay dearly for advances, choosing high advance options over low advance, high piece rate alternatives worth much more in present value terms. However, they vastly underestimate future earnings differences across contracts, suggesting these costs are not fully internalised. Round 2 was designed to collect more granular data on the nature of the calculation errors producing these distortions, and to test whether an information intervention can overcome them and shift preferences in favour of low advances. It embeds a second recruitment experiment with the original sample and partner kilns ahead of the 2026-27 work season, randomly giving half of respondents accurate earnings projections for the low and high advance contracts which applicants can choose between.

4.4.1 Sample and Timeline

The sample for the information experiment includes 1,078 households across 65 villages from the baseline sample of 1,494. Villages were drawn sequentially from the baseline list of 71 based on convenience, attempting all households, until the target 1,000 surveys were completed. In total, 1,364 households were attempted, implying attrition of 20.6%. Attrition is unrelated to treatment both in practice and by construction, as treatment is revealed only mid-way through the survey, and is uncorrelated with baseline characteristics (Table A3).[16] I do however detect minor imbalances between the Round 1 and Round 2 samples, and reject the test for overall balance with a p value of 0.048 (Table A4). For example, relative to the full Round 1 sample, those surveyed in Round 2 have 0.16 additional years of education. This works against my results: calculation errors are larger amongst less educated respondents, and so the Round 2 sample would understate the effect of the intervention, if anything.

4.4.2 Experimental Contracts

The information experiment tests whether accurate information about the costs of advances can affect workers' willingness to pay them. Respondents are therefore given the opportunity to apply for a job under a choice of contracts embedding the trade-off between upfront and overall pay: a high advance contract pairing an advance of 50,000 Rs per worker with a piece rate of 650 Rs per 1,000 bricks, and an alternative paying 5,000 Rs upfront and 900 Rs per 1,000 bricks. Both carry lower piece rates relative to the baseline offers in Table 1, placing them in line with outside options

[16] Attrition stands at 20.5% in the control arm and 20.7% in the treatment arm, and a regression of an attrition indicator on treatment with randomisation-block fixed effects returns a coefficient of 0.3 percentage points ($p = 0.88$).

for similar advance levels. At typical interest rates and productivity, the higher piece rate attached to the low advance contract means that it dominates the high advance alternative in present value terms, and should therefore be preferred unless workers face market or behavioural distortions, including a failure to price the contracts.

4.4.3 The Pre-Treatment Calculation Module

A secondary role of Round 2 is to provide descriptive evidence on the nature of calculation errors. To this end, respondents in both arms complete an identical set of calculation questions. These include hypothetical kiln earnings questions: respondents estimate take-home pay for a group of three workers producing 500,000 bricks under four fixed contracts, net of advance repayments, assuming workers are paid fairly. The first contract pairs an advance of 50,000 Rs per worker with a piece rate of 650 Rs per 1,000 bricks. The remaining three hold the advance at 5,000 Rs whilst raising the piece rate from 650 to 900 and then 1,200 Rs, so that contracts 1 and 3 match the high and low advance options advertised later in the survey (Table A5). Comparisons of mistakes across questions 1 and 2 speak to whether households accurately estimate the earnings benefits of lower debt deductions, for a fixed piece rate. Comparisons across questions 2 to 4 instead speak to their perceptions of the earnings gains from higher piece rates, for a fixed advance. Relating all answers to past earnings will also speak to whether mistakes are in part driven by anchoring.

4.4.4 The Information Intervention

After the calculation module, the enumerator tells the respondent about the upcoming vacancy. Relevant information about the contracts, including payment terms, eligibility requirements and amenities on site, is explained in a short video. Workers then answer a series of questions about their expectations over features of the job under each contract, starting with high advance, outlined in full in Figure A4. It is in this module that treatment arms first diverge.

Those assigned to control are asked to estimate what they would earn at the brick kiln, net of advance repayments, under a given contract. They are asked to assume that they are paid fairly by the employer and that they produce a quantity of bricks across the season which they would consider typical, as reported in an earlier question. As set out in Section 3, this quantity has an objective value, given by:

$$f(p, d, o, n) = p \times \frac{o}{1000} - d \times n \tag{5}$$

where f is total family earnings, p is the piece rate per 1,000 bricks, o is family output on the job in bricks, and d and n are the advance per worker and the number of workers per family. The difference between the self-estimated and correct values, at the respondent's own productivity and group size, yields a contract-specific measure of calculation error. The treatment group is instead told this objective value, computed by the survey software from the respondent's own reported output and group size and read by the enumerator in neutral language from a fixed script.

After completing the full job expectations module for each contract, respondents are asked whether they would like to apply for the advertised job and, if so, under which contract, following the baseline procedure described in Section 4.2.3. As at baseline, these outcomes are incentivised because applicants are later invited to meetings with employers.

Alongside these incentivised application decisions, the treatment is tested on an adapted version of the baseline MPL. Respondents make a series of choices between the advertised high advance contract, held fixed across rows, and a low advance alternative whose piece rate rises across five rows from 650 to 3,500 Rs per 1,000 bricks, the second of which matches the advertised low advance contract (Table A6 in the appendix). They again assume their self-reported typical productivity level.[17] Those in the control group are presented with the raw contract terms and asked which they prefer, whereas those in the treatment group are also told the earnings they would receive under each. This yields a more granular measure of the effect of accurate earnings projections on willingness to forgo high advance contracts for more lucrative low advance alternatives.

The design would be threatened if treated respondents registered disproportionately for the low advance contract because they inferred that this is what the researcher wanted, rather than because their earnings beliefs changed as a result of the information shared (de Quidt et al., 2018). Several design features limit this risk. First, the control group is active: delivery of information about the job is symmetric across treatment and control, and the only scripted difference is told-versus-asked pay at the earnings nodes. Second, the information is objective arithmetic based on the respondent's own stated output, with no persuasion and no statement about which contract is better. Third, the application outcome is a real decision linked to a genuine job offer. Fourth, enumerators are made aware of a respondent's treatment status only when the divergent content emerges. Finally, treatment delivery and outcome recording take place during the same survey, in private, with only around 20 households surveyed per village, limiting the scope for spillovers.

4.4.5 Randomisation

All 1,494 baseline households were randomised into the information intervention prior to data collection, stratifying by village and a measure of advance payment preferences drawn from the baseline survey.[18] Table A7 tests for balance on key baseline demographics, correlates of contract preferences and likely dimensions of treatment effect heterogeneity. On the full assignment sample, a joint F-test did not reject the null that the baseline variables are jointly unpredictive of treatment assignment ($p = 0.13$). In the realised sample, this test rejects at the 5% level ($p = 0.036$), though

[17] The order in which the rows are presented, ascending or descending in the low advance piece rate, is randomised at the respondent level.

[18] The measure of baseline advance demand is taken from the baseline hypothetical MPL exercise (Table 2), from which I derive the switch point, i.e. the index of the least generous low advance piece rate for which the respondent prefers the low advance contract to the high advance one, so that a higher switch point indicates stronger demand for advances. My "advance demand" stratification variable is an indicator for a switch point above the sample median.

only one of the seventeen variables differs across arms at the 10% level.[19]

5 Empirical Specifications

This section presents the empirical specifications used to answer the research questions. Throughout, the unit of analysis is the household as responses are given by the household head on behalf of themselves and other members.

5.1 Effect of Contract Terms on Contract Preferences

To measure the costs that respondents incur in order to access advances, I use the two demand elicitation exercises in which respondents rank high advance contracts against low advance, higher paying alternatives: the incentivised Choice treatment arm and the hypothetical MPL.

In the Choice arm, I compare the shares of households applying with a preference for the high versus low advance vacancy. Formally, I run the following regression, in which each household contributes one observation per contract:

$$\text{Choice arm:} \qquad \text{Applied}_{hc} = \alpha_1 + \beta_1 \cdot \text{HighAdvance}_c + \mu_h + \epsilon_{hc} \tag{6}$$

Applied_{hc} is an indicator for household h applying for the vacancy with a preference for contract c. It equals zero for the contract the household does not prefer, and for both contracts if the household does not apply at all. I estimate the specification separately for the two outcomes defined in Section 4.2.3: expressed interest and successful registration. HighAdvance_c indicates the high advance contract (1A in Table 1), equal to zero for the low advance alternative (1B). Each respondent is made identical job offers, and so these are mechanically orthogonal to household characteristics. Household fixed effects μ_h are included nonetheless to make the within-household design explicit. β_1 therefore isolates the average effect of the payment terms on relative demand for the two contracts. Standard errors are clustered at the household level, the unit within which the contract variation exists, with village-level clustering reported for robustness.

The hypothetical MPL extends this comparison to a larger set of low advance contracts, and to the entire sample of 1,494 Round 1 households. Each respondent contributes one observation per row of Table 2, and I estimate:

$$\text{Multiple price list:} \qquad \text{PrefHA}_{hr} = \alpha_2 + \beta_2 \cdot \text{Piece Rate}_r + \mu_h + \epsilon_{hr} \tag{7}$$

PrefHA_{hr} is an indicator for respondent h preferring the high advance contract over the low advance alternative in row r. Piece Rate_r is the piece rate attached to the low advance contract in row r,

[19] The share of households reporting a negative income shock at baseline stands at 44.8% in the control arm and 50.7% in the treatment arm ($p = 0.061$).

measured in Rs per brick, so a 100 Rs increase per 1,000 bricks is a 0.1 unit increase in the regressor. Every household faces the same six rows, so the regressor is orthogonal to household characteristics by construction, and the household fixed effects μ_h are included only to make the within-subject design explicit, as in equation 6. β_2 is identified from each respondent's choices across rows, and estimates with and without the fixed effects coincide exactly. It measures the average change across households in the probability of preferring the high advance contract as the low advance piece rate rises, which is also the slope of the aggregate demand curve in Figure 6. Respondents whose choice never changes enter this average with a slope of zero, so β_2 equals the share of respondents who switch multiplied by the average slope among switchers, and should be read as a population-level response. I test robustness to replacing the piece rate with the row index (1 to 6), treating rows as equal ordered steps, and to restricting the sample to single switchers, i.e. respondents who never switch back to the high advance contract as the piece rate rises (88.8% of the sample).

5.2 Effects of Contract Terms in the Single-Offer Arms

The High Advance and Low Advance arms (1,060 households) measure the causal effect of contract terms on demand for the vacancy against the outside option, as well as on expectations over features of the advertised job. The latter is crucial for testing whether the difference in earnings across contracts is correctly perceived by respondents on average, a core part of my third research question. This will also allow me to verify that expectations over features such as job amenities, productivity and wage theft do not vary with contract terms, so that gaps in estimated pay can more safely be attributed to calculation errors, and gaps in demand to preferences over payment timing. For outcome Y_{hv} of household h in village v, I estimate the following regression across the single offer arms:

$$\text{Single offer arms:} \qquad Y_{hv} = \alpha_3 + \beta_3 \cdot \text{HighAdvance}_h + \gamma_v + \theta_3' Z_h + \epsilon_{hv} \tag{8}$$

HighAdvance_h is a binary indicator for assignment to the High Advance treatment, rather than Low Advance. Contracts are randomly assigned across households within villages, so β_3 identifies the causal effect of being offered the high advance rather than the low advance contract. Y_{hv} is either a measure of interest in the vacancy or an expectation over a feature of the advertised job. The specification includes village fixed effects, γ_v, to account for the stratified randomisation design, as well as controls Z_h for the variables on which Table A2 detects imbalances: age, sex and education of the household head and household size. As a further safeguard against imbalance, I check robustness to a post-double-selection lasso specification, which selects controls from a larger candidate set (Cilliers et al., 2024). Standard errors are clustered at the village level.

5.3 Heterogeneity Analysis

The second research question examines the role of credit constraints in demand for upfront advances. This has two components: heterogeneity analysis, exploiting variation in access to liquidity across households at baseline, and a triple difference which measures how financial shocks shift contract preferences across surveys. Specifications for the former are discussed here.

The heterogeneity analysis employs both incentivised demand outcomes, starting with the single offer arms in equation 9:

$$\text{SINGLE OFFER ARMS:} \quad \text{Applied}_{hv} = \alpha_4 + \beta_4 \cdot \text{HighAdvance}_h + \delta_4 \cdot X_h + \zeta_4 \cdot (\text{HighAdvance}_h \times X_h) + \gamma_v + \theta_4' Z_h + \epsilon_{hv} \quad (9)$$

X_h denotes household h's borrowing capacity, the maximum amount they could borrow for one month today, if needed. This is the direct empirical analogue of $-b$, the borrowing limit in the model of Section 3. I report results both for a continuous and an indicator measure.[20] ζ_4 measures how the causal effect of being offered the high rather than the low advance contract on the propensity to apply at all varies with the ability to access outside loans.

Equation 10 produces the analogous test for the Choice arm:

$$\text{CHOICE ARM:} \quad \text{Applied}_{hc} = \alpha_5 + \beta_5 \cdot \text{HighAdvance}_c + \zeta_5 \cdot (\text{HighAdvance}_c \times X_h) + \mu_h + \epsilon_{hc} \quad (10)$$

As in equation 6, each household contributes one observation per contract, and household fixed effects μ_h absorb X_h and controls Z_h. ζ_5 measures how the within-household gap in applications between the high and low advance contracts varies with the ability to access outside loans.

The MPL permits a higher powered complement to the incentivised analysis:

$$\text{MULTIPLE PRICE LIST:} \quad \text{PrefHA}_{hr} = \alpha_6 + \zeta_6 \cdot X_h + \nu_r + \gamma_v + \theta_6' Z_h + \epsilon_{hr} \quad (11)$$

Each respondent contributes one observation per row of Table 2. Row fixed effects ν_r hold the comparison low advance contract fixed, so ζ_6 compares the propensity to prefer high advance contracts for households facing different borrowing constraints and choosing between the same pair of contracts.[21] This is preferable to an alternative specification interacting borrowing capacity with the piece rate, the more direct extension of equation 7, but which would not order respondents by

[20] The continuous measure has a long right tail, reaching 500,000 Rs against a sample mean of 5,600 Rs. I therefore winsorise it at the 95th percentile.

[21] A more familiar summary of MPL responses is the switch point, the index of the first row at which the respondent prefers the low advance contract, taking a value of seven for those who always choose high advance. Amongst respondents who switch only once, equation 11 is that regression rescaled by a factor of six. I work with the row level indicator in the main heterogeneity specification, as for the 11.2% of respondents who switch more than once, the first switch discards their choices in later rows and understates demand for advances. A robustness check which instead uses the switch point index can be found in the appendix.

the strength of preference.[22]

I cluster standard errors at the village level in all three specifications. This matches equation 8 for the single offer arms. For equations 10 and 11, it is more conservative than the household level clustering used in equations 6 and 7, and I adopt it for consistency.

Heterogeneity analysis should be interpreted with caution for two reasons. First, with relatively small sample sizes (1,060 in the RCT arms and 434 in the Choice treatment arm), I have limited power. Second, I am only able to test for associations, and not causal relationships. I therefore complement these descriptives with a triple difference, to which I now turn.

5.4 Triple Difference: Income Shocks between Surveys

As discussed in Section 4.3, the Choice arm completed an additional survey around one month after baseline, during which respondents could revise their baseline application decisions and also reported any unexpected income shocks in the intervening period. In the triple difference specification 12, I test whether the incidence of these shocks affects transitions in relative contract preferences: within a broader context of credit constraints, income losses and gains should have opposite effects on a respondent's willingness to pay for advances. The three differences are across contracts, across waves, and between shocked and unshocked households.

$$\begin{aligned}\textsc{Choice arm revisit:}\quad \text{Applied}_{hct} = {} & \gamma_8 \cdot (\text{HighAdvance}_c \times \text{Post}_t) \\ & + \beta_8^{\text{neg}} \cdot (\text{NegShock}_h \times \text{HighAdvance}_c \times \text{Post}_t) \\ & + \beta_8^{\text{pos}} \cdot (\text{PosShock}_h \times \text{HighAdvance}_c \times \text{Post}_t) + \mu_{hc} + \lambda_{ht} + \epsilon_{hct} \quad (12)\end{aligned}$$

Each of the 348 households observed in both waves contributes one observation per contract per wave, meaning 1,392 observations in total. Applied_{hct} is defined as in equation 6. NegShock_h and PosShock_h indicate whether the respondent mentions any unexpected decreases and increases in income since baseline. Post_t indicates this revisit wave. Household-contract fixed effects μ_{hc} absorb each household's average preference for each contract across waves, including any general tendency of shocked households to prefer a specific contract. Household-wave fixed effects λ_{ht} absorb each household's overall propensity to apply in each wave, including any effect of shocks on changes in interest between waves. γ_8 then captures changes in relative contract preferences for households who report stable income streams. β_8^{neg} and β_8^{pos} measure the marginal effects of income fluctuations on these preference transitions. Standard errors are clustered at the household level, with village-level clustering reported for robustness.

I also estimate a difference-in-differences decomposition of equation 12, comparing shocked and unshocked households before and after the revisit for one contract at a time. Here, the outcome is

[22]The coefficient on the piece rate is largest in magnitude for those switching at the mean piece rate, and decreases for those switching at either higher or lower rates. Those always and never choosing the high advance would both be assigned a zero coefficient, despite being at opposite extremes of the preference distribution.

Y_{ht}, an indicator for applying with a preference for high and low advance respectively, and I control for household and wave fixed effects. This breaks changes in relative preferences into changes in interest in each individual contract, with coefficients summing to β_8^{neg} and β_8^{pos}.

The identifying assumption for the *causal* interpretation of coefficients is parallel trends: income shocks between surveys are uncorrelated with other features predicting changes in relative contract preferences over the same period. Present bias represents one potential violation, the focus of analysis in Section 7.2: preferences may shift in favour of higher advances due to the proximity of the revisit to the recruitment events, where advances will be disbursed. If time inconsistency is also correlated with income volatility, this will confound my estimates. A correlation between income shocks and changes in outside options could also invalidate parallel trends, for example if respondents are approached by other brick kiln recruiters between surveys, and these reduce interest in one contract disproportionately.[23] I return to these threats in Section 7.1.2, controlling for interactions between baseline predictors of preference changes and wave and contract indicators and re-estimating the specification on placebo shocks that predate the revisit window.

5.5 Reversal Test: Present Bias

To understand whether demand for advances is in part driven by present bias, part of my third research question, I test for systematic movement towards the high advance contract between the baseline and revisit waves for the Choice arm as receipt of the advance becomes imminent. I refer to this as the reversal test, and formalise it through the following regression, in which each household contributes one observation per contract per survey wave:

$$\text{CHOICE ARM REVISIT:} \qquad \text{Applied}_{hct} = \gamma_7 \cdot (\text{HighAdvance}_c \times \text{Post}_t) + \mu_{hc} + \lambda_{ht} + \epsilon_{hct} \tag{13}$$

Applied_{hct} is defined as in equation 6, equal to one for the contract that household h applies for with a preference in wave t and zero for the other, and zero for both if the household does not apply. Post_t indicates the revisit survey. Household-contract fixed effects μ_{hc} absorb each household's average preference for each contract across waves, and household-wave fixed effects λ_{ht} absorb each household's overall propensity to apply in each wave. Unlike the triple difference, there is no comparison group of households: γ_7 is the average within-household change in the relative ranking of the two contracts as receipt of the advance becomes imminent. If workers are present biased, preferences should shift towards the high advance contract when elicited closer to the pay-out date, i.e. $\gamma_7 > 0$. Standard errors are clustered at the household level, with village-level clustering reported for robustness.

Two key adjustments allow me to isolate changes in job application behaviour arising from changes in preferences rather than changes in circumstances. First, I restrict the sample to those who

[23] Note that if respondents accept a job from another recruiter because of an unexpected income shock, such as a particularly bad harvest, this will not confound the estimates. In this case, the change in the outside option is a direct response to the shock.

are not already committed to another employer and are able to form the minimum group size of three workers at both waves. Second, I exclude respondents who report an unexpected change in their income in the intervening period. These restrictions result in a sample of just 76 households. The estimate is therefore imprecise, and should be interpreted as pertaining only to households whose circumstances remained stable. Table A8 compares baseline characteristics by shock status. The two rows which matter here are the shares measured as time inconsistent and the borrowing capacity of the household: households which report a shock, and are therefore excluded, are if anything less likely to be time inconsistent and no less able to borrow than those retained, so the restriction does not select out the households in whom present bias could appear.

These restrictions hold fixed eligibility for the vacancy and unexpected changes in income, but not every change in circumstances. For example, borrowing capacity was not re-elicited at the revisit, so a change in access to outside loans which arrives without an unexpected income change is not screened out. Such a change would alter the ranking of the two contracts, but the resulting bias is signed. A tightening of liquidity between waves would shift preferences towards the high advance contract and could only produce a false positive. A loosening would shift preferences away from high advance and could mask present bias, an acknowledged limitation of the test.

5.6 Effect of the Information Intervention

The information intervention tests whether providing respondents with accurate earnings projections can shift preferences away from high advance contracts, whose costs I find that respondents vastly understate. This is the final component of my analysis of the role of cognitive distortions on advance demand under the third research question. Treatment effects are measured with the following specification:

$$\text{Information experiment:} \qquad Y_{hb} = \alpha_9 + \beta_9 \cdot \text{Info}_h + \gamma_b + \theta_9' X_h + \epsilon_{hb} \tag{14}$$

Y_{hb} is an outcome for household h in randomisation block b. Info_h indicates assignment to the information intervention, and γ_b are randomisation-block fixed effects, absorbing the village by baseline advance demand cells within which treatment was assigned (Section 4.4.5). X_h contains baseline demand controls: the switch point in the baseline MPL and baseline application behaviour interacted with indicators for the baseline treatment arm.[24] Standard errors are clustered at the village level. This is applied to two sets of contract preference outcomes. First, indicators for applying to the advertised job with a preference for the low and high advance contracts, equal to zero for households which do not apply, including those found ineligible. Second, the index of the first row at which the respondent prefers the low advance contract in the hypothetical MPL exercise.

[24] For households in the baseline single-offer arms, this is an indicator for applying to the advertised job. For households in the baseline Choice arm, it is an indicator for applying with a preference for the low advance contract.

Heterogeneity analysis for a moderator M_h uses the following specification:

$$\text{INFORMATION EXPERIMENT:} \quad Y_{hb} = \alpha_{10} + \beta_{10} \cdot \text{Info}_h + \delta_{10} \cdot M_h + \zeta_{10} \cdot (\text{Info}_h \times M_h) + \gamma_b + \theta'_{10} X_h + \epsilon_{hb} \quad (15)$$

ζ_{10} measures how the effect of the information varies with the moderator. With around 540 households per arm, interaction terms are imprecise, and I interpret heterogeneity results as exploratory.

6 The Cost of Advance Payments

This section answers my first research question: do workers pay to be paid upfront, through lower rates of pay on the job? Descriptive evidence suggests this is the case: advances and piece rates are negatively correlated in equilibrium, by enough to imply a 7% monthly interest rate. However, this correlation is measured from differences across jobs, which may differ in other respects. Moreover, workers are unlikely to be choosing from an identical set of offers. The Choice treatment arm removes these confounding factors by presenting every respondent with the same menu: a high advance contract and a low advance alternative worth more at any monthly interest rate below 19.5%, identical in all other respects.

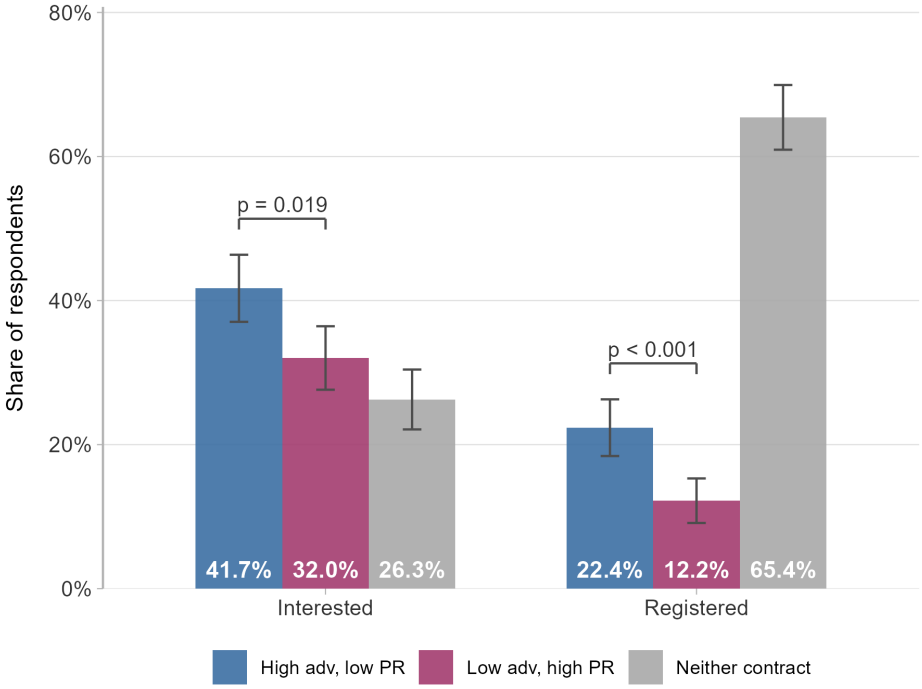
Figure 5 plots the breakdown of realised application decisions. Overall, 73.7% of households express interest in the advertised vacancy. The share doing so with a preference for high advance, at 41.7%, is higher than the share preferring low advance, at 32% ($p = 0.019$, estimated with specification 6 in Table A10). These results are robust to adopting the stricter interest outcome, i.e. successful registration. In the Choice arm, 53.1% of interested applicants reveal in subsequent checks that they do not meet one of the employer's requirements, meaning only 34.4% of respondents enter the list of registered applicants.[25] Again, the rate of successfully registering for high advance is 10 percentage points higher than for low advance ($p < 0.01$).

Even if the job advertisements are otherwise identical, payment terms could shift expectations about other features of the job, in which case these contract rankings would not reflect preferences over payment timing alone. Table A13 uses the single offer arms to test this. Columns 1 to 7 compare expected amenities across arms. Only the availability of a toilet differs ($p = 0.043$), and a joint test across all six amenities does not reject ($p = 0.232$). Columns 8 and 9 address two further motives which could be mistaken for a preference for early pay. Workers might avoid the high advance for fear of being unable to repay it on time, but almost all respondents in both arms expect to repay in time (Column 8). Alternatively, workers might choose a high advance intending to leave without repaying, so that the cost in future pay is never incurred. Anecdotal evidence from discussions with employers suggests reneging does occur, though rarely enough that large advances

[25] The eligibility checks are most commonly failed because the respondent reveals they have already committed to another employer (52.9% of Choice applicants, 32.6% and 31% of high and low advance applicants).

remain the dominant recruitment practice. Column 9 finds no difference across arms in the share expecting other workers to renege, an outcome which I acknowledge may be underreported.[26]

Figure 5: Share Applying with a Preference for High and Low Advance Contracts in the Choice Treatment Arm



Note: This figure plots the share of households in the Choice treatment arm ($N = 434$) applying for the vacancy with a preference for each contract. The left group uses the expressed interest outcome and the right group uses successful registration, both defined in Section 4.2.3. Error bars show 95% confidence intervals based on heteroskedasticity robust standard errors. Each p value tests the equality of the high and low advance shares, taken from the high advance coefficient in the stacked regression of equation 6, reported in Table A10.

These results from the Choice treatment arm concern applicants' rankings of two contractual arrangements under which my partner employers were hiring for the upcoming production season. To understand these trade-offs under a broader set of conditions, I turn to the hypothetical MPL. In this exercise, the full sample chooses between the experimental high advance contract and low advance alternatives at piece rates ranging from 950 to 1,500 Rs (Table 2). As the piece rate increases, so does the implied loss in future earnings from choosing the high advance contract. The left hand axis (blue) of Figure 6 plots the share of respondents expressing a preference for the high advance contract at each of these piece rates. The right hand axis (green) plots the monthly interest rates which would equate the income streams from these contracts, holding productivity constant across each contract at the average sample level.

A significant minority of respondents choose the low advance, even when this has a *lower* piece rate than the high advance option. In particular, 24.7% and 29.2% of respondents choose low advance

[26]The survey asked about the expected likelihood that other workers would renege if hired under the advertised contract, rather than the respondent's own intentions, which would be more likely to be underreported.

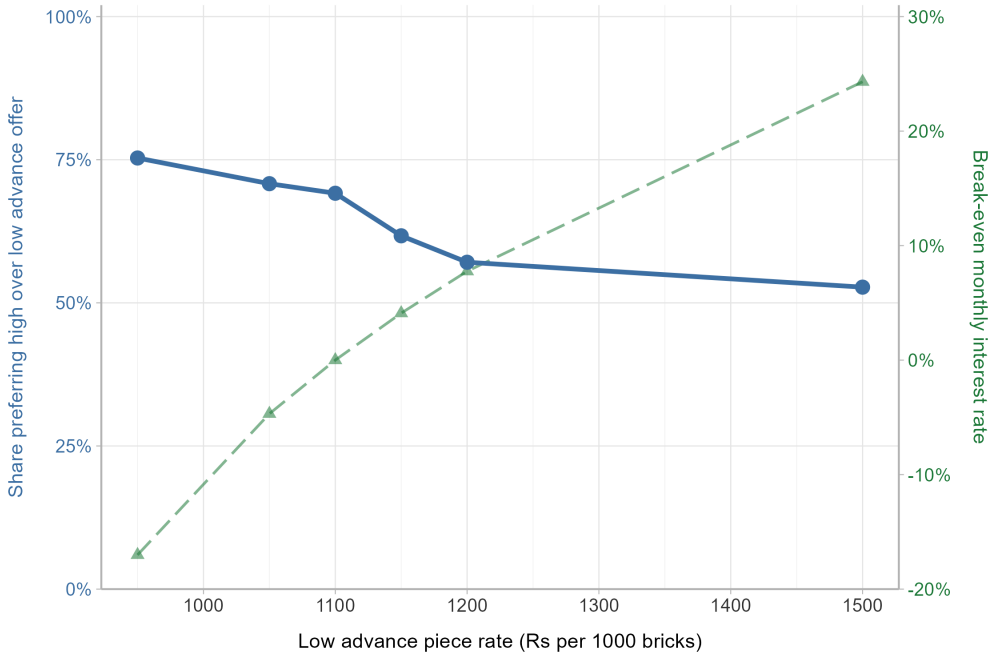
contracts with piece rates of 950 and 1,050, though these should imply lower pay both in advance and in the future, relative to the high advance alternative. As depicted on the right hand axis, respondents would have to be facing a *negative* interest rate in order to justify such a preference, assuming perfect credit markets. Potential explanations include distaste for being in debt to the employer, or concerns surrounding temptation spending under a lumpy income stream.

Demand for the high advance contract decreases as the low advance piece rate increases along the horizontal axis. Table A11 formalises this result using equation 7: a 100 Rs increase in the low advance piece rate lowers the likelihood of choosing high advance by 4.2 percentage points on average ($p < 0.01$), with similar estimates when the row index replaces the piece rate or the sample is restricted to single switchers. This response is gradual in an economic sense: only 33.1% of respondents are ever induced to switch contracts as the piece rate rises. Thus, even for the highest low advance piece rate of 1,500 Rs, 52.7% still express a preference for the high advance contract. At average productivity, this would imply a loss in income earned at the brick kiln of 84,462 Rs in exchange for an increase in advance of only 35,000 Rs, a trade-off which only a monthly interest rate of at least 24.3% could justify.

The interest rates in Figure 6 are illustrative, as they assume a common level of productivity across individuals and contracts. However, output would have to fall below around 5,400 bricks per worker per season, little more than a week of typical production, before the break-even rate on the most generous low advance contract fell below the 1.5% median among borrowers in my sample. Higher output under higher piece rates would only raise these rates further.

The Choice arm and the MPL therefore both show a substantial share of workers paying dearly to be paid upfront. Those in the single offer arms see only one contract, so applications now reveal how each compares to the respondent's outside option rather than to the other. The low advance offer is worth more at any monthly interest rate below 11% at average expected output (Section 4.2.2), so in the unconstrained model of Section 3 it should attract at least as many applicants. Instead, Table A12 presents results from regression equation 8, and finds that households are equally likely to apply under either contract: around two thirds express interest regardless of the offer, and the coefficient on high advance in Column 1 is small and insignificant. This is robust to the post-double-selection lasso specification described in Section 5, which selects additional controls from a larger candidate set (Cilliers et al., 2024) to adjust for chance imbalances (Column 2), as well as to using the outcome of successful registrations (Columns 3 and 4). A contract worth less in present value thus competes equally well against the outside option, which is likely to be partly attributable to the generosity of both piece rates relative to outside options (Section 4.2.2). Indeed, those who do not apply rarely cite pay terms as the reason (Figure A5). The aggregate null also conceals offsetting heterogeneous responses across households, which Section 7 documents: those able to borrow apply less often under the high advance offer, whilst those unable to borrow do not.

Figure 6: Share of Respondents Choosing High over Low Advance



Note: This figure compares two hypothetical contracts: a low advance contract (10,000 Rs) with varying piece rates, and a high advance contract (45,000 Rs with a piece rate of 1,100 Rs). For each low advance piece rate shown on the x-axis, the blue line plots the share of respondents who prefer the high advance contract ($N = 1,494$). The green dashed line, measured against the right axis, plots the monthly interest rate at which the two contracts would offer equal present value, assuming the average expected output of 123,000 bricks per worker per season and monthly compounding over the roughly four months between the receipt of the advance and the midpoint of the work season. This rate is negative for piece rates below 1,100 Rs, where the low advance contract offers both a lower advance and a lower piece rate.

7 Rational Borrowing or Bias?

The results so far suggest that a significant share of respondents pay dearly to access high advance contracts, forgoing alternatives which pay significantly more on the job. In this section, I test the three leading distortions which might explain these preferences, one market and two behavioural. The first posits that advances facilitate consumption smoothing in the absence of alternative borrowing opportunities, making them an efficiency enhancing solution to missing credit markets, even at steep interest rates. Next, I ask whether advances play into present bias, and whether workers possess the arithmetic skills to accurately price the trade-off, vulnerabilities which employers might exploit to hire workers under lower paying contracts than they would truly prefer.

7.1 The Role of Credit Constraints

As demonstrated theoretically in Section 3, workers may choose low paying, high advance contracts because these permit higher immediate consumption than is possible through incomplete credit markets. I first show descriptively that demand for advances correlates positively with credit constraints, in every elicitation. I then turn to a quasi-experimental approach, testing whether

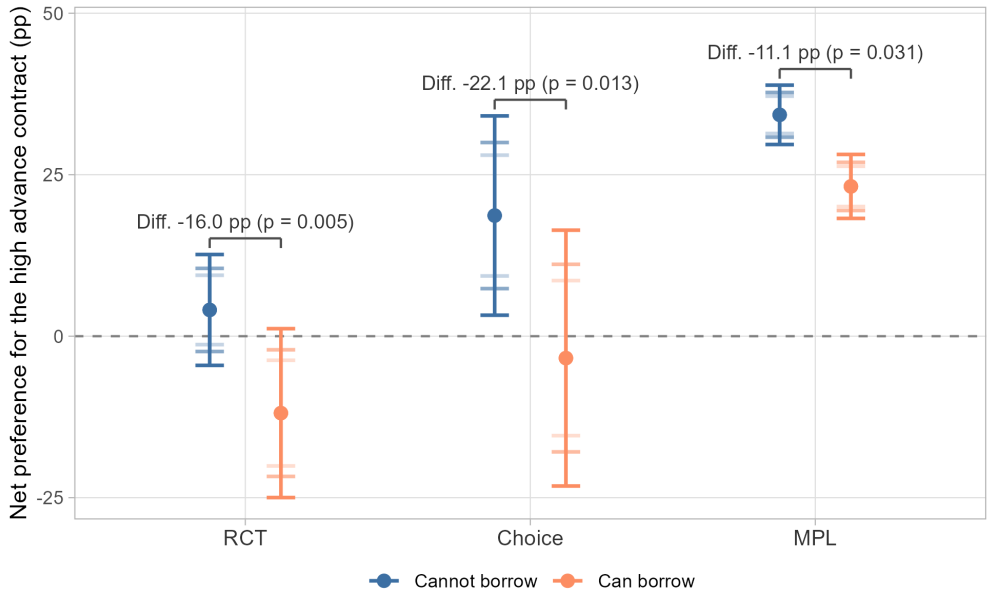
plausibly exogenous financial shocks between surveys shift preferences in favour of high advances.

7.1.1 Heterogeneity by Borrowing Capacity

Credit market failures are common in developing countries, but their incidence within a setting can vary by household: those with more assets and reliable income streams may access loans more easily. Indeed, 40% of my sample respondents *do* have debts, primarily from informal moneylenders (45.9%) or banks (21.6%), and face monthly interest rates of just 1.5% at the median. At the other extreme, 52% report that they could not borrow any amount from outside lenders if needed.

Figure 7 shows that this latter group has consistently stronger preferences for advances. The outcome in each panel is the net preference for the high advance contract over the low advance alternative, defined in the figure note. Among those who can borrow in the single offer arms, a high advance contract reduces the likelihood of applying to the job by 11.9 percentage points, significant at the 5 percent level. This negative effect is entirely absent for those without access to such loans, who are equally likely to apply under either contract, a difference of 16 percentage points ($p = 0.005$, Column 1 of Panel A in Table A14, estimated with specification 9).

Figure 7: Net Preference for the High Advance Contract by Access to Credit



Note: The outcome in each panel is net preference for the high advance contract, the share of choices favouring the high advance contract minus the share favouring the low advance alternative, in percentage points. In the single offer arms (RCT, $N = 1,060$) this is measured as the difference in application rates between households offered the high and the low advance contract, adjusted for village fixed effects and controls. In the Choice arm ($N = 434$) it is the mean within-household difference in applications between the two contracts. In the multiple price list (MPL, $N = 1,494$) it is the share of row-level choices selecting the high over the low advance contract, adjusted for contract row and village fixed effects and controls and recentred at the sample mean. Can borrow indicates a positive reported maximum one month loan amount. Error bars show 90, 95 and 99% confidence intervals in increasingly light shades, from standard errors clustered at the village level. Brackets report the difference between the can and cannot borrow groups, with p values taken from the corresponding interaction coefficients in Panel A of Table A14.

The same pattern appears in the Choice elicitation, where applicants rank the high and low advance offers. Those unable to borrow now respond *positively* to high advance, applying with a preference for this contract 18.7 percentage points more often ($p < 0.01$). No such response is observed for those with access to loans, and the difference in coefficients is again large and significant at 22.1 percentage points ($p = 0.013$, Column 3). Finally, in the MPL exercise, when households trade off high advance contracts against multiple low advance alternatives, high advances are preferred on net by both groups. However, this net preference is significantly weaker among those able to borrow from outside sources ($p = 0.031$, Column 5).[27]

These results hold with the continuous measure of borrowing capacity (Panel B), when the MPL sample is restricted to single switchers (Column 6), and for the stricter outcome of successful registration (Columns 2 and 4), where the interaction coefficients remain negative but shrink and the single offer estimate loses significance.[28]

Together, these results point in a clear direction: the appeal of the high advance contract is much weaker for households with access to outside lenders, as the model of Section 3 predicts. However, households who cannot borrow are likely to differ in other ways which shape contract preferences, so these correlations cannot be interpreted causally. Section 7.1.2 exploits plausibly random variation in a proxy for the tightness of borrowing constraints.

7.1.2 Evidence from Income Shocks between Surveys

Income volatility is a well known feature of developing countries. Reliance on agriculture leaves livelihoods vulnerable to seasonality and weather fluctuations (Breza and Kaur, 2025), and frequent health shocks raise expenditures and reduce labour supply, particularly in the absence of health insurance (Banerjee and Duflo, 2007). Both prevail in my setting: at baseline, 29.6% report death or illness of a family member within the past four months, and 16.5% that the harvest has been worse than expected. Such shocks raise the likelihood that a household's credit demand outstrips what it can borrow from outside lenders, and may therefore push it towards pay-in-advance contracts at the cost of depressed future earnings. Positive income shocks should have the opposite effect, creating a cushion which makes it easier to forgo a large advance in favour of higher future income.

As with credit constraints, those with more volatile incomes may have different contract preferences for unrelated reasons, such as differences in abilities. To address this, I adopt a triple difference specification using the revisit of the Choice arm around one month after baseline, at which respondents reconfirmed their interest in the vacancy and contract rankings, and were asked whether they had experienced any unexpected changes in income in the intervening period. Overall, interest

[27] The table drops the MPL rescaling applied in Figure 7 for consistency of interpretation with the other outcomes, so its coefficient of 0.055 on the indicator for choosing the high advance contract equals half the 11.1 percentage point net gap shown in the figure.

[28] Interested applicants who cannot borrow are negatively selected on eligibility, which narrows the gap between groups under the registration outcome.

dropped from 73% to 57%, and more so for low advance (32.2% to 19.3%) than high advance (40.8% to 37.9%). At least one negative shock is mentioned by 60.6% of respondents, most commonly crop failure (37.9%), followed by rises in the prices of household goods (28.7%) and of agricultural inputs (22.7%) (Figure A6). A positive shock is mentioned by 33%, with the leading categories being an improvement in a family member's health (10.1%), falls in the prices of household goods (9.2%) and the receipt of government benefits (8.6%).

Column 1 of Table 3 shows results from the triple difference specification. In Panel A, those with stable incomes directly prior to the survey exhibit no differential shift in their rate of application to the high advance job. By contrast, those who experience an unexpected decrease in their income shift towards high advance. Positive shocks have the opposite effect, shifting preferences away from high advance. Coefficients are large and of similar magnitude, both significant at the one percent level. Columns 2 to 4 attribute this change in net preferences to a positive impact of both shocks on rates of application, but for opposite contracts depending on the direction of the shock. In particular, negative shocks increase applications by 30.1 percentage points ($p < 0.01$), over 40% of the baseline mean of 73%. This is driven by the high advance contract, applications for which rise by 36.8 percentage points ($p < 0.01$), with low advance applications statistically unchanged. This is in line with high advances serving as a coping mechanism in the aftermath of financial difficulties. Positive shocks instead increase applications for the low advance contract only, again with a large estimated effect of 25.8 percentage points ($p < 0.01$).

The direction and significance of results are preserved for the more restrictive outcome of successful registration (Panel B) and when standard errors are clustered at the village level (Column 4 of Table A15). Village-wave-contract fixed effects, which absorb village-specific trends in relative contract preferences between surveys, shrink the coefficients, but these remain significant for the negative shock ($p = 0.03$) and only marginally insignificant for the positive shock ($p = 0.11$) under expressed interest. They lose significance under successful registration (Column 5).

A remaining concern is that shocks are correlated with household characteristics which would shift preferences in favour of one contract over time, even in the absence of a shock. Indeed, covariates jointly predict both negative and positive shock status ($p < 0.01$ in both cases, Table A8): households reporting a negative shock know more middlemen ($p < 0.01$) and expect higher Kharif incomes ($p = 0.026$), whereas those reporting a positive shock are less likely to be time inconsistent ($p < 0.01$).[29] Controlling for interactions between each imbalanced characteristic and the wave and contract indicators, so that shock effects are identified only from variation within these groups, leaves coefficients highly stable for both outcomes, whether categories of controls are added one at a time or all together (Table A16).

As an additional test, I re-estimate the specification with placebo shocks: negative and positive income shocks reported at baseline for the 2024 Kharif season, roughly twelve months before the revisit (Table A17). These more distant shocks should have much less bearing on current liquidity,

[29] Time inconsistency is measured with the hypothetical choice lists described in Section 7.2.

Table 3: Income Shocks and Contract Preferences Between Survey Waves

	Net high advance preference (1)	Applies (2)	Applies for high advance (3)	Applies for low advance (4)
Panel A: Expressed interest				
Negative shock × Post	0.434***	0.301***	0.368***	−0.067
	(0.119)	(0.073)	(0.076)	(0.063)
Positive shock × Post	−0.373***	0.144**	−0.115	0.258***
	(0.144)	(0.070)	(0.084)	(0.075)
High advance × Post	−0.039			
	(0.072)			
Baseline mean	0.086	0.730	0.408	0.322
Panel B: Successful registration				
Negative shock × Post	0.249***	0.276***	0.262***	0.014
	(0.088)	(0.077)	(0.067)	(0.049)
Positive shock × Post	−0.180*	0.094	−0.043	0.137**
	(0.100)	(0.082)	(0.072)	(0.056)
High advance × Post	−0.097*			
	(0.058)			
Baseline mean	0.092	0.351	0.221	0.129
Specification	Triple diff	DiD	DiD	DiD
Num. obs.	1392	696	696	696

Note: Standard errors clustered at the household level. The sample is the 348 Choice arm households surveyed in both waves. Shocks are unexpected income changes between the surveys, reported at the revisit, and enter jointly. Column 1 estimates equation 12 on the household by contract by wave panel, where the outcome equals one for the contract a household applies with a preference for and zero for the other, and zero for both if it does not apply. Household by contract and household by wave fixed effects leave the coefficients measuring the between-wave change in net preference for the high advance contract, relative to households reporting no shock of that sign. Columns 2 to 4 estimate the household by wave analogue with household and wave fixed effects, the outcome being application to the vacancy at all, with a preference for high advance, and with a preference for low advance, so columns 3 and 4 sum to column 2 and difference to column 1. Panel B applies the stricter registration outcome used elsewhere in the paper. Baseline mean gives each column's outcome in the baseline wave. Alternative shock definitions, village clustering and village trends are reported in Table A15. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

but would still correlate with any characteristics predicting shock-specific preference shifts, and so would reproduce my estimates if those characteristics were driving them. Instead, replacing the contemporary shocks with the placebos yields coefficients which are small and insignificant (Column 2), and including both sets leaves the contemporary shock coefficients essentially unchanged at 0.42 and -0.40, both significant at the 1 percent level (Column 3). This supports the claim that the estimates capture the causal effect of changes in household liquidity.

Column 4 addresses the remaining possibility of mean reversion, whereby shocks correlated with initial preferences could produce mechanical shifts between waves. Controlling for baseline application and baseline net preference for high advance, each interacted with contract and wave indicators, identifies shock coefficients only from households starting from the same baseline preferences. Both shrink towards zero but preserve their direction, with the negative shock interaction remaining significant at the 5 percent level and the positive interaction at the 10 percent level.

Together, these results show that in the face of an unexpected deficit in income, households increasingly apply for the high advance contract. These contracts provide large immediate cash injections with which to overcome this shortfall, particularly within a broader context of incomplete credit markets, but come at a steep cost in the form of lower overall earnings. The experience of unexpected boosts in income has the opposite effect, causing an increase in applications under low advance. In this way, advances might operate as a poverty trap: low advance contracts are investments in future income which only those experiencing periods of relative prosperity can entertain, leaving the most vulnerable households in a high advance cycle where immediate needs are met but poverty is sustained long term.

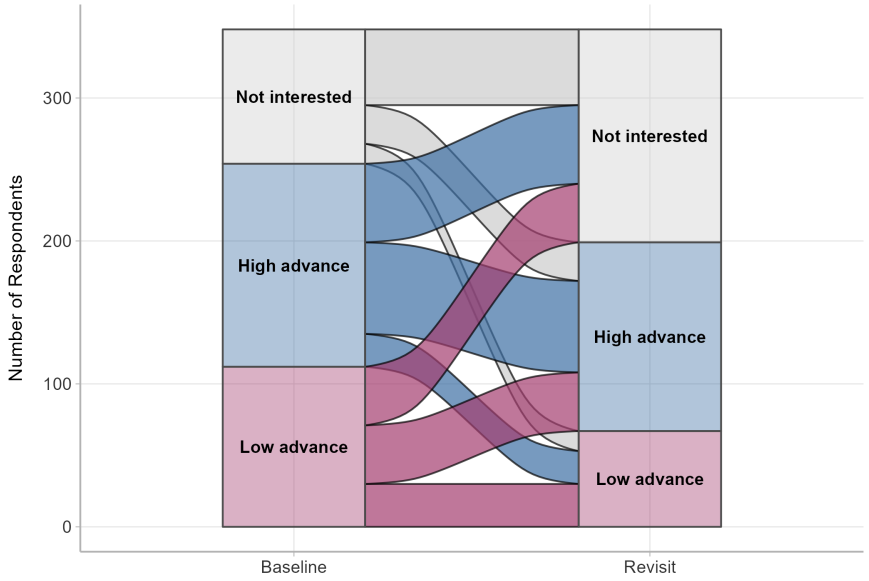
7.2 Do Advances Trigger Present Bias?

In the previous section, I argued that advances support consumption smoothing in the face of income volatility, suggesting these may be welfare enhancing, in spite of the loss in overall earnings. Weighed against this is the risk that advances trigger present bias: workers may be enticed by immediate cash transfers and accept larger interest rates than their ex ante selves would prefer. I study the role of this bias in demand for advances by testing for shifts in contract preferences in the Choice arm between the baseline and revisit surveys. The latter takes place in the days directly prior to advance receipt. Present bias would thus predict a revision of preferences in favour of high advance, given the imminence of the transfer.

As discussed in Section 7.1.2, I do observe a drop on aggregate in low advance applications which exceeds that for high advance. Figure 8 furthermore shows that there is cross-contract churn, including around one third of low advance applicants shifting towards high advance at the revisit. However, changes in circumstances between surveys mean that these cannot be read directly as evidence of present bias. Income shocks are prevalent and shift applications, as the previous section showed, and households may be disqualified from applying at the revisit, having recently

made an agreement with a middleman or no longer being able to form the required group of three workers. I therefore restrict the sample to always eligible households who report no income shock, as described in Section 5.5. This leaves just 76 respondents, so estimates are imprecise and apply only to this subset of stable households.

Figure 8: Transitions in Contract Preferences Between the Baseline and Revisit Surveys



Note: The left and right bars break down contract preferences among the 348 Choice arm households surveyed at both waves. Each household is assigned to the contract it applies with a preference for, as defined in Section 4.2.3, and to Not interested if it does not apply. Flows between the bars record transitions between the two surveys.

Results are displayed in Table 4, with Panels A and B presenting results for expressions of interest and successful registration respectively. Column 1 uses specification 13, the outcome for which is relative preferences for the high advance contract. I detect no differential shift towards high advance between surveys in the stable subsample. If anything, coefficients are negative, indicating a shift away from high advance, matching the sign of the stable income coefficient in Column 1 of Table 3, which omits the eligibility restriction. Columns 2 to 4 break this coefficient into changes in applications for each contract. Interest falls sharply for both: applications for high advance drop by 30.3 percentage points from a baseline mean of 44.7%, and for low advance by 15.8 percentage points from a baseline mean of 18.4%. The larger fall for high advance is what produces the negative coefficient in Column 1. Results are robust to the outcome of successful registration in Panel B.

An analysis of heterogeneity in baseline job application behaviour permits further exploration of the relationship between time preferences and demand for upfront pay. At baseline, survey respondents traded off 100 Rs at a sooner date against 150, 200 or 250 Rs two weeks later, first with the sooner date set to today and then set to two weeks. I use these to construct two impatience indicators, each equal to one for respondents who never take the later payment in the list in question.[30] I then

[30]Switching points are heavily concentrated at this corner, with under 8% of respondents switching at an interior

Table 4: Contract Preferences as Receipt of the Advance Becomes Imminent

	Net high advance preference (1)	Applies (2)	Applies for high advance (3)	Applies for low advance (4)
Panel A: Expressed interest				
High advance × Post	−0.145			
	(0.089)			
Post		−0.461***	−0.303***	−0.158***
		(0.069)	(0.062)	(0.050)
Baseline mean	0.263	0.632	0.447	0.184
Panel B: Successful registration				
High advance × Post	−0.079			
	(0.083)			
Post		−0.342***	−0.211***	−0.132***
		(0.069)	(0.060)	(0.047)
Baseline mean	0.197	0.513	0.355	0.158
Specification	Stacked DiD	DiD	DiD	DiD
Num. obs.	304	152	152	152

Note: Standard errors clustered at the household level. The sample is the 76 Choice arm households eligible for the vacancy at both waves who report no unexpected income change between surveys. Column 1 estimates equation 13 on the household by contract by wave panel, with household by contract and household by wave fixed effects. Columns 2 to 4 estimate the household by wave analogue with household and wave fixed effects, the outcomes being application to the vacancy at all, with a preference for high advance, and with a preference for low advance, so columns 3 and 4 sum to column 2 and difference to column 1. Panel B applies the stricter registration outcome. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

classify respondents as present biased if they exhibit more impatience when the sooner payment is today. Many respondents exhibit strong preferences for earlier payments: 67.8% (71.4%) always choose the earlier payment when the early date is today (in two weeks). However, only 9% behave *more* impatiently when the early date is today.

Table A18 reproduces the heterogeneity analysis in Table A14 using these three moderators. In Panels B and C, both measures of impatience predict stronger preferences for high advance contracts, significantly so in both MPL columns and in one of the two Choice outcomes in each panel, with coefficients for the single offer elicitation small and insignificant throughout. This is consistent with impatience in these lists capturing the same tightness of liquidity documented in the credit constraints analysis above. However, the present bias measure does not predict higher rates of application for high advance contracts in Panel A for either of the incentivised outcomes, and the estimates are imprecise. In the MPL exercise, respondents classified as present biased exhibit *weaker* preferences for high advance contracts.

These results are inconsistent with the predictions of present bias: preferences do not move towards high advance as its receipt approaches, and respondents classified as present biased show no stronger demand for advances in either incentivised elicitation. The welfare concern introduced at the start of this section, that workers accept effective interest rates which their ex ante selves would reject, therefore finds little support.

The absence of present bias may be partially related to the process through which advances are

row of either list, so I do not model the switching point as a linear index.

given, both in the typical setting and in my experiment. In qualitative interviews, participants described a process whereby respondents have an initial discussion with a brick kiln recruiter. They then discuss the terms of the agreement with family members over several days, after which consenting migrants will be given a portion of the advance. Similarly, during the revisit survey, respondents express preferences over advance payments which would ultimately be delivered at a recruitment event in a few days. This separates advances from the consumer credit products for which present bias has been shown to sustain costly borrowing (Meier and Sprenger, 2010), where funds are disbursed on the spot. A further explanation is that time inconsistency has been found to be considerably weaker over money than over effort (Augenblick et al., 2015).

Finally, these results do not speak to present bias in the use of the money once received. What the evidence in this section does argue against is that present bias drives the choice of pay-in-advance contracts before these advances are in hand. The competing behavioural account, that workers misjudge the costs of advances in terms of the loss in future earnings, is the subject of Section 7.3.

7.3 Do Workers Know What Advances Cost?

This section tests the remaining risk introduced by bonded labour contracts: workers may misvalue their costs, which here take the form of higher debt repayments and lower piece rates, due to these being implicit in a calculation which exceeds their numeracy. Consider, for example, a family of three offered a typical contract of 45,000 Rs in per worker advance and 650 Rs per 1,000 bricks. For a modest production of 369,000 bricks across the season (the average expectation of surveyed workers), future earnings would be calculated as piece rate earnings ($369,000 \times \frac{650}{1000}$) net of the total advance ($45,000 \times 3$), i.e. 104,850 Rs. To obtain losses, this would then need to be differenced against earnings from an alternative contract. Low levels of education, averaging at just 3.6 years in my sample, likely reinforce this cognitive burden. On the other hand, people with low formal numeracy can excel in complex arithmetic applied at work (Banerjee et al., 2025).

If workers are proficient in forecasting earnings, then random assignment to the high advance contract should have a negative effect on estimated earnings in the single offer arms, the magnitude of which should roughly reflect output projections. For a family producing the average expected output of 123,000 bricks per worker across the season, this difference stands at around 53,000 Rs per worker.[31] Table 5 tests this hypothesis by estimating the treatment effect of high advance assignment on earnings per worker across the production season, a figure that respondents were asked to estimate for the contract to which they were assigned. In Column 1, I estimate a negative coefficient on the high advance contract, statistically significant at the 5 percent level, suggesting the negative effect of higher advances and lower piece rates is directionally understood on average. However, the size of the coefficient stands at only around 4,000 Rs, an order of magnitude smaller

[31] The additional cost of the debt repayments is the $105,000$ Rs difference in the total advance ($3 \times [45,000 - 10,000]$), and the $55,350$ Rs loss in piece rate earnings is the difference in piece rates multiplied by total output ($369,000 \times [1.25 - 1.1]$). Together these give a loss of 53,450 Rs per worker.

Table 5: Effect of Contract on Expected Earnings, Productivity and Underpayment

	(1)	(2)	(3)	(4)
	Earnings	Bricks	Hours	Expects underpay
High Advance	−4.060**	−1.820	0.226	0.067**
	(1.670)	(2.104)	(1.012)	(0.029)
Low advance mean	37.227	123.390	61.669	0.340
Num. obs.	1060	1060	1060	1060
R^2	0.206	0.174	0.274	0.221
Adj. R^2	0.163	0.130	0.235	0.179

Note: Standard errors are clustered at the village level. All specifications include village fixed effects and controls for age, gender and education of household head and household size. In column 1, the dependent variable is expected household Rabi season earnings per worker in 1000 Rs, net of the advance repayments. In columns 2 and 3, the dependent variables are expected brick output per worker across the season (1000 bricks) and expected weekly working hours. In column 4, the dependent variable is an indicator for whether the respondent says that underpayment by the owner is either likely or very likely in the advertised job.

than the difference implied by average expected productivity levels. This is also less than the difference in advance across contracts (35,000 Rs), suggesting that respondents interpret the high advance contract as paying more upfront and overall. Comparing average estimates in each contract, this is driven by respondents underestimating earnings in both contracts, but more so under low advance. In particular, the low advance mean stands at just 37,227 Rs, significantly less than the 144,100 Rs that an average productivity worker would expect to earn under that contract, or even the 90,600 Rs for the high advance contract.

An alternative explanation for the small average estimated gain in earnings under low advance is that workers expect to be less productive under this contract. Columns 2 and 3 test this hypothesis, replacing the outcome with respondents' estimates of brick production and work hours. Along both margins, expectations are identical across contracts, with coefficients small and insignificant. Finally, if random assignment to the high advance contract reduced expectations of wage theft, this could also depress the expected difference in earnings across contracts. This is tested in Column 4. Respondents were asked for their perceived likelihood of being underpaid by the employer in the advertised contract, answering on a scale from one to five. Strikingly, in the low advance arm, a third of respondents report that underpayment is either "likely" or "very likely". This share is 6.7 percentage points *higher* in the high advance arm ($p = 0.026$), the opposite direction to that required to explain the small difference in earnings relative to low advance.

This miscalculation of the effects of advances and piece rates on earnings may have significant welfare consequences. Indeed, if workers cannot accurately assess the terms on which they supply their labour, they may choose contracts which make them worse off. Two questions follow. First, what is the source of these calculation errors? Second, can an information intervention correct these beliefs and shift contract choices in favour of low advance options? I examine both directly through Round 2, the information experiment, described in Section 8.

8 Can Workers Be Told What Advances Cost?

8.1 Where Do Calculation Errors Come From?

Calculation errors affect workers at two points. On the job, workers must verify that the employer has paid what was owed, a real risk in developing countries where contracts are often only weakly enforced (Kapoor, 2026). When choosing a new contract, they must project what each option would pay. Round 2 was designed to study both, collecting data on terms, output and pay from the prior work season alongside pay estimates for hypothetical and advertised contracts.

8.1.1 Undetected Underpayment on the Job

The vast majority of workers report that, rather than simply taking the employer's pay figure at face value, they undergo the cognitively difficult process of counting bricks and inferring correct pay from these.[32] Asked how confident they are in their ability to detect underpayment through these calculations, most said they were either confident (65.3%) or very confident (17.8%). For most respondents, audits have a positive outcome: 76% believe that last season's pay was fair given the terms they were promised.

This confidence is difficult to reconcile with the pay workers report receiving. For the 55% of my Round 2 sample who migrated to a kiln in the 2025-26 season, earnings were low: 35.4% of migrants took home nothing beyond the advance, and a further 3.0% left the kiln still owing the employer, not having produced enough to repay their debts. Median pay stands at just 6,000 Rs. Low pay partly reflects the contracts themselves. Most migrants take large advances, so little remains to be paid out once debts are deducted.[33] However, a large share of migrants also took home less than those terms imply. Comparing take-home pay, with deductions for food, travel and other expenses added back, against recalled output multiplied by the piece rate net of the advance, 52% of respondents were paid over 10,000 Rs less than those terms imply, and the median respondent took home 20,000 Rs less than the imputed fair amount. Discrepancies of this size are hard to attribute to misremembered terms or output, and instead suggest wages are being siphoned by employers. This largely goes unnoticed: perceived underpayment is rare and unrelated to imputed underpayment, and the average respondent was underpaid by 36,000 Rs more than they perceive.

8.1.2 Anchoring on Past Pay

Pricing a new offer requires the same arithmetic, applied to output which has not yet been produced. This section asks how workers arrive at a pay figure for a contract they have not yet worked under, using the hypothetical calculation questions and the pay projections which control respondents give

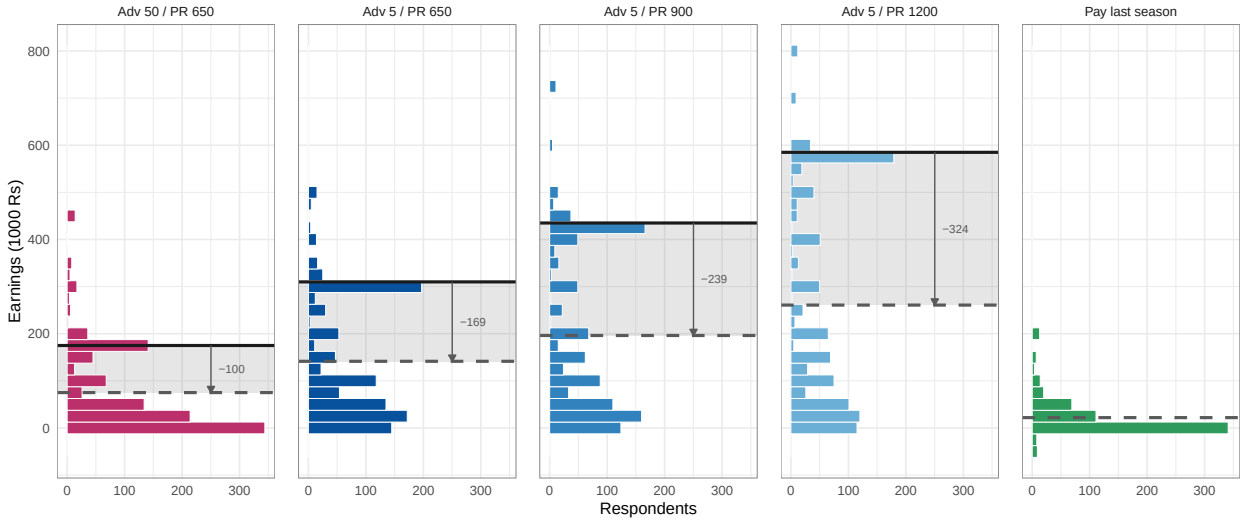
[32] 78% of brick kiln workers paid on a piece rate basis report keeping track of their output on site, and 94% of those report using this to ascertain what their pay should be.

[33] The median migrant was paid 45,000 Rs per worker in advance and 650 Rs per 1,000 bricks last season.

for the two advertised contracts. Whereas the Round 1 earnings calculation results presented in Section 7.3 risked being confounded by differences in wage theft and output expectations, output is held fixed and pay is fair by assumption for these questions. This means that calculations should purely reflect arithmetic ability. Figure 9 plots the distribution of estimated pay for each hypothetical contract shown, together with the distribution of pay last season. The solid black lines show the correct answers, and the dashed black lines the mean estimates. In every question, the vast majority of responses understate the correct value: a small mass of estimates sits at each correct value, very few lie above it, and a large mass sits near zero, much closer to the distribution of pay reported last season by those who migrated (Panel 5).

Average errors are larger for better paying contracts. First, when the advance shrinks across panels 1 and 2, holding the piece rate constant, the average respondent adjusts estimated take-home pay upward by just 50% of the correct amount (Column 1 of Table A19).[34] Second, as the piece rate rises across panels 2 to 4, holding the advance constant, the average respondent uprates earnings by just 43.4% of the correct value (Column 3).[35] Both biases appear exacerbated by low education: 45% of respondents report having no formal education whatsoever, and understate the positive effects of lower advances and higher piece rates by 57% and 65%, respectively (Columns 2 and 4).

Figure 9: Estimates of Pay in Hypothetical Contracts and Last Season's Take-Home Pay



Note: The first four panels plot estimated group take-home pay for each hypothetical contract in the pre-treatment calculation battery, in thousands of rupees. The solid line marks the correct answer and the dashed line the mean estimate, with the shaded band and arrow showing the gap between the two. Both arms are pooled, since the battery precedes any treatment content. The final panel plots reported take-home pay from kiln work over the 2025-26 season on top of the advance, before any other debts the employer deducted, for the households which migrated to a kiln last season, with the dashed line marking the mean. All amounts are winsorised at the 1st and 99th percentiles, within contract for the estimates and across households for last season's pay, so no observation is dropped. $N = 1{,}078$ households in each of the first four panels and $N = 596$ in the final panel.

[34]The correct upward adjustment is 135,000 Rs in total, i.e. 45,000 Rs per worker.

[35]The correct upward adjustment is 500 Rs per 1 Rs increase in the piece rate per 1,000 bricks.

Table 6: Anchoring of hypothetical contract estimates on last season's pay

	(1)	(2)	(3)
Objective value	0.466***	0.466***	0.466***
	(0.021)	(0.021)	(0.021)
Last-season take-home		0.394***	0.360**
		(0.145)	(0.138)
Years of education			7.416***
			(1.605)
Constant	−3.024	−11.672**	−35.410***
	(5.382)	(5.374)	(7.128)
Households	596	596	596
Num. obs.	2384	2384	2384
R^2	0.152	0.159	0.183
Adj. R^2	0.151	0.159	0.182

Note: One observation per respondent and hypothetical contract, pooling the four questions in which respondents estimate season take-home for a group of three producing 500,000 bricks. Amounts are in 1,000 Rs, so the coefficients on the objective value and on last-season take-home are both rupees per rupee. The objective value is the correct answer for that contract, which takes the four values in Table A5 and is the same for every respondent. Last-season take-home is what the household reports taking home from kiln work over the 2025-26 season on top of the advance, before any other debts the employer deducted from earnings. Estimates are winsorised at the 1st and 99th percentiles within contract and take-home across households. The sample is the households which migrated to a kiln last season. Column (3) adds the respondent's years of education. The columns include no fixed effects, so the constant is reported, with standard errors clustered at the village level.

Table 6 shows that anchoring appears at an individual level. Pooling the four calculation questions among the 596 households which migrated to a kiln last season, I regress each estimate on the correct answer for that question and on the household's own take-home pay last season, clustering standard errors at the village level. In Column 2, holding the question fixed, each additional 10,000 Rs taken home last season raises the estimate by 3,900 Rs ($p < 0.01$). This relationship is not explained by a correlation between pay and education, as shown in Column 3.

These estimates can be summarised by a simple representation of anchoring, in which projections of pay in new vacancies are biased in the direction of previous jobs. Formally, perceived earnings are a convex combination of correct earnings and an anchor f_a: $\tilde{f}_j = (1-\theta)f_j + \theta f_a$ with $\theta \in [0,1]$. The additive mistake $\theta(f_a - f_j)$ is decreasing in f_j, meaning this bias disproportionately penalises contracts paying more in the future: a high anchor gives a larger relative uplift to low pay, and a low anchor gives a larger relative penalty to high pay. Thus, in a comparison of equal present value contracts, the high advance option is now perceived to strictly dominate:

$$\hat{W}_{ha} - \hat{W}_{la} = \theta \frac{f_{la} - f_{ha}}{1+r} > 0 \tag{16}$$

Table 7: Estimated and correct take-home pay at the advertised contracts (1,000 Rs)

	High advance	Low advance	Difference
Estimated take-home	31.4	112.6	81.3
	(3.3)	(5.4)	(5.6)
Correct take-home	32.7	257.5	224.9
	(3.4)	(5.5)	(2.7)
Estimated minus correct	−2.0	−145.3***	−143.6***
	(4.1)	(5.6)	(5.8)
Num. obs.	536	536	536

Note: Control households only, who estimate their own pay rather than being told it. Each cell is a mean from a regression of the outcome on a constant, with standard errors clustered at the village level. Estimated take-home is the household's own figure for what its group would earn at the kiln over the season, net of the advance, at its typical output scenario. Correct take-home applies the contract terms to that same reported output and group size. The difference column is low advance minus high advance. Stars mark the test that the estimate departs from the correct value, the only row in which zero is a meaningful null. Amounts are winsorised at the 1st and 99th percentiles within series, so rows need not difference exactly.

The same pattern appears when respondents turn from hypothetical contracts to the real vacancy advertised later in the Round 2 survey. Just as the calculation battery and the anchoring model predict, the earnings gain from its low advance option barely registers. These pay projections, in which respondents estimate pay under each contract at self-reported typical productivity, are given only by the control group ($N = 536$), as treated respondents are instead *told* the correct values. The first row of Table 7 shows average estimates, as well as average individual-level cross-contract differences. The second and third rows report average correct earnings and the average individual-level gap between estimates and correct values. On average, respondents accurately value the high advance contract, and errors are statistically indistinguishable from zero. Under the low advance contract, however, the average worker would accept as fair a take-home pay amount which is less than half of the true fair value, and understates the gain relative to high advance by 143,600 Rs. For 73.9% of respondents, the estimated gain in take-home pay under low advance is smaller than the extra upfront pay under high advance, implying that the high advance contract pays both more upfront *and* more overall, reproducing the Round 1 result in Section 7.3. Whether these errors can be resolved by doing the arithmetic on a worker's behalf is the question I turn to next.

8.2 Does Telling Workers the Answer Help?

Given the size and direction of these errors, providing the treatment group with correct earnings projections under each contract should raise their perceived gain from the low advance contract, relative to control. I test this using a question at the end of the module on expectations of the advertised vacancy, before the job application, which asks respondents in both arms for their best

Table 8: Treatment effects on expected seasonal pay (1,000 Rs)

	Low advance		High advance		Low − high	
	Fair pay	Take-home	Fair pay	Take-home	Fair pay	Take-home
Treatment	140.7***	11.3	8.4	−0.2	132.3***	11.5
	(8.5)	(13.7)	(5.0)	(6.0)	(6.7)	(9.3)
Control mean	112.8	125.1	25.0	43.2	87.8	81.9
R^2	0.408	0.186	0.158	0.207	0.476	0.200
Adj. R^2	0.319	0.062	0.030	0.086	0.396	0.078
Num. obs.	525	525	525	525	525	525

Note: Outcomes in 1,000 Rs, winsorised at the 1st and 99th percentiles. Fair pay is take-home at normal output with full payment, estimated by control and told to treatment. Take-home is what the respondent thinks the group would really take home over the season, taking everything together. The outcome in the last two columns is the within-respondent difference between the low and high advance amounts. All columns include randomisation-block fixed effects, with standard errors clustered at the village level.

guess of what they would take home in each contract. This is separate from earlier questions in the expectations module, asked to the control group only, which elicit estimates under specific productivity and wage theft scenarios, discussed further in Section 8.3.2. This final question was added part way through data collection to investigate the pattern of application decisions, so it is available for 525 households and should be treated as exploratory. Treatment assignment is balanced within this subsample, but villages were surveyed sequentially, and those reached after the question was added have weaker ties to kiln work (Table A20). If anything, this biases towards finding an effect, so the null which follows is conservative: households less attached to advances should be more open to news about their cost.

Table 8 reports the effect of the information intervention on this final earnings estimate. Columns 1 and 3 compare pay estimates given *to* treated respondents to those given *by* control respondents for each contract, earlier in the expectations module. In line with the earlier finding that respondents underestimate the earnings gains from low advance, the coefficient in Column 1 is large and significant: treated respondents are given a low advance pay projection which is 140,700 Rs larger on average than that which control respondents estimate for themselves. No such difference is detected for high advance in Column 3, and in Column 5, the estimated cross-contract difference is large and significant at around 130,000 Rs. If treated households fully internalise the earnings projections provided by enumerators, Columns 1 and 3 should closely match Columns 2 and 4, for which the outcomes are self-reported pay estimates from the end-of-module question asked to treatment and control alike. Instead, treated households give low advance pay projections no larger than control, despite having been informed that they would be paid much higher amounts just minutes prior. In Column 6, the effect on the estimated earnings gain over high advance falls to an insignificant 11,500 Rs.

In line with the null effect on the believed earnings gain from low advance, the applications of

Table 9: Application for the advertised job, by preferred contract

	Low advance		High advance	
	Block FE	+ controls	Block FE	+ controls
Treatment	−0.040**	−0.041**	0.042**	0.040**
	(0.019)	(0.019)	(0.018)	(0.019)
Control mean	0.233	0.233	0.125	0.125
Num. obs.	1078	1078	1078	1078
R^2	0.169	0.178	0.148	0.158
Adj. R^2	0.056	0.061	0.032	0.038

Note: The outcome is an indicator for registering as an applicant for the advertised job and preferring the contract shown in the column header, either accepting only that contract or accepting both and naming it as preferred. It is 0 for households that do not register, including the ineligible. Baseline demand controls are the baseline MPL switch point and baseline application behaviour interacted with baseline arm indicators. All columns include randomisation-block fixed effects, and Columns 2 and 4 add baseline demand controls, with standard errors clustered at the village level.

treated households do not shift towards this contract. Results are displayed in Table 9. First, at the level of applications in control, 23.3% apply with a preference for the low advance job, relative to 12.5% with a preference for high advance and 64.2% not applying. Over a third of applicants therefore choose the option offering more upfront but far less in present value, though their propensity to do so is weaker than in Round 1. This could be because Round 2 contracts carry lower piece rates, leaving less reason to forgo the more lucrative low advance alternative for those without an acute liquidity need.[36] Moreover, Round 2 took place *before* the 2026 harvest, whereas the baseline took place soon *after* the 2025 harvest. If demand for advances lies partly in smoothing consumption over a bad harvest, as Section 7.1.2 documented, this may be weaker before the harvest is realised.

In Column 1, I find that applications for the low advance contract are 4 percentage points lower for treated households, significant at the 5 percent level, an effect robust to including additional baseline demand controls (Column 2). This is almost exactly counterbalanced by a 4 percentage point increase in the rate at which respondents apply with a preference for the high advance contract (Columns 3 and 4).

The information intervention therefore appears to have backfired: informing respondents about the size of the earnings gain from the low advance contract failed to move beliefs, and shifted applications in the opposite direction, towards the lower paying option. An exploration of the mechanisms underlying these results is the focus of the next section.

[36] At self-reported output, the average respondent would take home only 37,000 Rs over the season under the Round 2 high advance contract, against 193,000 Rs under its baseline counterpart.

8.3 Mechanisms

The projections which treated respondents were given lay far outside their experience. For the median respondent, the low advance figure stood at the 99.5th percentile of take-home pay in the previous season, and those expecting higher productivity were told larger amounts still. Section 8.3.1 shows that the size of the figure matters: the more a respondent was told they would earn under the low advance contract, the less likely they were to apply for it. The remaining two subsections ask why a large projection fails to raise demand. Respondents may have believed the projections to be accurate, but expected not to receive them: by raising expectations of what they should be paid under a low advance contract, the intervention may simultaneously have raised expectations of the amount which employers must be siphoning to explain why the pay they have actually received falls so far below this. Alternatively, they may not have believed the figures at all, sticking to a mental model in which fair pay is close to what they have received before, a rejection which may itself be motivated by a strong preference for the liquidity which an advance provides.

8.3.1 The Size of the Told Figure

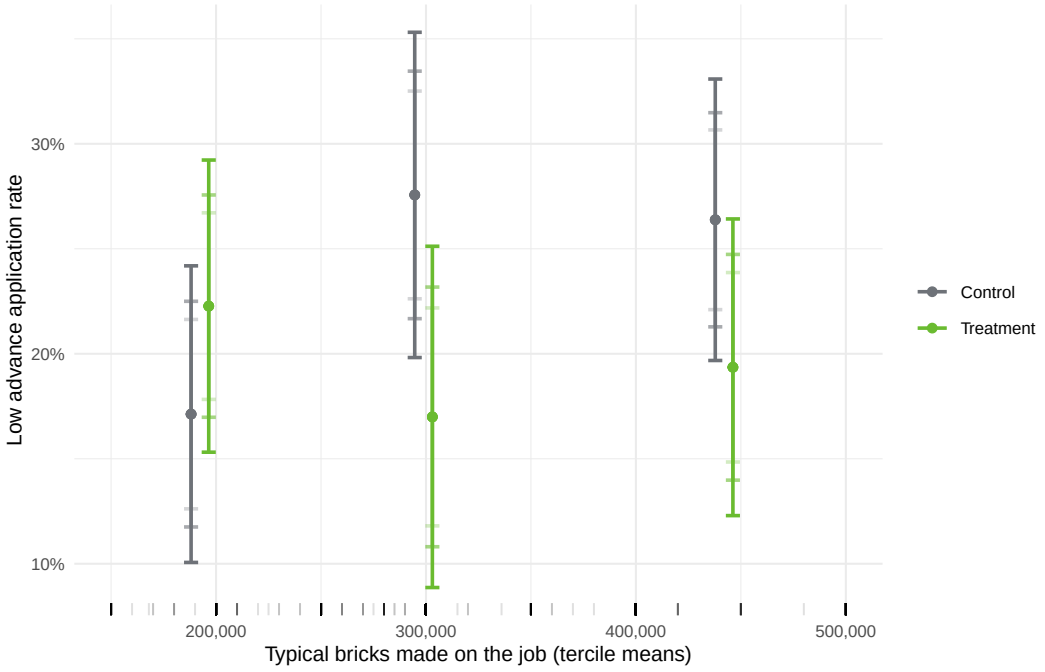
If the told figures were dismissed as implausible, the information should have deterred most where the figure was largest. Returns to the higher piece rate are increasing in output, so demand for the low advance contract should rise with expected productivity. Section 8.1 showed that, when pricing contracts for themselves, respondents register only a fraction of this gain. Treated respondents are told the correct amounts, computed from their own expected output. If they internalised this information, the relationship between productivity and low advance applications should be at least as steep in treatment as in control. Figure 10 tests this, plotting the rate of application to the low advance contract by tercile of expected productivity and treatment arm.

In the control group, the relationship is as expected: respondents in the upper two terciles of expected productivity are more likely to apply with a preference for the low advance contract, a pattern not observed for the high advance contract (Column 5 of Table A21). In the treatment group, it breaks down: those in the upper terciles apply at lower rates than the bottom tercile, though these differences are not statistically significant (Column 2). Being told of a larger gain deterred, rather than encouraged, applications for the contract offering it. These results are robust to using continuous productivity measures (Column 1), and to allowing productivity predictors to moderate treatment effects (Column 3).

8.3.2 Wage Theft

A rational account of the null is that respondents believed the projections but did not expect to receive them, attributing the gap to wage theft. Two pieces of evidence argue against this. First, wage theft expectations under the low advance contract do not rise by enough to counteract the

Figure 10: Applications for the Low Advance Contract by Productivity and Arm



Note: Regression-adjusted share applying for the low advance contract, by productivity tercile and treatment arm. Estimates come from a single regression on arm-by-tercile indicators with baseline demand controls and randomisation-block fixed effects, centred on the sample application rate. Points sit at the mean output of their tercile, nudged apart by arm for legibility. Error bars show 90, 95 and 99% confidence intervals in increasingly light shades, with standard errors clustered at the village level. Ticks along the bottom show the productivity distribution over the middle 90% of the sample. $N = 1{,}078$ households.

told earnings gain from low advance. These results come from questions in which the respondent allocates ten stones to the fair, underpayment and overpayment scenarios in proportion to their perceived relative probabilities, and gives estimated pay amounts under each scenario. The likelihood of being paid below the correct amount, the figure told to treated respondents and estimated by control respondents themselves, rises from 9% to 13% (Table A22). The loss conditional on underpayment rises more sharply, but a large loss suffered with small probability does little to move expected pay. Taking these beliefs at face value and calculating expected pay as the probability-weighted average of these scenarios, treated respondents should still expect roughly 125,000 Rs more than control when asked what they would really take home, but Table 8 shows that the effect is null in practice.

Second, the same information intervention has no effect on contract rankings in a hypothetical MPL which abstracts from wage theft. In particular, respondents state their preference between each pair of high and low advance contracts in Table A6, assuming that they make their self-reported typical number of bricks and are paid fairly for this work. Before deciding, treated households are given accurate pay projections for each contract. If the absence of a positive effect on low advance applications in the incentivised elicitation in Table 9 was driven by increased wage theft concerns, a positive effect should emerge here, where those concerns are removed. However, I detect no differences in the propensity to rank the high advance contract first at any rung of the MPL, including at the rung which reproduces the contract choice from the incentivised elicitation (Column 2 of Table A23). Nor do I detect an effect on the first rung at which low advance is chosen (Column 6).

8.3.3 Is the Rejection Motivated?

If the absence of a treatment effect is not driven by increased concerns about wage theft, the more plausible explanation is that the intervention simply failed to correct the misspecified mental model underlying earnings calculations. This would be in line with existing evidence that misspecified models can persist even in the presence of accurate information (Hanna et al., 2014; Esponda et al., 2024). In this final section of the results, I explore whether this rejection was in any way motivated. Motivated reasoning posits that the desire to reach a particular conclusion distorts the cognitive process which produces it (Kunda, 1990). Households with a strong underlying preference for high advances, or for their outside option, may appeal to doubt in the projections to justify forgoing the low advance contract, much as people discount information which contradicts their political views (Thaler, 2024) or appeal to confusion to justify selfish choices (Exley and Kessler, 2024). Information which people are not motivated to believe has even been shown to entrench initial positions (Fryer et al., 2019).

Isolating this cleanly would require exogenous variation in whether a decision is attached to a motive. As suggestive evidence, I instead rely on cross-household variation in features which would predict a motive for applying to the lower paying, high advance option: access to credit, income

shocks and the intention to take an advance at least as large as the advertised one during the upcoming season. Treatment effects on high advance applications by liquidity demand are estimated using equation 15 and plotted in Figure 11. Across all three measures, the increase in applications for the contract which the intervention underlined as less profitable is concentrated amongst those with the strongest motive to choose it. Treatment leaves high advance applications unchanged for those who can access outside loans, but increases them by 8.4 percentage points ($p = 0.001$) for those who cannot, a difference across groups of 13.6 percentage points ($p = 0.004$). The same holds for the 31.6% reporting a negative income shock in the past three months, most commonly death or illness in the family: they raise high advance applications by 10.9 percentage points ($p = 0.003$), against no effect for stable-income households. Expected advance size next season, the most direct measure of liquidity demand, shows the same pattern at 10 percent significance. Table A24 shows that none of these variables predict treatment effects on low advance applications (Panel A).

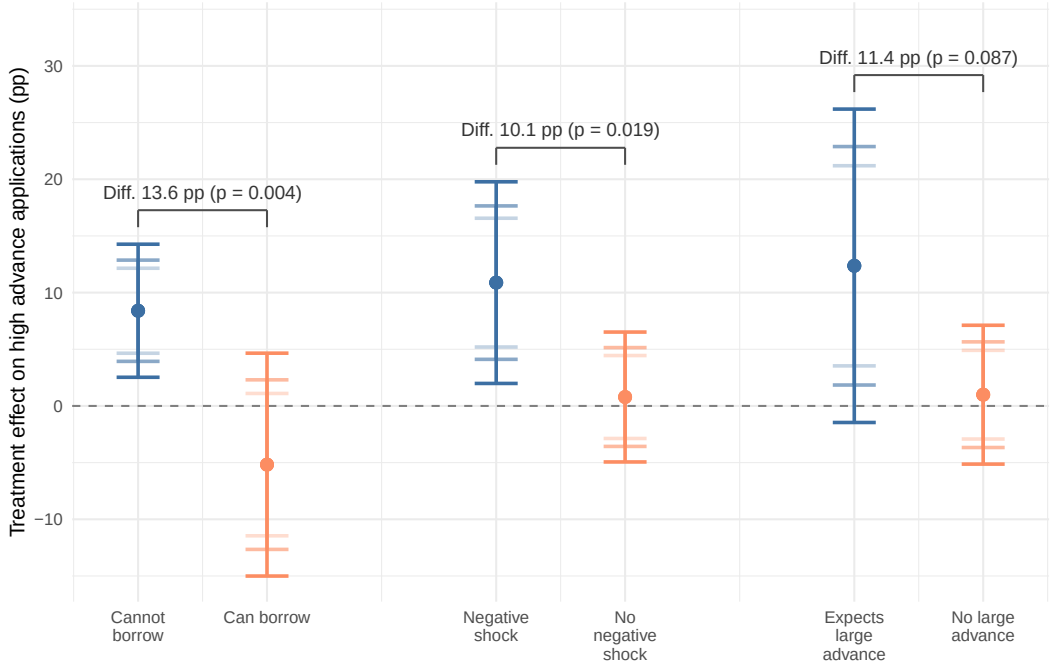
One concern is that liquidity demand proxies for cognitive ability, which could push treated households in either direction: those able to absorb the information towards low advance, those unable to follow it back towards high advance. Columns 4 and 5 argue against this. Treatment effects on either contract do not vary with education or with accuracy in the pre-treatment calculation questions, and the numeracy interaction runs the wrong way for confounding, with worse arithmetic if anything weakening the shift towards high advance. Taken together, the perverse response to the intervention appears best explained by respondents rejecting figures far outside their experience, and doing so most readily where an advance is most needed.

9 Discussion and Conclusion

Bonded labour arrangements in India's brick kilns draw widespread modern slavery allegations, yet empirical evidence on what these contracts cost workers has so far been absent. This paper fills this gap by asking how much income workers forgo in order to access the advance payments which bind them to employers until these debts are repaid out of earnings. I then ask where demand comes from: is it a rational response to missing credit markets, or a mistake rooted in behavioural biases from which workers need protecting? In doing so, I speak to the broader question of whether bonded labour arrangements are exploitative, a perspective written into international law and advocated by human rights campaigns, or instead a vital source of relief which government interventions and credit markets have failed to provide for some of the poorest households.

My results show that workers pay a steep cost in order to be paid upfront: job applicants are significantly more likely to choose a high advance option over a low advance alternative worth more at monthly interest rates well below those which borrowers face in local credit markets. One caveat is that respondents could withdraw at no cost after the survey, so interest may be over-reported. However, in the choice elicitation, this should affect only the level of applications, rather than the ranking of the two contracts.

Figure 11: Treatment Effects on High Advance Applications by Demand for Liquidity



Note: Estimated effects of the information treatment on high advance applications within each moderator group, in percentage points. Each pair of points comes from a single regression of the outcome on treatment, the moderator and their interaction, with baseline demand controls and randomisation-block fixed effects, and the bracket reports the difference between the two effects, the interaction term, with its p value. Error bars show 90, 95 and 99% confidence intervals in increasingly light shades, with standard errors clustered at the village level. Cannot borrow indicates that the household reports being unable to borrow any amount for one month. Expects large advance indicates an expected advance of at least 50,000 Rs per worker for the coming season, the amount offered under the advertised high advance contract. $N = 1{,}078$ households.

Across all incentivised and hypothetical elicitations, demand for high advances is concentrated amongst the half of my sample which cannot access outside loans. It also increases by 37 percentage points after an unexpected income shortfall between surveys. Though advance receipt and spending outcomes are beyond the scope of this study, these results are consistent with respondents intending to use advances to smooth consumption over income fluctuations, as a rational decision maker would. Such fluctuations can otherwise have devastating consequences for credit constrained households. Moreover, whilst the standard caveats surrounding the endogeneity of these fluctuations apply, my large and significant effects survive several robustness tests. These findings extend existing evidence that credit constraints push workers towards jobs which pay sooner (Fink et al., 2020; Scarelli and Margolis, 2025; Kapoor, 2026; Scarelli, 2025) to bonded labour contracts, which had until now been studied only theoretically (Bardhan, 1983; Mukherjee and Ray, 1995; Genicot, 2002).

What also sets my research apart from existing studies is that I distinguish the consumption smoothing benefits of payment structures within labour contracts from their potential risks. I find no evidence that workers succumb to the temptation of immediate cash when an advance's disbursal is imminent, casting doubt on the present bias hypothesis. As Section 7.2 argues, this likely reflects a recruitment process in which agreement and disbursal are separated by several days, which sets advances apart from payday loans and Earned Wage Access products that transfer cash immediately. Two caveats apply. The preference reversal test rests on just 76 households whose circumstances were stable between surveys, so the estimate is imprecise. It also concerns the choice of contract alone, and says nothing about self-control over how the advance is spent once received. That distinction may matter, since advances are disbursed during a major festive period in the region, when self-control may be under greater strain.

My data instead point to a failure to quantify what high advance contracts cost workers. The loss in future pay associated with choosing a high advance can only be quantified by multiplying the piece rate across hundreds of thousands of bricks, and then deducting the advance. This stands out as likely to be far beyond the numeracy of a sample with average education of just 3.6 years. On the other hand, workers have significant experience in the industry, and those with low formal numeracy can excel in complex arithmetic applied at work (Banerjee et al., 2025). My results show that people fail badly at these calculations, and in a direction that causes them to underestimate how much higher future pay would be under higher piece rate, low advance contracts. Cognitive limitations of this kind have been shown to confound choices in consumer credit (Stango and Zinman, 2009) and health insurance (Bhargava et al., 2017), and more recently in the spending forecasts of Zambian farmers (Augenblick et al., 2023). My results place them in the labour contracts through which poor households earn most of their income, an informal market with none of the disclosure rules or consumer protections which govern credit and insurance in higher income countries.

The most direct remedy for these errors is to do the arithmetic on workers' behalf. However, evidence from the information experiment suggests that doing so fails to correct the mental model

behind these errors: pay estimates do not change, and job applications shift towards the high advance contract, that which the information reveals to be less profitable. This is in line with evidence that misspecified models can persist even under abundant feedback (Hanna et al., 2014; Esponda et al., 2024). The rejection also looks partly motivated, with the caveat that this rests on cross-household comparisons rather than experimental variation. In particular, the rise in high advance applications is concentrated amongst those without access to loans and those recently hit by a negative income shock, the same households for whom Section 7 showed the liquidity of an advance to be most important. The existing literature on preferences over payment timing studies market failures and behavioural distortions apart, but here they reinforce one another: the households with the most to gain from an advance are the least willing to learn what it costs.

Bonded labour arrangements therefore appear to facilitate consumption smoothing for credit-constrained households. However, the price of that relief is steep, and more worrying still, workers do not internalise it. This allows employers to raise the price, and hence their profits, without workers' knowledge. Making the price more visible, rather than banning these contracts altogether, could preserve the relief element whilst ensuring that choices are well informed. However, interventions of this kind must be mindful of the behavioural biases they might trigger. Indeed, the news that an alternative contract is far better than any a worker has recently worked under might easily be dismissed as too good to be true, particularly by those motivated to do so because of a strong preference for the liquidity which an advance provides. An effective intervention might have to focus on correcting the underlying mental model rather than the valuations it produces, for example by giving workers the tools to calculate earnings themselves.

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Figure A1: Brick Production at a Traditional Clamp Kiln

(a) Brick Moulds



(b) Drying Bricks

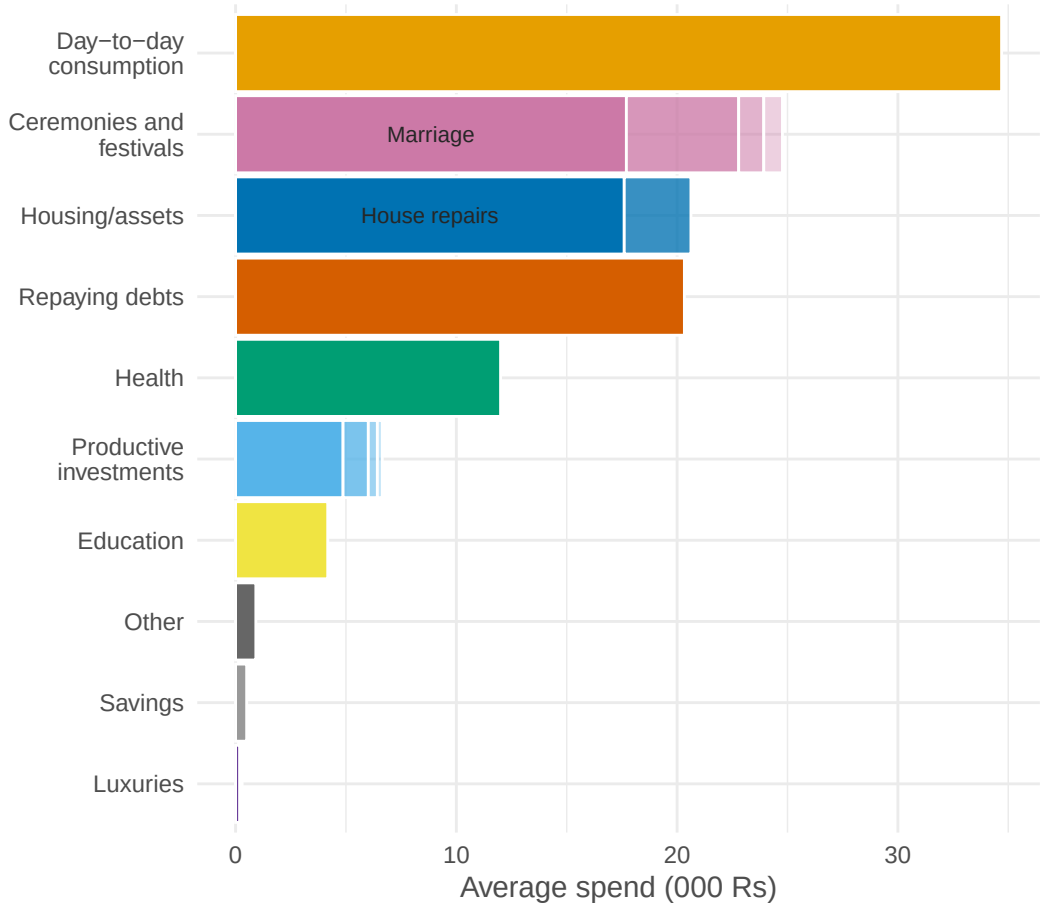


(c) Kiln



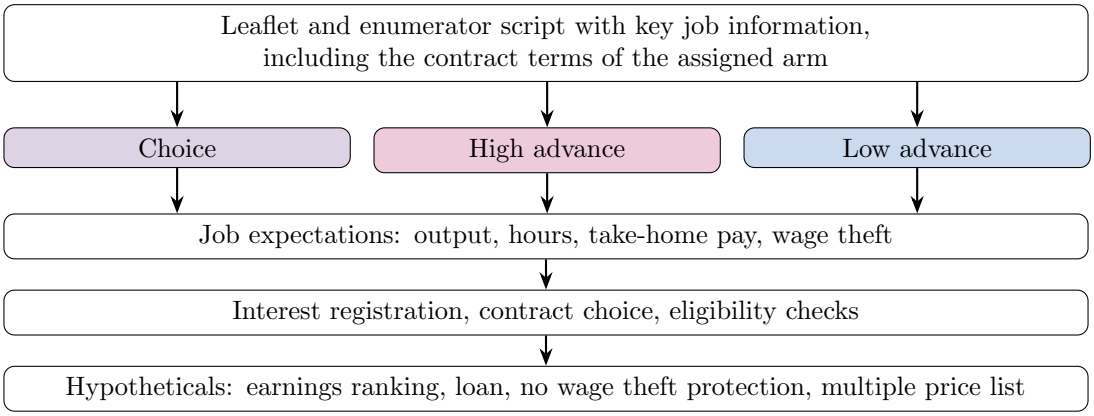
Note: Author's own photographs, all taken at a brick kiln in Hyderabad. This kiln is a traditional clamp kiln, a temporary structure in which unfired bricks are stacked with layers of fuel.

Figure A2: How Brick Kiln Advances Were Spent



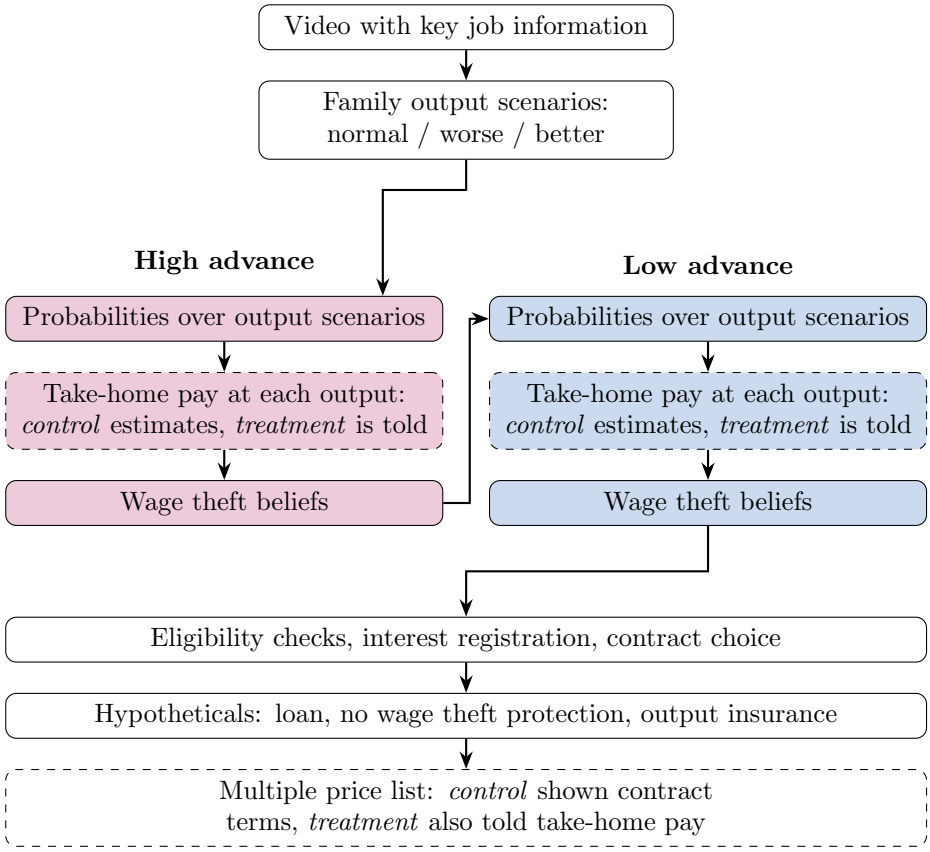
Note: Average rupee amount of the advance allocated to each spending area, among households who took an advance for the most recent brick kiln season, November 2025 to June 2026 ($N = 525$). Respondents selected up to five spending categories and divided ten stones across them in proportion to the amount spent on each. Households selecting a single category are assigned the full advance to it. A category's amount is its stone share multiplied by the household's advance (the advance per worker times the number of household members who migrated, winsorised at the 95th percentile). Bars group the detailed survey categories into broader areas. Within each bar, segments show the detailed categories from largest to smallest in shades of the bar colour, and segments averaging at least 10,000 Rs are labelled in bars with more than one category.

Figure A3: Question Sequence: Baseline Job Offer and Preferences over Contracts



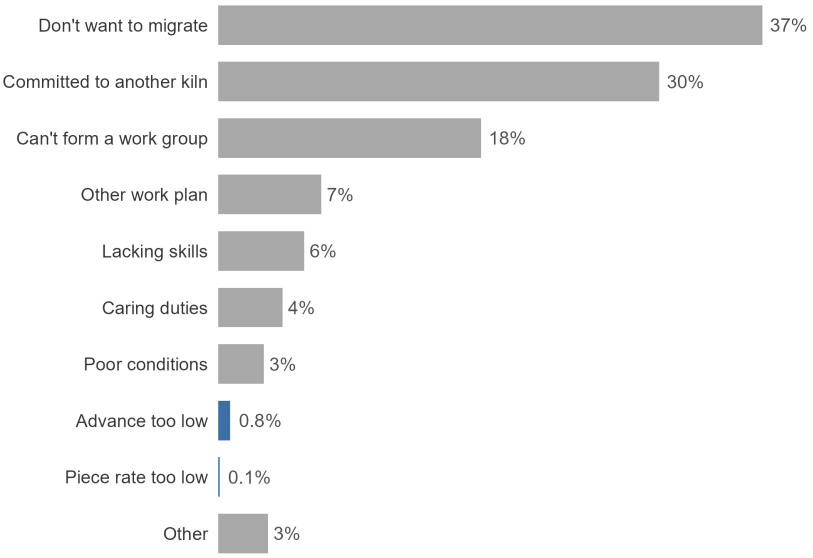
Note: This figure plots the sequence of questions in the baseline survey module covering respondents' expectations of and preferences over the advertised contracts, as well as additional hypothetical contracts. Shaded boxes mark the randomised treatment arms, whose contract terms are set out in Table 1.

Figure A4: Question Sequence: Round 2 Job Offer, Expectations and Application Decisions



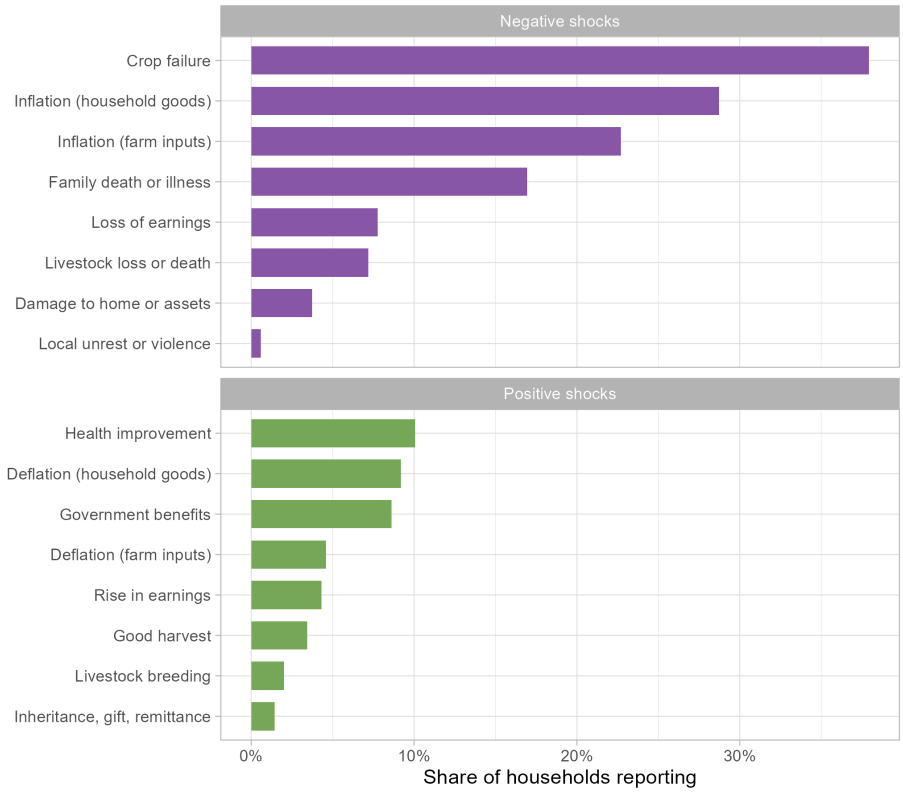
Note: This figure plots the sequence of questions in a survey module covering respondents' expectations of and preferences over the advertised contracts, as well as additional hypothetical contracts. Dashed boxes mark the points at which treatment and control diverge.

Figure A5: Reported Reason for not Applying to the Advertised Vacancy



Note: This figure plots the share of households citing each reason for not applying to the advertised vacancy, among the 1,105 households across all three treatment arms which did not successfully register interest, as defined in Section 4.2.3. Households could provide multiple reasons, so shares do not sum to 100%. The blue bars mark the two reasons relating to the advertised pay terms.

Figure A6: Unexpected Income Changes Between the Baseline and Revisit Surveys



Note: This figure plots the share of Choice arm households reporting each type of unexpected change in income between the baseline and revisit surveys ($N = 348$). Households could report more than one change, so shares do not sum to the share reporting any change.

Table A1: Effect of the Advertised Contract on Choices in the Hypothetical Multiple Price List

	Low advance piece rate						Index
	950	1,050	1,100	1,150	1,200	1,500	First switch
Panel A: Low advance vs. high advance arm							
Low advance arm	0.063***	0.057**	0.061*	0.047	0.042	0.061*	0.322*
	(0.024)	(0.028)	(0.030)	(0.033)	(0.033)	(0.032)	(0.162)
High advance arm mean	0.740	0.708	0.691	0.614	0.567	0.526	4.648
Fixed effects	Village	Village	Village	Village	Village	Village	Village
Num. obs.	1060	1060	1060	1060	1060	1060	1060
Panel B: Choice vs. single offer arms							
Choice arm	−0.077***	−0.112***	−0.124***	−0.092***	−0.087**	−0.126***	−0.656***
	(0.027)	(0.027)	(0.031)	(0.031)	(0.033)	(0.034)	(0.156)
Single offer arms mean	0.774	0.739	0.725	0.642	0.593	0.559	4.831
Fixed effects	District	District	District	District	District	District	District
Num. obs.	1494	1494	1494	1494	1494	1494	1494

Note: Household level regressions. In Columns 1 to 6, the dependent variable indicates preferring the high advance contract over the low advance alternative at the piece rate shown (Rs per 1,000 bricks) in the hypothetical multiple price list. In Column 7, it is the index of the first row at which the respondent prefers the low advance contract, equal to 7 if they never do. Panel A restricts the sample to the single offer arms, where contracts were randomised across households within villages, and includes village fixed effects. The high advance arm, which was shown no low advance contract before the multiple price list, is the omitted group. Panel B uses the full sample and compares the Choice arm, randomised across villages, to the two single offer arms combined, with district fixed effects. All regressions control for the gender, age and education of the respondent and for household size. Standard errors clustered at the village level. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table A2: Balance by Treatment Assignment

	RCT						Choice	
	Low advance		High advance					
Variable	N	Mean	N	Mean	Difference	P-value	N	Mean
Age of head	526	43.01	534	41.24	-1.77	0.032**	434	41.47
		(0.56)		(0.56)				(0.60)
Female head	526	0.36	534	0.33	-0.03	0.272	434	0.34
		(0.02)		(0.02)				(0.02)
Edu years of head	526	3.38	534	4.02	0.64	0.012**	434	3.48
		(0.16)		(0.18)				(0.18)
Household size	526	4.77	534	4.97	0.20	0.094*	434	5.25
		(0.08)		(0.08)				(0.09)
Scheduled Caste/Tribe	526	0.70	534	0.67	-0.03	0.345	434	0.74
		(0.02)		(0.02)				(0.02)
Expected Kharif income p.c.	526	5355.36	534	5614.45	259.09	0.671	434	6625.22
		(327.46)		(316.13)				(500.00)
Expected Rabi income p.c.	526	14595.26	534	14210.71	-384.55	0.673	434	15342.41
		(554.22)		(492.62)				(645.20)
Land (acres)	526	1.36	534	1.69	0.32	0.113	434	1.60
		(0.11)		(0.15)				(0.10)
Net savings p.c.	439	-3748.07	431	-4222.02	-473.95	0.682	368	-2544.02
		(589.02)		(599.61)				(714.80)
Migrating to kiln	526	0.65	534	0.62	-0.04	0.239	434	0.63
		(0.02)		(0.02)				(0.02)
Kiln years	526	8.10	534	7.59	-0.51	0.461	434	7.95
		(0.34)		(0.34)				(0.36)
Joint F-test p-value						0.013**		

Note: Standard errors are reported below each mean in parentheses. In the RCT panel, p-values are from OLS regressions of each baseline variable on the treatment indicator (high advance) with village fixed effects, clustering standard errors at the village level. The joint F-test reports the p-value from a Wald F-test of the null that all baseline variables jointly do not predict treatment assignment (high vs. low advance) in the RCT arm, estimated by regressing the treatment indicator on all baseline variables and village fixed effects, clustering standard errors at the village level. The Choice panel reports summary statistics for the choice treatment arm. This arm is omitted from the formal balance test, as it will not form part of RCT comparisons. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Attrition from the Baseline Sample: Surveyed and Non-Surveyed Households

	Not surveyed		Surveyed				
Variable	N	Mean	N	Mean	Baseline mean	Difference	P-value
Age of head	281	41.46	1083	42.12	41.93	0.66	0.427
		(0.70)		(0.40)		(0.83)	
Female head	281	0.41	1083	0.34	0.34	-0.07	0.006***
		(0.03)		(0.01)		(0.03)	
Edu years of head	281	3.32	1083	3.79	3.64	0.48	0.107
		(0.23)		(0.12)		(0.29)	
HH size	281	4.88	1083	4.99	4.98	0.11	0.370
		(0.11)		(0.06)		(0.12)	
Scheduled Caste/Tribe	281	0.68	1083	0.71	0.70	0.03	0.382
		(0.03)		(0.01)		(0.03)	
Land (acres)	281	1.56	1083	1.56	1.55	0.00	0.998
		(0.20)		(0.08)		(0.22)	
Debt p.c.	245	4833.45	971	5485.37	5266.20	651.92	0.414
		(660.26)		(369.39)		(792.31)	
Savings p.c.	242	1801.18	948	1920.84	1915.92	119.66	0.712
		(263.02)		(158.55)		(323.18)	
Max. personal loan available	281	6286.48	1083	5540.63	5568.95	-745.84	0.550
		(1260.69)		(559.00)		(1241.35)	
Any negative shock	281	0.51	1083	0.48	0.52	-0.04	0.413
		(0.03)		(0.02)		(0.04)	
Any positive shock	281	0.81	1083	0.82	0.83	0.01	0.666
		(0.02)		(0.01)		(0.03)	
Ever worked at kiln	281	0.90	1083	0.90	0.90	-0.00	0.961
		(0.02)		(0.01)		(0.02)	
Kiln experience	281	7.44	1083	8.23	7.87	0.79	0.195
		(0.44)		(0.24)		(0.60)	
Migrating to kiln	281	0.64	1083	0.66	0.64	0.02	0.547
		(0.03)		(0.01)		(0.03)	
Joint F-test p-value							0.181

Note: Sample: the 1364 baseline households in the 65 villages reached by the 2026 follow-up survey, of which 1083 were surveyed. Standard errors are reported below each mean in parentheses. Baseline mean is the mean over the full baseline sample of 1494 households. Difference is the coefficient from an OLS regression of each baseline variable on an indicator for being surveyed, with standard errors clustered at the village level, reported in parentheses below. The joint F-test reports the p-value from a Wald F-test of the null that all baseline variables jointly do not predict being surveyed, estimated by regressing the surveyed indicator on all variables, clustering standard errors at the village level. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table A4: Baseline Characteristics of the Round 1 and Round 2 Samples

	Round 1		Round 2			
Variable	N	Mean	N	Mean	Difference	P-value
Age of head	1494	41.93	1078	42.10	0.17	0.397
		(0.33)		(0.40)		
Female head	1494	0.34	1078	0.34	-0.01	0.397
		(0.01)		(0.01)		
Edu years of head	1494	3.64	1078	3.80	0.16	0.026**
		(0.10)		(0.12)		
HH size	1494	4.98	1078	4.99	0.00	0.877
		(0.05)		(0.06)		
Scheduled Caste/Tribe	1494	0.70	1078	0.71	0.01	0.478
		(0.01)		(0.01)		
Land (acres)	1494	1.55	1078	1.56	0.01	0.782
		(0.07)		(0.08)		
Debt p.c.	1307	5266.20	967	5439.55	173.34	0.470
		(320.50)		(368.61)		
Savings p.c.	1274	1915.92	944	1924.65	8.74	0.911
		(129.81)		(159.19)		
Max. personal loan available	1494	5568.95	1078	5547.78	-21.17	0.935
		(474.88)		(561.57)		
Any negative shock	1494	0.52	1078	0.48	-0.04	0.028**
		(0.01)		(0.02)		
Any positive shock	1494	0.83	1078	0.82	-0.00	0.532
		(0.01)		(0.01)		
Ever worked at kiln	1494	0.90	1078	0.90	-0.00	0.797
		(0.01)		(0.01)		
Kiln experience	1494	7.87	1078	8.23	0.36	0.049**
		(0.20)		(0.24)		
Migrating to kiln	1494	0.64	1078	0.66	0.02	0.104
		(0.01)		(0.01)		
Baseline MPL switch point	1494	4.66	1078	4.68	0.02	0.602
		(0.07)		(0.08)		
Choice arm at baseline	1494	0.29	1078	0.25	-0.04	0.115
		(0.01)		(0.01)		
Assigned high advance (RCT villages)	1060	0.50	807	0.50	0.00	0.949
		(0.02)		(0.02)		
Applied for baseline job (RCT villages)	1060	0.67	807	0.65	-0.01	0.312
		(0.01)		(0.02)		
Applied preferring low advance (Choice villages)	434	0.32	271	0.30	-0.02	0.368
		(0.02)		(0.03)		
Joint F-test p-value						0.048**

Note: Round 1 is the full 2025 baseline sample of 1494 households in 71 villages. Round 2 is the subset of 1078 households in 65 villages surveyed in the 2026 information experiment. All variables are measured at baseline. Standard errors are reported below each mean in parentheses. P-values are from OLS regressions of each variable on a Round 2 indicator in a stacked sample in which Round 2 households appear once in each round, clustering standard errors at the village level. The joint F-test reports the p-value from a Wald F-test of the null that the baseline variables jointly do not predict membership of the Round 2 sample, estimated by regressing a Round 2 indicator on all variables over the Round 1 sample, clustering standard errors at the village level. The final three variables are excluded from the joint test, as they are not defined for all baseline arms, and are compared on the subsamples on which they are defined (either RCT or Choice villages). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Brick Kiln Contracts and Payouts in Hypothetical Calculation Questions

	Advance per worker (Rs)	**Piece rate per 1,000 bricks (Rs)**	**Correct answer (Rs)**
1	50,000	650	175,000
2	5,000	650	310,000
3	5,000	900	435,000
4	5,000	1,200	585,000

Note: Respondents are asked to calculate future earnings under each of the four contracts, net of advance repayments, for a group of three workers producing 500,000 bricks across the season. They are told to assume workers are paid fairly. The order in which questions are answered (ascending vs descending) is randomised.

Table A6: Hypothetical Multiple Price List in Round 2

	High advance (50,000 Rs per worker)	Low advance (5,000 Rs per worker)
	Piece rate (Rs per 1,000 bricks)	
1		650
2		900
3	650	1,200
4		1,700
5		3,500

Note: In each row, the respondent chooses between the two contracts shown. Whereas the high advance contract is fixed across rows at the real advertised counterpart, the low advance piece rate changes. Row 2 corresponds to the advertised low advance contract.

Table A7: Balance by Treatment Assignment in the Follow-Up Sample

	Control		Treatment			
Variable	N	Mean	N	Mean	Difference	P-value
Age of head	536	41.86	542	42.33	0.47	0.435
		(0.56)		(0.56)		
Female head	536	0.35	542	0.32	-0.03	0.184
		(0.02)		(0.02)		
Edu years of head	536	3.96	542	3.63	-0.33	0.150
		(0.18)		(0.16)		
HH size	536	5.00	542	4.98	-0.02	0.874
		(0.08)		(0.08)		
Scheduled Caste/Tribe	536	0.70	542	0.72	0.01	0.683
		(0.02)		(0.02)		
Land (acres)	536	1.60	542	1.52	-0.08	0.497
		(0.12)		(0.12)		
Debt p.c.	483	5358.50	484	5520.42	161.92	0.953
		(530.72)		(512.24)		
Savings p.c.	470	2048.15	474	1802.20	-245.95	0.545
		(229.66)		(220.69)		
Max. personal loan available	536	5803.18	542	5295.20	-507.98	0.506
		(1014.77)		(491.28)		
Any negative shock	536	0.45	542	0.51	0.06	0.061*
		(0.02)		(0.02)		
Any positive shock	536	0.81	542	0.83	0.02	0.400
		(0.02)		(0.02)		
Ever worked at kiln	536	0.88	542	0.91	0.03	0.111
		(0.01)		(0.01)		
Kiln experience	536	7.91	542	8.55	0.64	0.139
		(0.34)		(0.35)		
Migrating to kiln	536	0.64	542	0.67	0.03	0.506
		(0.02)		(0.02)		
Assigned high advance at baseline	405	0.52	402	0.49	-0.02	0.602
		(0.02)		(0.02)		
Applied for baseline job (RCT villages)	405	0.64	402	0.67	0.02	0.532
		(0.02)		(0.02)		
Applied preferring low advance (Choice villages)	131	0.31	140	0.29	-0.01	0.675
		(0.04)		(0.04)		
Joint F-test p-value						0.036**

Note: Sample: the 1078 households surveyed in the 2026 follow-up. Standard errors are reported below each mean in parentheses. P-values are from OLS regressions of each baseline variable on an indicator for assignment to the information treatment, with randomisation-block (village × baseline-MPL stratum) fixed effects, clustering standard errors at the village level. The joint F-test reports the p-value from a Wald F-test of the null that all baseline variables jointly do not predict treatment assignment, estimated by regressing the treatment indicator on all variables and randomisation-block fixed effects, clustering standard errors at the village level. The baseline high advance indicator and the final two variables are excluded from the joint test, as they are not defined for all baseline arms. These are tested on the subsamples on which they are defined (either single-offer or Choice arms). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Balance of Baseline Characteristics by Income Shock Between Survey Waves

		Negative shock		Positive shock	
	Mean	Diff.	p	Diff.	p
Demographics					
Age of head	41.882	1.611	0.304	1.215	0.462
Female head	0.351	-0.168	0.029**	0.087	0.331
Edu years of head	3.402	0.447	0.241	0.503	0.451
Household size	5.336	0.001	0.997	-0.022	0.918
Scheduled Caste/Tribe	0.730	-0.132	0.039**	0.001	0.989
Household finances					
Expected Kharif income p.c. (000s)	5.164	2.238	0.026**	2.345	0.037**
Expected Rabi income p.c. (000s)	14.203	-0.935	0.378	0.291	0.780
Land (acres)	1.518	-0.254	0.386	0.296	0.340
Savings p.c. (000s)	1.739	-0.622	0.178	0.589	0.194
Debt p.c. (000s)	3.699	1.023	0.400	-1.271	0.157
Borrowing capacity (000s)	3.674	0.869	0.139	0.499	0.490
Baseline job preferences					
Interested (base)	0.730	0.168	0.086*	0.105	0.110
Net high adv preference (base)	0.086	-0.171	0.276	0.131	0.290
Kiln experience and middlemen					
Migrating to kiln	0.649	-0.073	0.216	-0.048	0.427
Kiln years	8.152	0.154	0.863	-1.131	0.212
Middlemen known	3.078	1.186	0.008***	0.236	0.350
Kiln years with main middleman	3.790	0.115	0.819	-0.688	0.228
Time preferences					
Time inconsistent	0.078	-0.017	0.642	-0.077	< 0.001***
Switch point now vs 2wk	3.204	0.048	0.754	0.007	0.961
Switch point 2wk vs 4wk	3.382	0.197	0.116	0.260	0.038**
Past shocks					
Neg shock Kharif 25	0.399	0.201	0.039**	0.183	0.029**
Pos shock Kharif 25	0.750	0.153	0.012**	0.049	0.388
Neg shock Kharif 24	0.583	0.071	0.297	0.142	0.051*
Pos shock Kharif 24	0.649	0.192	< 0.001***	0.199	< 0.001***
Survey					
Days between interviews	17.466	-4.228	0.022**	-4.527	0.008***
Joint test (p)			< 0.001		< 0.001

Note: The sample is the 348 Choice arm households surveyed in both waves. Shocks are unexpected income changes between the two surveys, reported at the revisit. Each p value comes from a regression of the characteristic on the shock indicator, with standard errors clustered at the village level and no village fixed effects. Money amounts are in thousands of Rs. Income, savings, debt, borrowing capacity, kiln years, middlemen known and kiln years with main middleman are winsorised at the 95th percentile of the full 1,494 household sample. Item nonresponse reduces the sample for Savings p.c. ($N = 313$), Debt p.c. ($N = 315$), Middlemen known ($N = 322$) and Kiln years with main middleman ($N = 329$), which enter the joint test mean imputed with a missing indicator alongside. The joint test regresses the shock indicator on all characteristics at once and is heteroskedasticity-robust, the village clustered variance matrix being rank deficient at this number of parameters. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table A9: Baseline Characteristics of Attriting and Revisited Households in the Choice Arm

Variable	Attrited	Revisited	Difference	P-value
Age of head	39.83	41.88	-2.06	0.223
	(1.33)	(0.68)		
Female head	0.30	0.35	-0.05	0.418
	(0.05)	(0.03)		
Edu years of head	3.78	3.40	0.38	0.338
	(0.40)	(0.20)		
Household size	4.90	5.34	-0.44	0.140
	(0.20)	(0.10)		
Scheduled Caste/Tribe	0.78	0.73	0.05	0.377
	(0.04)	(0.02)		
Expected Kharif income p.c.	9770.81	5847.86	3922.95	0.063*
	(1710.23)	(450.78)		
Expected Rabi income p.c.	15934.42	15196.11	738.30	0.724
	(1698.17)	(687.71)		
Land (acres)	1.92	1.52	0.40	0.006***
	(0.20)	(0.11)		
Net savings p.c.	-5643.25	-1916.07	-3727.18	0.203
	(2616.37)	(674.02)		
Migrating to kiln	0.57	0.65	-0.08	0.313
	(0.05)	(0.03)		
Kiln years	5.95	8.45	-2.49	0.010***
	(0.61)	(0.41)		
Neg shock Kharif 25	0.60	0.40	0.21	0.061*
	(0.05)	(0.03)		
Pos shock Kharif 25	0.83	0.75	0.08	0.129
	(0.04)	(0.02)		
Time inconsistent	0.10	0.08	0.03	0.430
	(0.03)	(0.01)		
Switch point now vs 2wk	3.23	3.20	0.03	0.846
	(0.13)	(0.07)		
Joint F-test p-value				0.001***

Note: The sample is the 434 Choice arm households at baseline, of which 86 lack a revisit observation. The first two columns report group means with standard errors in parentheses. Differences and p-values are from regressions of each variable on an attrition indicator without fixed effects, clustering standard errors at the village level. The joint F-test regresses the attrition indicator on all variables at once, mean-imputing missing values and including a missing indicator per affected covariate, with heteroskedasticity-robust errors. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Contract Preferences in the Choice Treatment Arm

	Interested	Registered
High advance	0.10**	0.10***
	(0.04)	(0.03)
Household FE	Yes	Yes
Low advance mean	0.320	0.122
Num. obs.	868	868
Num. groups: hh_id	434	434
R^2	0.22	0.41
Adj. R^2	−0.57	−0.17

Note: Choice treatment arm households, one observation per household per contract. The dependent variable indicates applying for the vacancy with a preference for the contract, equal to zero for the non-preferred contract and for both contracts if the household does not apply. All regressions include household fixed effects. Standard errors clustered at the household level. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table A11: Preferences over Contracts in the Hypothetical Multiple Price List

	Piece rate	Piece rate	Row index	Row index
Low advance piece rate	−0.424***	−0.469***		
	(0.023)	(0.023)		
Row index			−0.046***	−0.051***
			(0.002)	(0.003)
Respondent FE	Yes	Yes	Yes	Yes
Sample	All	Single switchers	All	Single switchers
Outcome mean	0.645	0.666	0.645	0.666
Num. obs.	8964	7956	8964	7956
Num. groups: hh_id	1494	1326	1494	1326
R^2	0.739	0.808	0.743	0.812
Adj. R^2	0.687	0.769	0.692	0.775

Note: One observation per respondent per row of the hypothetical multiple price list. The dependent variable indicates preferring the high advance contract over the low advance alternative in that row. All regressions include respondent fixed effects. Standard errors clustered at the respondent level. In Columns 1 and 2, the regressor is the piece rate attached to the low advance contract, expressed in Rs per brick. In Columns 3 and 4, the regressor is instead the index of the hypothetical low advance contract. Columns 2 and 4 restrict the sample to single switchers, i.e. respondents who never switch back to the high advance contract as the low advance piece rate rises. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table A12: Effect of High Advance on Applications in the Single Offer Arms: Robustness to Post-Double-Selection Lasso

	Interested	Interested	Registered	Registered
High advance	−0.04	−0.02	−0.04	−0.02
	(0.03)	(0.03)	(0.03)	(0.03)
PDS controls	No	Yes	No	Yes
Control mean	0.684	0.684	0.243	0.243
Num. obs.	1060	1060	1060	1060
R^2	0.20	0.40	0.10	0.19
Adj. R^2	0.14	0.36	0.04	0.14

Note: Standard errors clustered at village level. All regressions include village fixed effects and controls for age, gender, and education of household head and household size. Piece rates in the advertised vacancy were 1100 Rs (high advance contract) and 1250 (low advance). In columns marked Yes for PDS controls, additional controls are selected by Post-Double Selection Lasso following Cilliers, Elashmawy & McKenzie (2024), using the plug-in penalty. The lasso selects 5 and 3 additional controls in columns 2 and 4, respectively. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table A13: Effect of Contract on Expected Amenities and Repayment in the Single Offer Arms

	Expected amenities							Repayment	
	Education	Electricity	Toilet	Free to leave	Medical	Water	Count	Repay on time	Others abscond
High advance	0.004	−0.018	−0.063**	−0.012	−0.014	−0.005	−0.108	−0.009	−0.003
	(0.027)	(0.014)	(0.030)	(0.029)	(0.016)	(0.007)	(0.075)	(0.009)	(0.021)
Low advance mean	0.506	0.913	0.785	0.409	0.935	0.989	4.536	0.985	0.209
Num. obs.	1060	1060	1060	1060	1060	1060	1060	1060	1060

Note: Standard errors are clustered at the village level. All specifications include village fixed effects and controls for age, gender and education of household head and household size. In columns 1 to 6, the dependent variable is an indicator for whether the respondent expects the amenity to be available at the kiln, and in column 7 the number of these six amenities expected. A joint test of the high advance coefficients across columns 1 to 6, from a stacked regression, gives F = 1.35 (p = 0.232). In column 8, the dependent variable is an indicator for expecting to repay the advance early or on time. In column 9, it is an indicator for rating it likely or very likely that other workers leave the kiln without repaying their advance. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table A14: Demand for High Advance Contracts by Borrowing Capacity

	Single offer		Choice		MPL	
	Interested	Registered	Interested	Registered	Prefers high advance	Prefers high advance
Panel A: Can borrow						
High advance	0.041	−0.011	0.187***	0.160***		
	(0.032)	(0.035)	(0.054)	(0.031)		
High advance x Can borrow	−0.160***	−0.056	−0.221**	−0.143**		
	(0.054)	(0.047)	(0.081)	(0.053)		
Can borrow	0.137***	−0.032			−0.055**	−0.057**
	(0.045)	(0.039)			(0.025)	(0.027)
R^2	0.182	0.096	0.231	0.422	0.111	0.140
Adj. R^2	0.137	0.046	-0.544	-0.160	0.103	0.131
Panel B: Maximum loan						
High advance	0.016	−0.011	0.170***	0.137***		
	(0.029)	(0.032)	(0.048)	(0.025)		
High advance x Max loan	−0.014***	−0.007	−0.020***	−0.010**		
	(0.004)	(0.004)	(0.007)	(0.004)		
Max loan	0.016***	0.003			−0.005**	−0.005**
	(0.003)	(0.003)			(0.002)	(0.002)
R^2	0.188	0.094	0.233	0.419	0.111	0.140
Adj. R^2	0.143	0.043	-0.540	-0.166	0.103	0.131
Sample	Single offer	Single offer	Choice	Choice	All	Single switch
Low advance mean	0.684	0.243	0.320	0.122	0.355	0.334
Num. obs.	1060	1060	868	868	8964	7956

Note: Standard errors clustered at the village level. Columns 1 and 2 estimate equation 9 on the single offer arms, with village fixed effects and controls for the gender, age and education of the respondent and for household size. Columns 3 and 4 estimate equation 10 on the Choice arm, with one observation per household per contract and household fixed effects. Columns 5 and 6 estimate equation 11 on the multiple price list rows, with contract row and village fixed effects, the same controls and an indicator for ascending elicitation order. Column 6 restricts to respondents who switch contracts at most once. Can borrow (Panel A) indicates a positive reported maximum one month loan amount. Max loan (Panel B) is the reported maximum one month loan amount in thousands of Rs, winsorised at the 95th percentile (20,000 Rs). The low advance mean row reports the application rate in the low advance arm (columns 1 and 2), the share applying with a preference for the low advance contract (columns 3 and 4) and the share of row choices favouring the low advance contract (columns 5 and 6). $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table A15: Income Shocks and Net Contract Preferences: Alternative Shock Definitions, Clustering and Fixed Effects

	Negative only (1)	Positive only (2)	Joint (3)	Village clustering (4)	Village trends (5)
Panel A: Expressed interest					
Negative shock × High advance × Post	0.298***		0.434***	0.434**	0.300**
	(0.108)		(0.119)	(0.175)	(0.138)
Positive shock × High advance × Post		−0.202	−0.373***	−0.373*	−0.233
		(0.130)	(0.144)	(0.188)	(0.147)
High advance × Post	−0.080	0.167***	−0.039	−0.039	
	(0.072)	(0.063)	(0.072)	(0.107)	
Panel B: Successful registration					
Negative shock × High advance × Post	0.183**		0.249***	0.249*	0.143
	(0.082)		(0.088)	(0.143)	(0.107)
Positive shock × High advance × Post		−0.082	−0.180*	−0.180*	−0.071
		(0.093)	(0.100)	(0.095)	(0.106)
High advance × Post	−0.117**	0.021	−0.097*	−0.097	
	(0.057)	(0.050)	(0.058)	(0.085)	
Clustering	Household	Household	Household	Village	Household
Village × wave × contract FE	No	No	No	No	Yes
Num. obs.	1392	1392	1392	1392	1392

Note: Every column estimates equation 12 on the household by contract by wave panel of the 348 Choice arm households surveyed in both waves, with household by contract and household by wave fixed effects throughout, and measures the between-wave change in net preference for the high advance contract relative to households reporting no shock of that sign. Panel A uses expressed interest and Panel B successful registration. Columns 1 and 2 enter one shock indicator at a time. Column 3 enters both jointly and reproduces column 1 of Table 3. Column 5 adds village by wave by contract fixed effects. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table A16: Income Shocks and Net Contract Preferences: Interacted Baseline Controls

	Interacted controls							
	None	Middlemen	Time preferences	Past shocks	Demographics	Survey gap	Baseline interest	All
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Expressed interest								
Negative shock × Post	0.434***	0.353***	0.451***	0.395***	0.347***	0.382***	0.449***	0.330***
	(0.119)	(0.122)	(0.119)	(0.120)	(0.123)	(0.122)	(0.121)	(0.126)
Positive shock × Post	−0.373***	−0.358**	−0.359**	−0.385***	−0.374**	−0.431***	−0.368**	−0.321**
	(0.144)	(0.143)	(0.145)	(0.148)	(0.145)	(0.147)	(0.144)	(0.152)
High advance × Post	−0.039	0.005	−0.054	−0.012	0.014	0.011	−0.050	0.007
	(0.072)	(0.074)	(0.073)	(0.075)	(0.074)	(0.076)	(0.077)	(0.081)
Panel B: Successful registration								
Negative shock × Post	0.249***	0.214**	0.254***	0.219**	0.197**	0.216**	0.266***	0.180*
	(0.088)	(0.091)	(0.088)	(0.092)	(0.094)	(0.092)	(0.089)	(0.098)
Positive shock × Post	−0.180*	−0.170*	−0.188*	−0.193*	−0.188*	−0.217**	−0.175*	−0.192*
	(0.100)	(0.101)	(0.102)	(0.101)	(0.102)	(0.100)	(0.100)	(0.109)
High advance × Post	−0.097*	−0.079	−0.097*	−0.075	−0.063	−0.065	−0.109*	−0.051
	(0.058)	(0.060)	(0.059)	(0.061)	(0.060)	(0.061)	(0.061)	(0.066)
Num. obs.	1392	1392	1392	1392	1392	1392	1392	1392

Note: Standard errors clustered at the household level. Every column estimates equation 12 on the household by contract by wave panel of the 348 Choice arm households surveyed in both waves, with household by contract and household by wave fixed effects. Panel A uses expressed interest and Panel B successful registration. Column 1 reproduces the headline estimates of Table 3. Columns 2 to 7 each add one block of baseline characteristics interacted with the high advance and post indicators, and column 8 adds all six blocks together. Controls are taken from Table A8, each block containing at least one characteristic that predicts shock status there: middlemen known, kiln years, migrating to kiln and kiln years with main middleman (2), time inconsistency and the two switch points (3), negative and positive shocks in the 2024 and 2025 Kharif seasons (4), female head, Scheduled Caste or Tribe and expected Kharif income (5), days between interviews (6), baseline interest in the vacancy (7). $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table A17: Income Shocks and Net Contract Preferences: Placebo Shocks and Baseline Outcome Trends

	Headline	Placebo alone	Placebo and actual shocks	Baseline outcome trends
	(1)	(2)	(3)	(4)
Negative shock × High advance × Post	0.434***		0.419***	0.203**
	(0.119)		(0.119)	(0.080)
Positive shock × High advance × Post	−0.373***		−0.396***	−0.169*
	(0.144)		(0.148)	(0.101)
Negative shock (Kharif 24) × High advance × Post		0.045	0.070	
		(0.118)	(0.115)	
Positive shock (Kharif 24) × High advance × Post		0.108	0.095	
		(0.117)	(0.116)	
High advance × Post	−0.039	0.004	−0.125	0.033
	(0.072)	(0.098)	(0.104)	(0.044)
Baseline outcomes × High advance × Post	No	No	No	Yes
Num. obs.	1392	1392	1392	1392

Note: Standard errors clustered at the household level. Every column estimates equation 12 on the household by contract by wave panel of the 348 Choice arm households surveyed in both waves, with household by contract and household by wave fixed effects, for the expressed interest outcome. Column 1 reproduces column 1 of Table 3. Placebo shocks are negative and positive income shocks from the Kharif 24 season, resolved roughly twelve months before the revisit window. Column 2 replaces the interim shocks with the placebo shocks and column 3 enters the two sets jointly. Column 4 controls for baseline interest and baseline net preference for the high advance contract, each interacted with the high advance and revisit indicators, so the shock coefficients compare households starting from the same baseline outcome. The difference in differences specification and conditioning on the lagged outcome rest on opposite assumptions about selection into shocks, and the two estimates bound the effect. Conditioning on the lagged outcome differs from controlling for the overall baseline interest level, which leaves the headline estimates essentially unchanged. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table A18: Demand for High Advance Contracts by Time Preferences

	Single offer		Choice		MPL	
	Interested	Registered	Interested	Registered	Prefers high advance	Prefers high advance
Panel A: Present biased						
High advance	−0.040	−0.043	0.113**	0.116***		
	(0.033)	(0.030)	(0.044)	(0.024)		
High advance × Present biased	−0.019	0.044	−0.196	−0.171		
	(0.082)	(0.080)	(0.159)	(0.150)		
Present biased	0.016	−0.099			−0.195***	−0.221***
	(0.052)	(0.059)			(0.048)	(0.055)
R^2	0.173	0.095	0.221	0.417	0.122	0.153
Adj. R^2	0.127	0.044	-0.563	-0.169	0.114	0.144
Panel B: Always chooses earlier, sooner date today						
High advance	−0.037	−0.045	−0.030	−0.000		
	(0.060)	(0.041)	(0.109)	(0.053)		
High advance × Always earlier	−0.006	0.009	0.183	0.146*		
	(0.068)	(0.048)	(0.134)	(0.072)		
Always earlier	0.036	0.039			0.099***	0.107***
	(0.040)	(0.035)			(0.028)	(0.030)
R^2	0.174	0.094	0.226	0.421	0.117	0.148
Adj. R^2	0.128	0.043	-0.554	-0.161	0.109	0.139
Panel C: Always chooses earlier, sooner date in two weeks						
High advance	−0.079	−0.070	−0.096	0.017		
	(0.051)	(0.043)	(0.101)	(0.072)		
High advance × Always earlier	0.053	0.043	0.262**	0.114		
	(0.058)	(0.058)	(0.123)	(0.085)		
Always earlier	−0.016	0.052			0.116***	0.128***
	(0.044)	(0.049)			(0.030)	(0.033)
R^2	0.174	0.098	0.232	0.418	0.120	0.151
Adj. R^2	0.128	0.048	-0.540	-0.168	0.112	0.142
Sample	Single offer	Single offer	Choice	Choice	All	Single switch
Low advance mean	0.684	0.243	0.320	0.122	0.355	0.334
Num. obs.	1060	1060	868	868	8964	7956

Note: Standard errors clustered at the village level. Columns 1 and 2 estimate equation 9 on the single offer arms, with village fixed effects and controls for the gender, age and education of the respondent and for household size. Columns 3 and 4 estimate equation 10 on the Choice arm, with one observation per household per contract and household fixed effects. Columns 5 and 6 estimate equation 11 on the multiple price list rows, with contract row and village fixed effects, the same controls and an indicator for ascending elicitation order. Column 6 restricts to respondents who switch contracts at most once. All three moderators come from two hypothetical lists in which respondents traded 100 Rs at a sooner date against 150, 200 or 250 Rs two weeks later, with the sooner date set first to today and then to two weeks. Always earlier indicates never taking the later payment in the list named in the panel heading. Present biased indicates requiring a larger later payment to wait when the sooner payment is available today than when it arrives in two weeks, so Panel A isolates the differential between the two horizons rather than the sum of Panels B and C. The low advance mean row reports the application rate in the low advance arm (columns 1 and 2), the share applying with a preference for the low advance contract (columns 3 and 4) and the share of row choices favouring the low advance contract (columns 5 and 6). $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table A19: Calculation mistakes and years of education

	Advance deduction		Piece-rate slope	
	(1)	(2)	(3)	(4)
Constant	−0.496***	−0.556***	−0.566***	−0.647***
	(0.028)	(0.031)	(0.020)	(0.024)
Years of education		0.017***		0.023***
		(0.005)		(0.004)
Num. obs.	1078	1078	1078	1078

Note: Estimates are derived from hypothetical questions in which individuals estimate season take-home for a group of three producing 500,000 bricks. The advance deduction is the estimate at the 5,000 Rs advance minus the estimate at 50,000 Rs, both at 650 Rs per 1,000 bricks (correct value 135,000 Rs). The piece-rate slope is the respondent-level slope of the estimate on the piece rate across the three 5,000 Rs advance contracts (correct value 500). Each outcome is the deviation from the correct value as a share of the correct value, so 0 is no mistake and a positive coefficient means a smaller one. The constant is the average mistake in Columns 1 and 3, and the mistake for a respondent with no schooling in Columns 2 and 4. The columns include no fixed effects, so the constant is reported, with standard errors clustered at the village level. Adding village fixed effects leaves the education coefficients at 0.014 and 0.024.

Table A20: Characteristics of Households With and Without the Final Pay Estimate

	Not asked		Asked			
Variable	N	Mean	N	Mean	Difference	P-value
Age of head	553	42.48	525	41.70	-0.79	0.348
		(0.57)		(0.55)		
Female head	553	0.34	525	0.33	-0.01	0.846
		(0.02)		(0.02)		
Edu years of head	553	3.65	525	3.95	0.31	0.276
		(0.17)		(0.17)		
HH size	553	5.17	525	4.80	-0.37	0.007***
		(0.08)		(0.07)		
Scheduled Caste/Tribe	553	0.70	525	0.71	0.01	0.863
		(0.02)		(0.02)		
Land (acres)	553	1.38	525	1.75	0.37	0.083*
		(0.08)		(0.15)		
Debt p.c.	552	6423.52	415	4130.75	-2292.77	0.007***
		(538.83)		(466.35)		
Savings p.c.	553	1840.19	391	2044.12	203.93	0.551
		(221.88)		(222.10)		
Max. personal loan available	553	4922.97	525	6205.90	1282.93	0.255
		(279.58)		(1114.71)		
Any negative shock	553	0.35	525	0.61	0.26	0.000***
		(0.02)		(0.02)		
Any positive shock	553	0.81	525	0.83	0.02	0.549
		(0.02)		(0.02)		
Ever worked at kiln	553	0.95	525	0.84	-0.10	0.001***
		(0.01)		(0.02)		
Kiln experience	553	9.91	525	6.46	-3.45	0.000***
		(0.35)		(0.33)		
Migrating to kiln	553	0.76	525	0.55	-0.21	0.000***
		(0.02)		(0.02)		
Baseline MPL switch point	553	4.92	525	4.43	-0.49	0.024**
		(0.10)		(0.12)		
Assigned information treatment	553	0.49	525	0.51	0.02	0.266
		(0.02)		(0.02)		
Calculation error (mean abs. % error)	553	0.62	525	0.58	-0.03	0.161
		(0.01)		(0.02)		
Expected typical output (bricks)	553	320885.17	525	300476.19	-20408.98	0.150
		(9887.77)		(5443.33)		
Migrated to kiln last season	553	0.58	525	0.53	-0.05	0.192
		(0.02)		(0.02)		
Negative shock, past 3 months	553	0.31	525	0.32	0.00	0.933
		(0.02)		(0.02)		
Assigned high advance at baseline	385	0.51	422	0.50	-0.01	0.526
		(0.03)		(0.02)		
Applied for baseline job (RCT villages)	385	0.60	422	0.71	0.11	0.041**
		(0.03)		(0.02)		
Applied preferring low advance (Choice villages)	168	0.29	103	0.31	0.02	0.842
		(0.04)		(0.05)		
Take-home pay last season (migrants)	320	29159.69	276	18971.56	-10188.13	0.105
		(5322.10)		(2900.20)		
Joint F-test p-value						0.000***

Note: Sample: the 1078 households surveyed in the 2026 follow-up, of which 525 were asked the end-of-module take-home pay question, added to the survey mid-way through data collection. Villages were surveyed sequentially, so the two groups differ almost entirely between villages. The first fourteen variables and the baseline high advance indicator are measured at baseline. The remaining variables are measured in the follow-up survey before treatment content is delivered, except for the negative shock indicator, which refers to the three months before the follow-up. Standard errors are reported below each mean in parentheses. Differences and p-values are from OLS regressions of each variable on an indicator for being asked the question, without fixed effects, clustering standard errors at the village level. The joint F-test reports the p-value from a Wald F-test of the null that the variables jointly do not predict being asked, estimated by regressing the indicator on all variables, clustering standard errors at the village level. The baseline high advance indicator and the final three variables are excluded from the joint test, as they are not defined for all households, and are tested on the subsamples on which they are defined. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A21: Advance contract applications and productivity

	Low advance			High advance		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	0.170**	0.051	0.143*	−0.014	0.008	−0.018
	(0.068)	(0.038)	(0.074)	(0.067)	(0.033)	(0.061)
Productivity	0.049***		0.037**	0.005		0.010
	(0.015)		(0.017)	(0.016)		(0.014)
Treatment x productivity	−0.070***		−0.060**	0.018		0.018
	(0.022)		(0.024)	(0.021)		(0.020)
Tercile 2		0.104**			−0.025	
		(0.046)			(0.038)	
Tercile 3		0.093**			0.005	
		(0.045)			(0.044)	
Treatment x tercile 2		−0.157**			0.044	
		(0.069)			(0.044)	
Treatment x tercile 3		−0.122*			0.052	
		(0.066)			(0.053)	
Moderator interactions	No	No	Yes	No	No	Yes
Control mean	0.233	0.233	0.233	0.125	0.125	0.125
R^2	0.187	0.185	0.197	0.160	0.160	0.181
Adj. R^2	0.069	0.065	0.073	0.039	0.037	0.055
Num. obs.	1078	1078	1078	1078	1078	1078

Note: Productivity is the number of bricks which the respondent estimated their group would produce in a typical year in the advertised job, in units of 100,000. Columns 3 and 6 add four moderators which correlate with productivity, each entering in levels and interacted with treatment: the number of seasons the household has worked at brick kilns, the number of workers in the job group, the pre-treatment kiln calculation error and the maximum loan the respondent reports being able to obtain, all standardised. None of these interactions is significant at the 5 percent level in either column, the largest being that with the maximum loan in Column 3, at -0.039 ($p = 0.08$). Baseline demand controls are the baseline MPL switch point and baseline application behaviour interacted with baseline arm indicators. All columns include baseline demand controls and randomisation-block fixed effects, with standard errors clustered at the village level.

Table A22: Wage theft beliefs against the told pay amounts, low advance contract

	Control	Treatment	Difference	(SE)	N
Fair pay benchmark (1,000 Rs)	112.7	261.5	148.7***	(6.5)	1078
Any underpayment expected (%)	22.4	28.1	5.7**	(2.4)	1078
Probability of underpayment (%)	9.4	13.4	4.0***	(1.2)	1078
Loss if underpaid (1,000 Rs)	46.2	146.5	100.2***	(14.9)	277
Expected net loss to theft (1,000 Rs)	2.3	17.8	15.5***	(3.0)	1078
Implied expected pay (1,000 Rs)	111.6	243.7	132.1***	(6.9)	1078
Final take-home estimate (1,000 Rs)	125.1	136.4	11.3	(13.7)	525

Note: Each row is a separate regression of the outcome on treatment with randomisation-block fixed effects, with standard errors clustered at the village level. The treatment column is the control mean plus the estimated coefficient. All outcomes refer to the low advance contract. The fair pay benchmark is take-home at normal output under full payment, estimated by control and told to treatment. The probability rows come from the ten-stone allocation across pay scenarios, classified by whether the stated amount falls below the fair benchmark. Loss if underpaid is the fair benchmark minus the stated take-home in underpayment years, among households placing any stone there. The expected net loss weights under- and overpayment gaps by their stone shares, and implied expected pay subtracts it from the fair benchmark. The final take-home estimate is a single question asked after the theft module, available for the later cohort only. Rupee amounts are in 1,000 Rs, winsorised at the 1st and 99th percentiles. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A23: Treatment effect on the choice at each rung of the price list and on the switch point

	Chose high advance at low-advance piece rate					Switch point
	650	900	1,200	1,700	3,500	1 to 6
Treatment	0.024	−0.038	−0.020	0.005	0.006	−0.024
	(0.024)	(0.026)	(0.025)	(0.026)	(0.025)	(0.102)
Control mean	0.707	0.287	0.224	0.207	0.190	2.535
Num. obs.	1078	1078	1078	1078	1078	1078
R^2	0.148	0.203	0.234	0.235	0.224	0.194
Adj. R^2	0.026	0.090	0.125	0.126	0.114	0.080

Note: The outcome in Columns 1 to 5 is an indicator for preferring the high advance contract at the low-advance piece rate shown in the column header. The outcome in Column 6 is the switch point, the index of the first rung at which the respondent prefers the low advance contract, from 1 to 5, with 6 assigned to respondents who prefer the high advance contract at every rung. Baseline demand controls are the baseline MPL switch point and baseline application behaviour interacted with baseline arm indicators. All columns include baseline demand controls and randomisation-block fixed effects, with standard errors clustered at the village level.

Table A24: Heterogeneity in information treatment effects on job applications

	Demand for liquidity			Ability	
	Cannot borrow	Any neg. shock	Expects large advance	Numeracy	Education
	(1)	(2)	(3)	(4)	(5)
Panel A: Low advance application					
Treatment	-0.061	-0.028	-0.038	-0.041**	-0.021
	(0.037)	(0.027)	(0.023)	(0.019)	(0.033)
Moderator	-0.022	0.017	0.138***	-0.052***	0.006
	(0.042)	(0.040)	(0.046)	(0.018)	(0.005)
Treatment × moderator	0.029	-0.042	-0.021	0.024	-0.005
	(0.047)	(0.055)	(0.059)	(0.028)	(0.007)
Panel B: High advance application					
Treatment	-0.052	0.008	0.010	0.040**	0.049
	(0.038)	(0.022)	(0.024)	(0.018)	(0.030)
Moderator	-0.050	-0.028	0.068	0.040***	-0.005
	(0.037)	(0.035)	(0.045)	(0.015)	(0.004)
Treatment × moderator	0.136***	0.101**	0.114*	-0.022	-0.004
	(0.046)	(0.042)	(0.065)	(0.021)	(0.006)

Note: Each column reports one regression per panel of the application outcome on treatment, the moderator and their interaction, with baseline demand controls and randomisation-block fixed effects. Baseline demand controls are the baseline MPL switch point and baseline application behaviour interacted with baseline arm indicators. Standard errors are clustered at the village level. Cannot borrow indicates that the household reports being unable to borrow any amount for one month. Expects large advance indicates an expected advance of at least 50,000 Rs per worker for the coming season, the amount offered under the advertised high advance contract and the median among positive expected advances. Numeracy is the mean absolute percentage error in the pre-treatment kiln calculation battery, standardised, and Education is measured in years. $N = 1{,}078$ households in every column. Control means are 0.233 for low advance and 0.125 for high advance applications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

A Model Appendix

A.1 Solution to the Worker's Problem

Under contract j, the worker chooses savings s_j to maximise

$$u_j(s_j) = \log(a + d_j - s_j) + \delta \log(f_j + (1 + r)s_j)$$

where $f_j = p_j o - d_j$ denotes period 2 pay. The first order condition yields

$$s_j^* = \frac{\delta(a + d_j) - \frac{f_j}{1+r}}{1 + \delta} \tag{A1}$$

Substituting into the budget constraint delivers the consumption rule reported in the main text, $c_1^j = \frac{W_j}{1+\delta}$ and $c_2^j = \frac{\delta(1+r)W_j}{1+\delta}$, and indirect utility

$$u_j(s_j^*) = (1 + \delta)\log(W_j) + K, \qquad K = \delta \log(\delta) + \delta \log(1 + r) - (1 + \delta)\log(1 + \delta) \tag{A2}$$

The cross-contract utility difference is $(1 + \delta)\log\left(\frac{W_{ha}}{W_{la}}\right)$, yielding the present value comparison stated in the main text.

A.2 Borrowing Constraint Cases

I now impose $s_j \geq b$ with $b \leq 0$, so that optimal savings become $\max\{b, s_j^*\}$. Under the benchmark where the two contracts are equivalent in present value, $s_{ha}^* > s_{la}^*$. This means that the borrowing constraint cannot bind under the high advance contract only, leaving three cases.

Case 1: the constraint never binds.

This is equivalent to the unconstrained case, where a worker is indifferent between contracts which are equivalent in present value terms.

Case 2: the constraint binds under the low advance contract only.

This is the case presented in the main text. Indirect utility under the high advance contract retains its unconstrained form, whilst under low advance it becomes $u_{la}(b) = \log(a + d_{la} - b) + \delta \log(f_{la} + (1 + r)b)$. The welfare loss from the binding constraint, defined in equation 3, is

$$\Lambda_{la} \equiv u_{la}(s_{la}^*) - u_{la}(b) = (1 + \delta)\log(W_{la}) + K - \log(a + d_{la} - b) - \delta \log(f_{la} + (1 + r)b) > 0 \tag{A3}$$

The worker chooses the high advance contract whenever $u_{ha}(s_{ha}^*) > u_{la}(b)$, which rearranges to the rule in equation 4. Where contracts have equal present value, this condition holds strictly.

Case 3: the constraint binds under both contracts.

Utility under contract j is now $u_j(b) = \log(a + d_j - b) + \delta \log(f_j + (1 + r)b)$. To see whether high advance contracts are favoured in this scenario, consider a one unit increase in the advance, met by a decrease in future pay of $1 + r$ so that the contract's present value is preserved. With savings fixed at b, consumption rises by one unit in period 1 and falls by $1 + r$ in period 2, yielding the following change in utility:

$$\frac{\partial u}{\partial d} = \frac{1}{c_1} - \frac{\delta(1 + r)}{c_2} > 0$$

The inequality follows from the binding constraint: a worker who wanted to borrow but could not consumes less today, and more tomorrow, than they would optimally choose, so the marginal utility of period 1 consumption exceeds its discounted period 2 counterpart. Since this holds at every point along the equal present value locus, the worker strictly prefers the high advance contract to a low advance alternative of equal present value.

A.3 The Wedge and the Discount Factor

The main text states that the wedge $W_{la}\left[1 - \exp\left(-\frac{\Lambda_{la}}{1+\delta}\right)\right]$ in Case 2 is decreasing in the discount factor if the worker is credit constrained under low advance. This is the result underlying the effect of present bias on contract choice. The wedge is increasing in $\frac{\Lambda_{la}}{1+\delta}$, so it suffices to show that this ratio falls with δ. Differentiating K gives $\frac{\partial K}{\partial \delta} = \log\left(\frac{\delta(1+r)}{1+\delta}\right)$, and hence

$$\begin{aligned}\frac{\partial \Lambda_{la}}{\partial \delta} &= \log(W_{la}) + \frac{\partial K}{\partial \delta} - \log(f_{la} + (1 + r)b) \\ &= \log\left(\frac{\delta(1 + r)W_{la}}{(1 + \delta)\big(f_{la} + (1 + r)b\big)}\right) = \log\left(\frac{c_2^{la}}{f_{la} + (1 + r)b}\right) < 0\end{aligned}$$

where the final equality uses the unconstrained consumption rule. The inequality holds as the binding constraint forces the worker to save more than is optimal, pushing period 2 consumption $f_{la} + (1 + r)b$ above its unconstrained level c_2^{la}. By the quotient rule,

$$\frac{\partial}{\partial \delta}\left[\frac{\Lambda_{la}}{1 + \delta}\right] = \frac{(1 + \delta)\frac{\partial \Lambda_{la}}{\partial \delta} - \Lambda_{la}}{(1 + \delta)^2} < 0$$

as the numerator sums two negative terms. The wedge therefore shrinks with patience. A present biased worker deciding in the present applies the effective factor $\beta\delta < \delta$, widening the wedge.

A.4 The Borrowing Limit and Initial Wealth

In Case 2, the utility gain from choosing the high advance contract is $D \equiv u_{ha}(s^*_{ha}) - u_{la}(b) = (1 + \delta)\log\left(\frac{W_{ha}}{W_{la}}\right) + \Lambda_{la}$. Only Λ_{la} depends on b, and

$$\frac{\partial D}{\partial b} = \frac{\partial \Lambda_{la}}{\partial b} = \frac{1}{a + d_{la} - b} - \frac{\delta(1 + r)}{f_{la} + (1 + r)b} > 0$$

The inequality follows from the binding constraint, as in Case 3. A tighter borrowing limit, with b closer to zero, therefore raises the relative value of the high advance contract.

Turning to initial wealth, the envelope theorem gives

$$\frac{\partial D}{\partial a} = \frac{1}{c_1^{ha}} - \frac{1}{a + d_{la} - b}$$

This is negative whenever $D \geq 0$. To see this, suppose instead that $c_1^{ha} \leq a + d_{la} - b$. The unconstrained consumption rule gives $c_2^{ha} = \delta(1+r)c_1^{ha}$, whilst the binding constraint implies $f_{la} + (1+r)b > \delta(1+r)(a + d_{la} - b)$. Together these give $c_2^{ha} < f_{la} + (1+r)b$. The high advance contract would then deliver weakly lower consumption in period 1 and strictly lower consumption in period 2, contradicting $D \geq 0$. A fall in initial wealth, such as a negative income shock realised before the contract is chosen, therefore raises the relative value of the high advance contract for any worker who weakly prefers it.

A.5 Stochastic Output

Suppose output is stochastic with mean $\bar{o}$ and variance σ_o^2, realised in period 2 after savings have been chosen. This captures shocks to individual productivity, such as illness, and to conditions at the kiln, such as adverse weather. Stochastic period 2 income eliminates the closed form solution for savings, so I rely on a second order Taylor expansion of period 2 utility around expected consumption, $E[\log(c_2)] \approx \log(\bar{c}_2) - \frac{\sigma_{c_2}^2}{2\bar{c}_2^2}$.[37] With $\sigma_{c_2}^2 = p_j^2 \sigma_o^2$, expected indirect utility becomes

$$E[u_j] \approx (1+\delta)\log(W_j) + K - \frac{\delta\, p_j^2 \sigma_o^2}{2\,[c_2^j]^2} \tag{A4}$$

where W_j is evaluated at expected output. The final term is the utility cost of consumption risk. Only piece rate earnings depend on realised output, the advance having been paid beforehand, so this cost rises with the piece rate. Under the equal present value benchmark, the low advance contract therefore carries the larger penalty, and a risk averse worker strictly prefers the high advance alternative.

[37] This approximation abstracts from the precautionary savings motive, wherein the worker would raise period 1 savings in response to period 2 risk.